



US012405089B2

(12) **United States Patent**
Underwood

(10) **Patent No.:** **US 12,405,089 B2**

(45) **Date of Patent:** **Sep. 2, 2025**

(54) **LOCATING FASTENER FOR FIREARM OPTIC**

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(71) Applicant: **AXTS INC**, Redmond, OR (US)

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(72) Inventor: **Joshua A. Underwood**, Redmond, OR (US)

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(73) Assignee: **AXTS, INC.**, Redmond, OR (US)

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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(21) Appl. No.: **18/480,425**

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(22) Filed: **Oct. 3, 2023**

Southwest Precision; Instagram; at least as early as Oct. 2, 2022.

(Continued)

(65) **Prior Publication Data**

US 2024/0110767 A1 Apr. 4, 2024

Primary Examiner — Bret Hayes

(74) *Attorney, Agent, or Firm* — Schwabe Williamson & Wyatt

Related U.S. Application Data

(60) Provisional application No. 63/417,236, filed on Oct. 18, 2022, provisional application No. 63/412,758, filed on Oct. 3, 2022.

(51) **Int. Cl.**
F41G 11/00 (2006.01)
F41A 21/36 (2006.01)

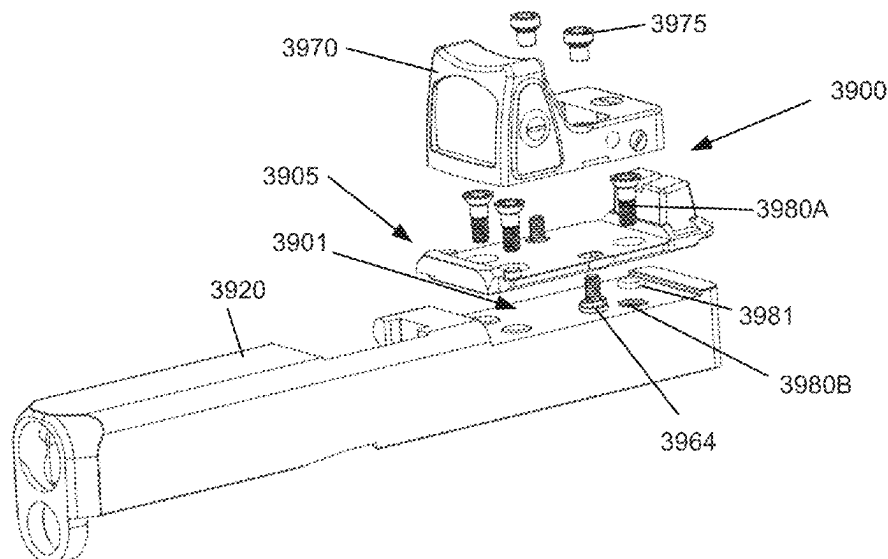
(52) **U.S. Cl.**
CPC **F41G 11/001** (2013.01); **F41A 21/36** (2013.01); **F41G 11/003** (2013.01)

(58) **Field of Classification Search**
CPC F41G 1/30; F41G 11/001; F41G 11/002; F41G 11/003
USPC 42/113
See application file for complete search history.

(57) **ABSTRACT**

In various embodiments, an apparatus comprises a locating optic assembly mount arranged to receive an optic or a bracket on which the optic is installed, wherein the locating optic assembly mount includes: one or more locating fastener openings to receive one or more locating fasteners with threaded sections and non-threaded shoulders, wherein the non-threaded shoulders extend through the optic or bracket and below a surface located below the optic or bracket, each locating fastener opening including: a threaded hole to mate with the threaded section of the corresponding locating fastener; and a counter opening located above the threaded hole, wherein the counter opening is arranged to slip fit with the non-threaded shoulder of the corresponding locating fastener to align a center axis of the locating fastener with a center axis of the locating fastener openings. Other embodiments may be disclosed and/or claimed.

30 Claims, 37 Drawing Sheets



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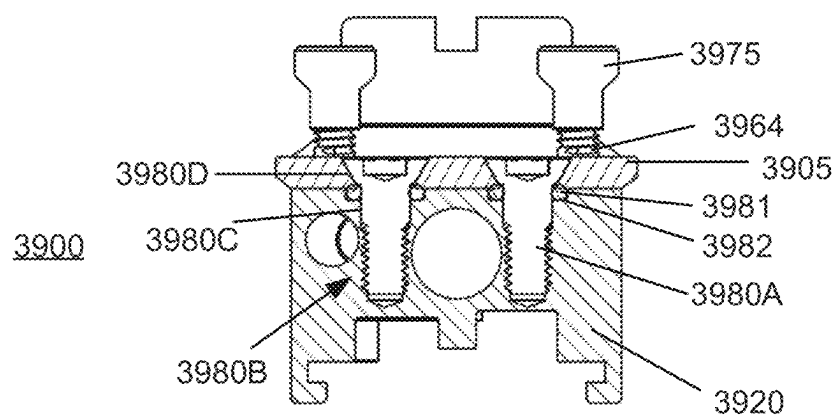
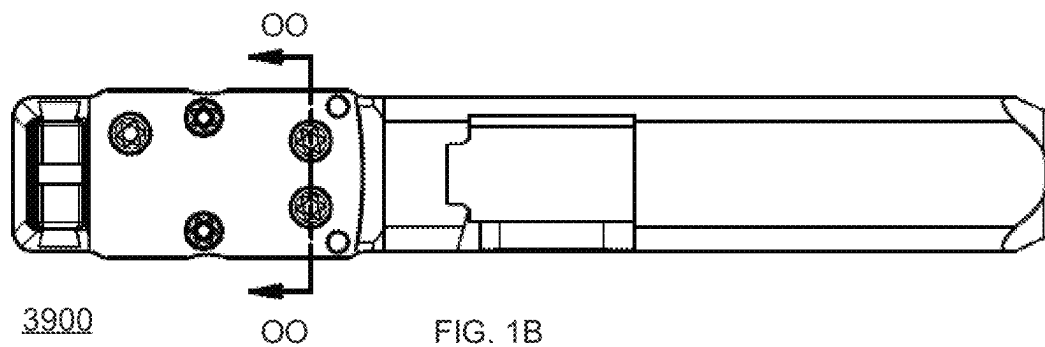
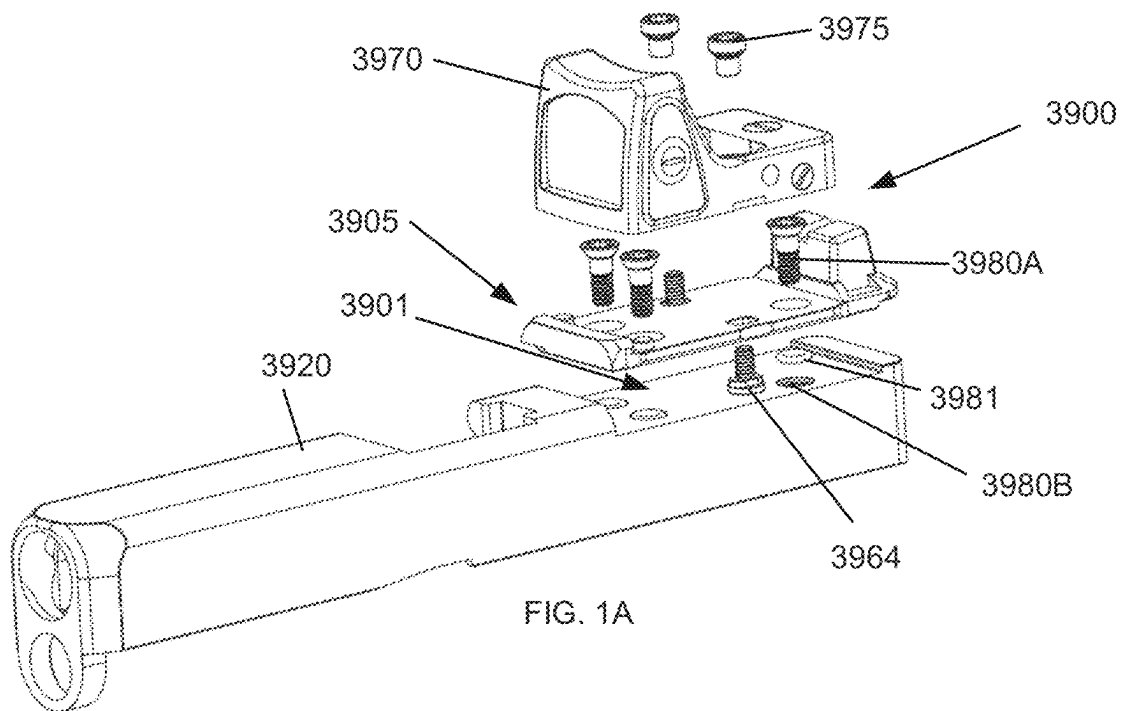


FIG. 1C

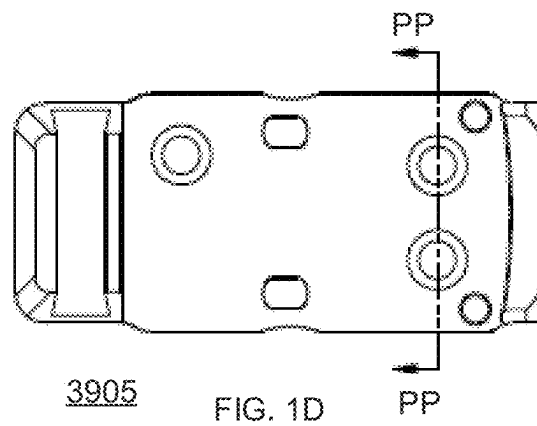


FIG. 1D

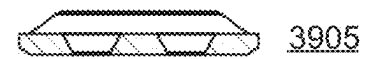


FIG. 1E

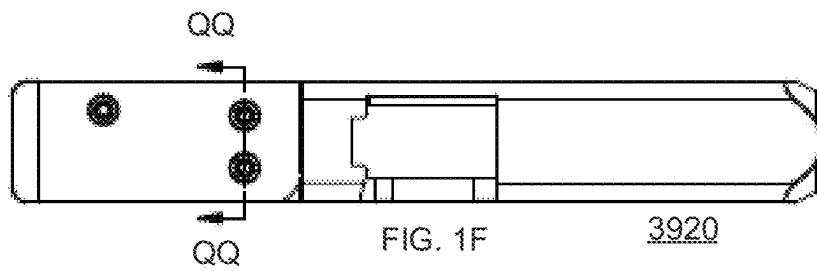


FIG. 1F

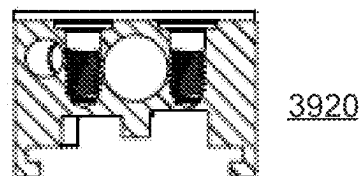


FIG. 1G

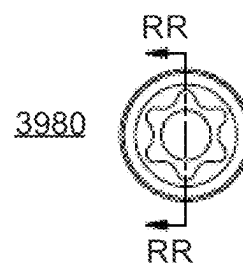


FIG. 1H

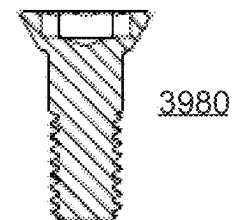


FIG. 1I

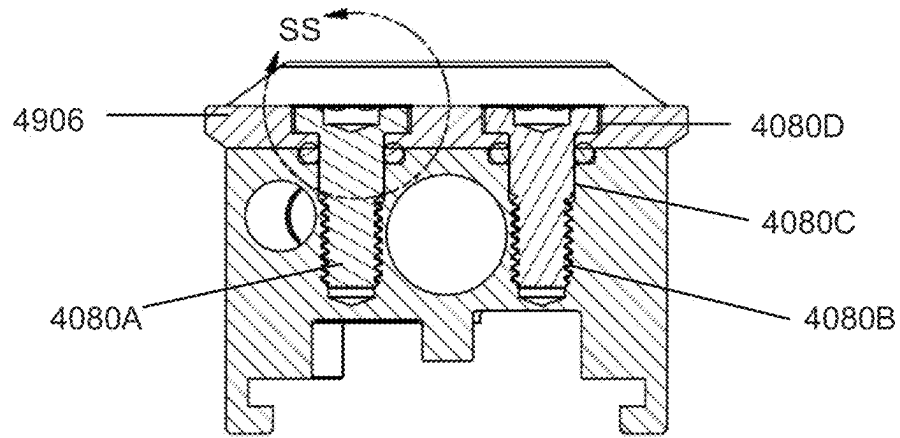


FIG. 2A

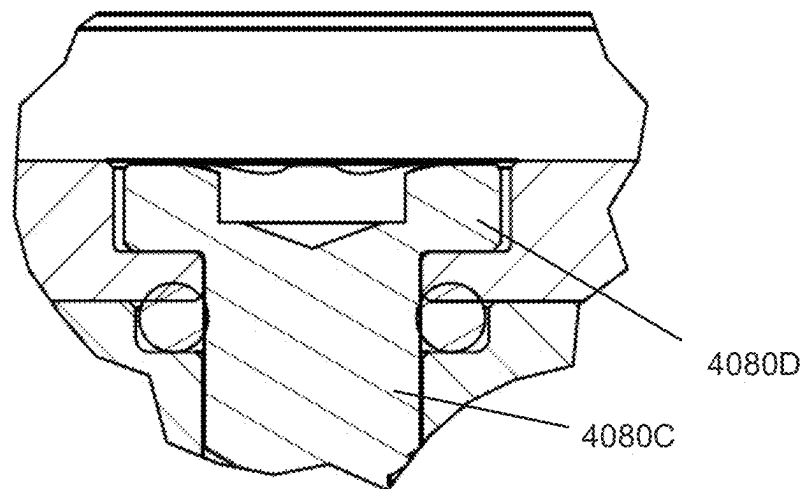
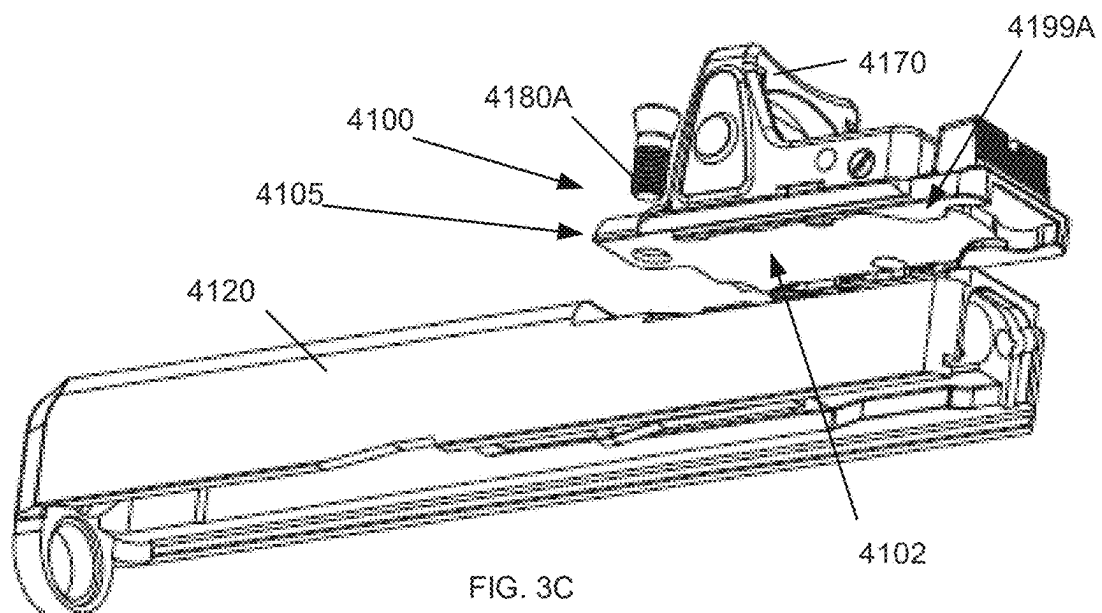
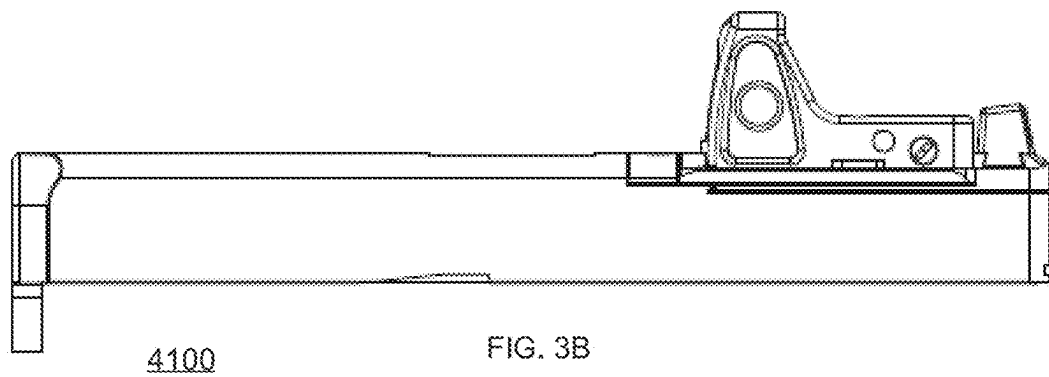
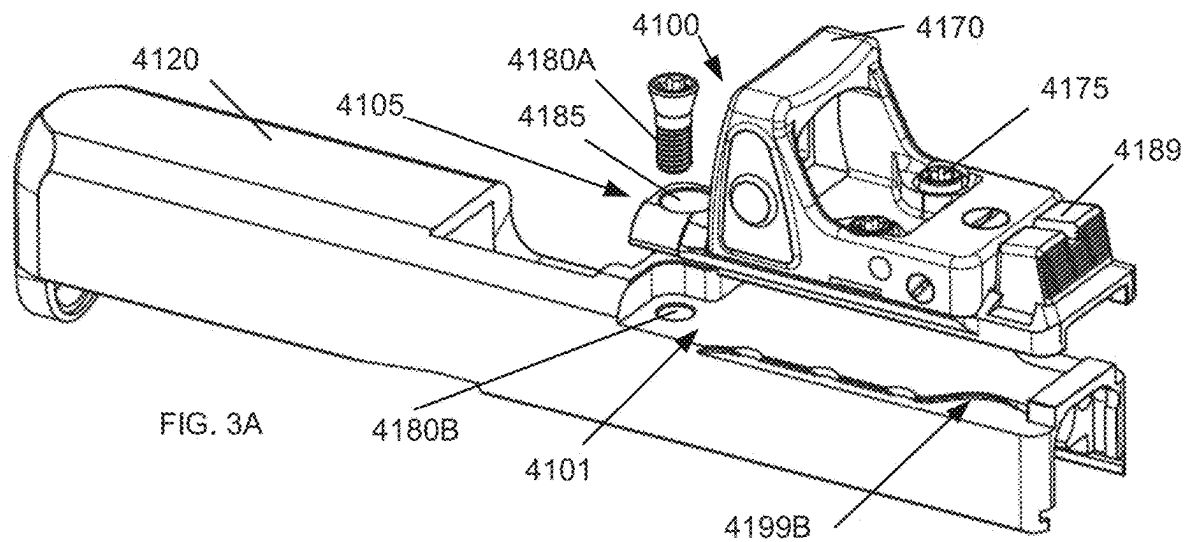
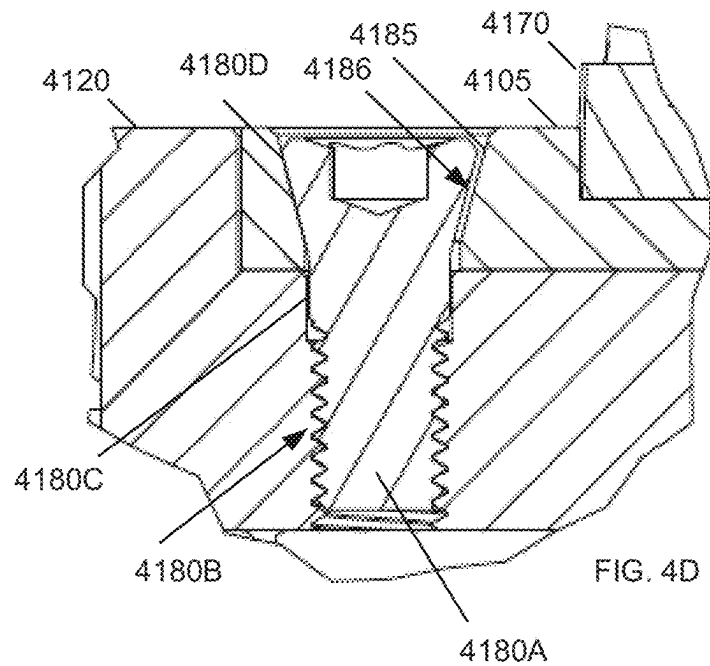
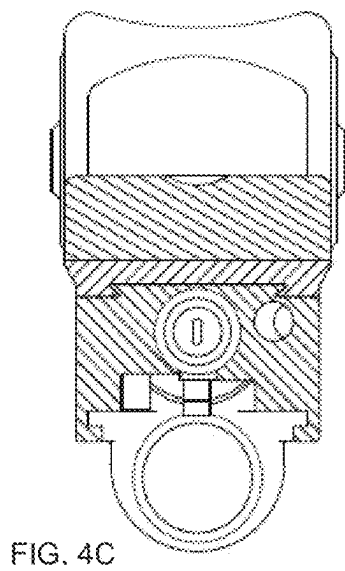
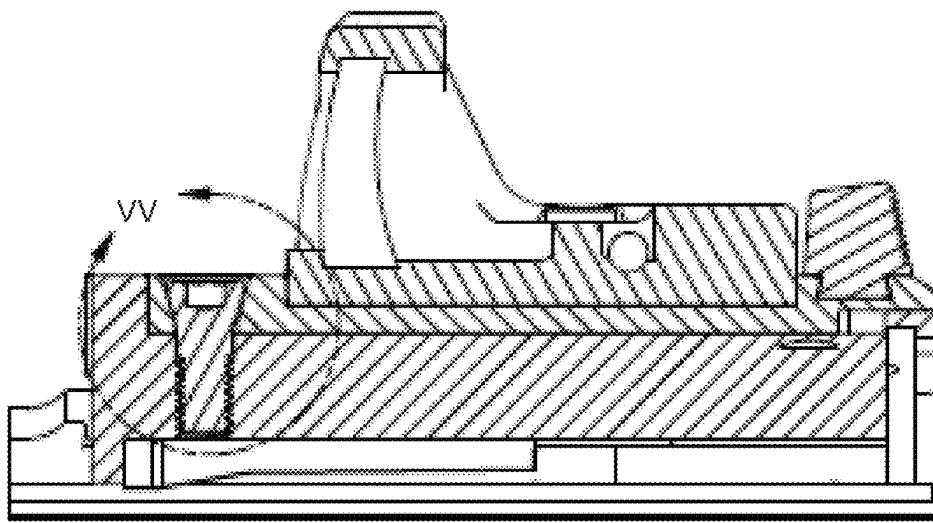
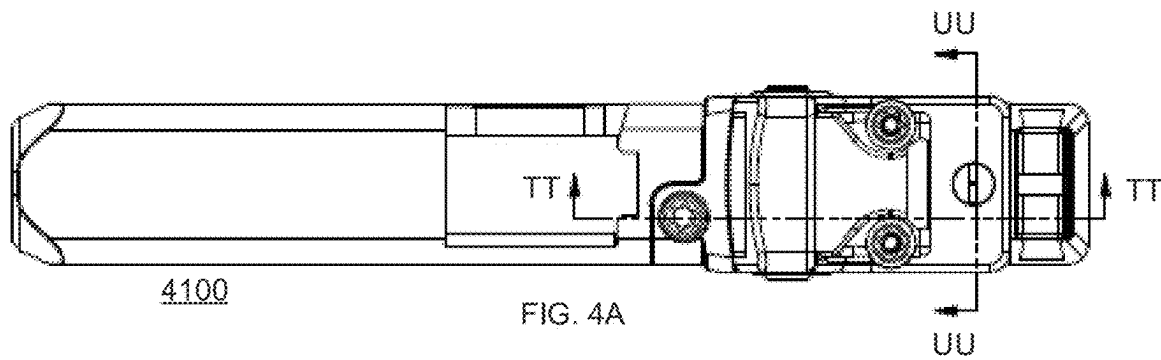
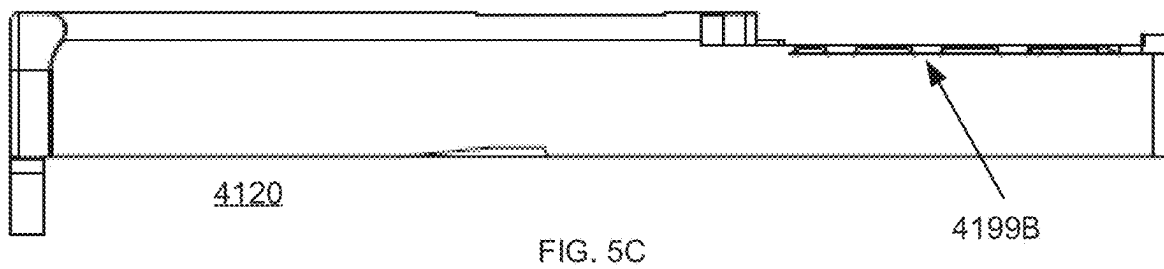
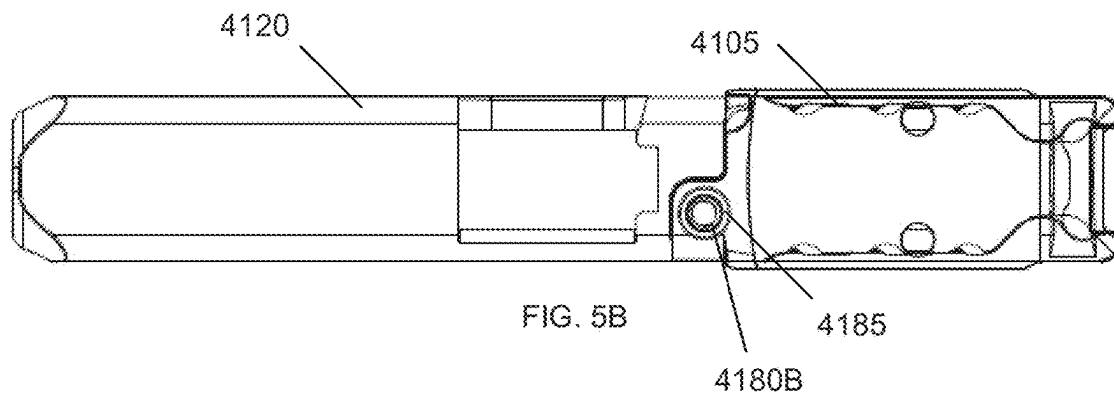
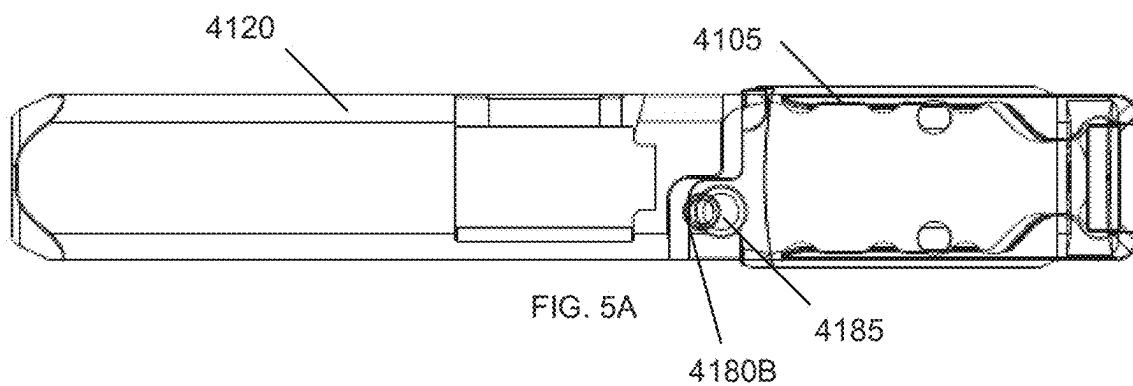
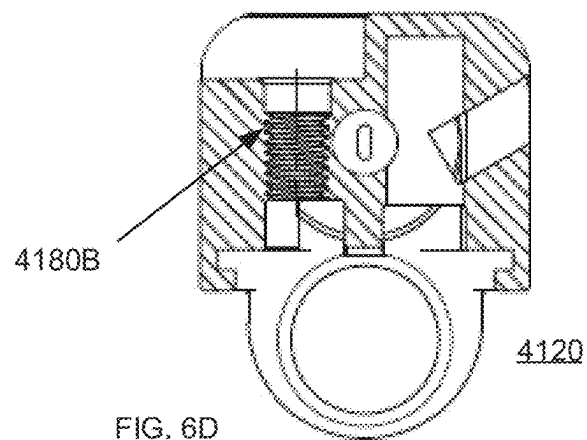
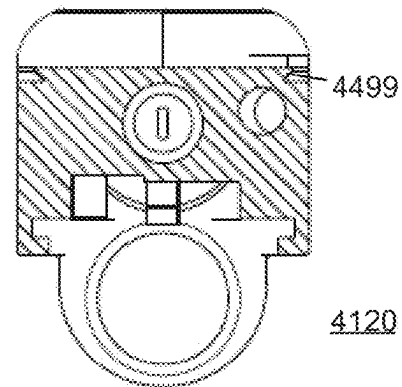
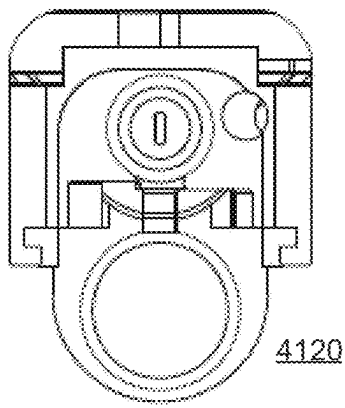
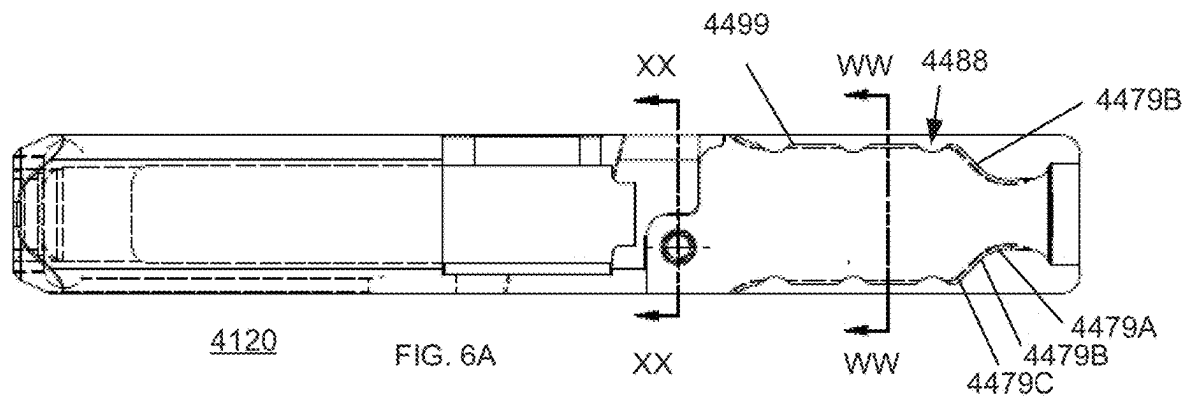


FIG. 2B









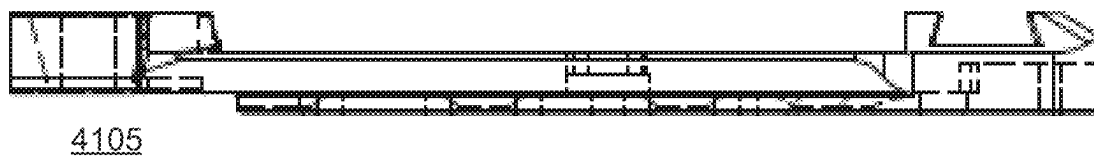


FIG. 7A

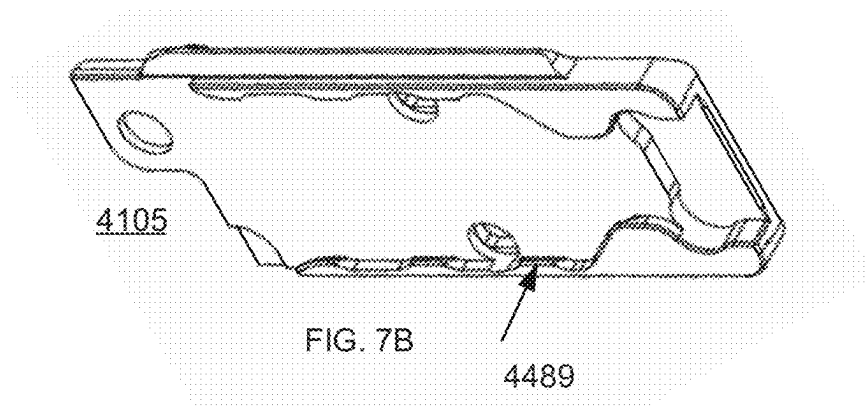


FIG. 7B

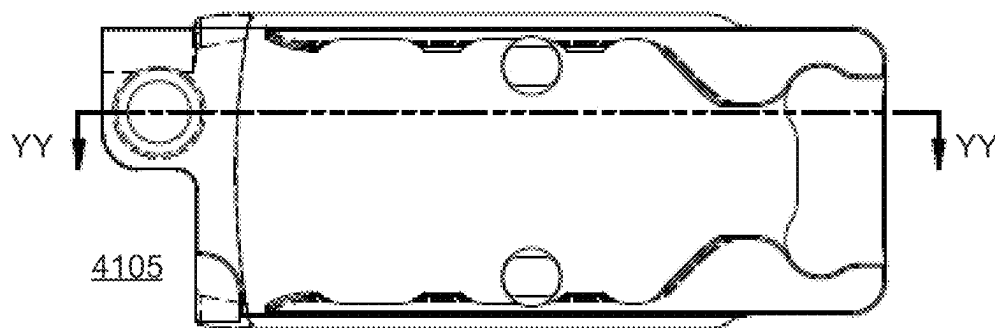


FIG. 7C

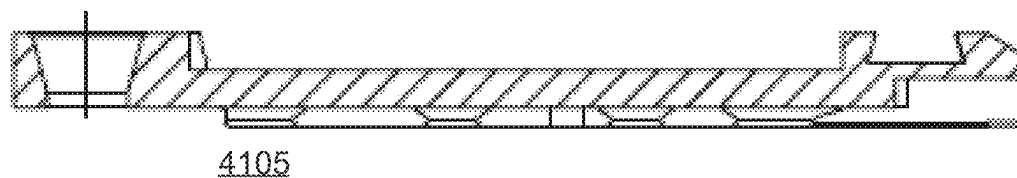


FIG. 7D

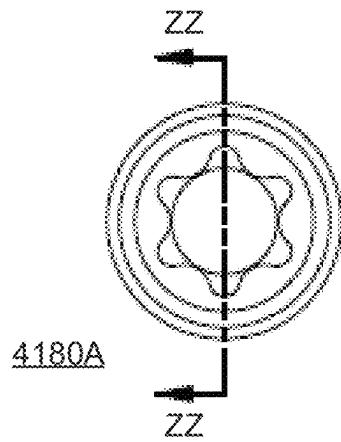
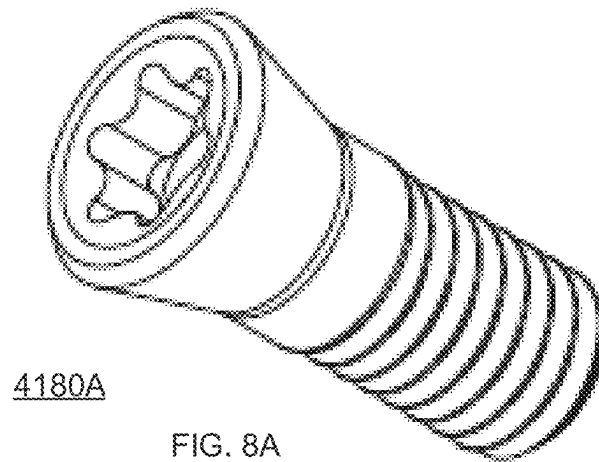


FIG. 8B

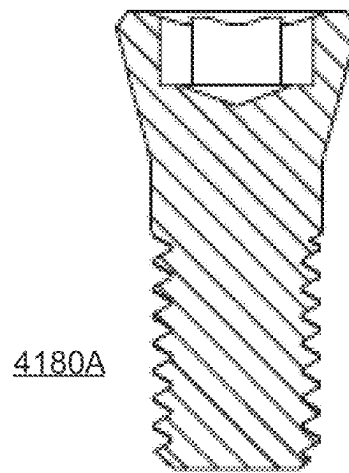


FIG. 8C

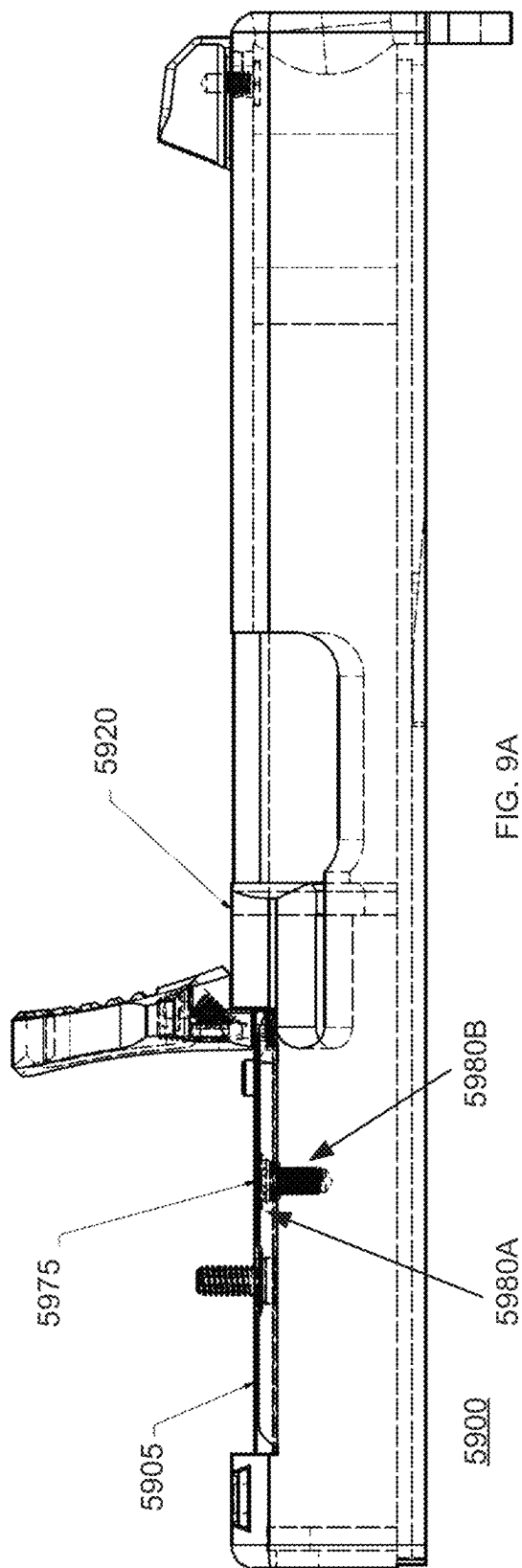


FIG. 9A

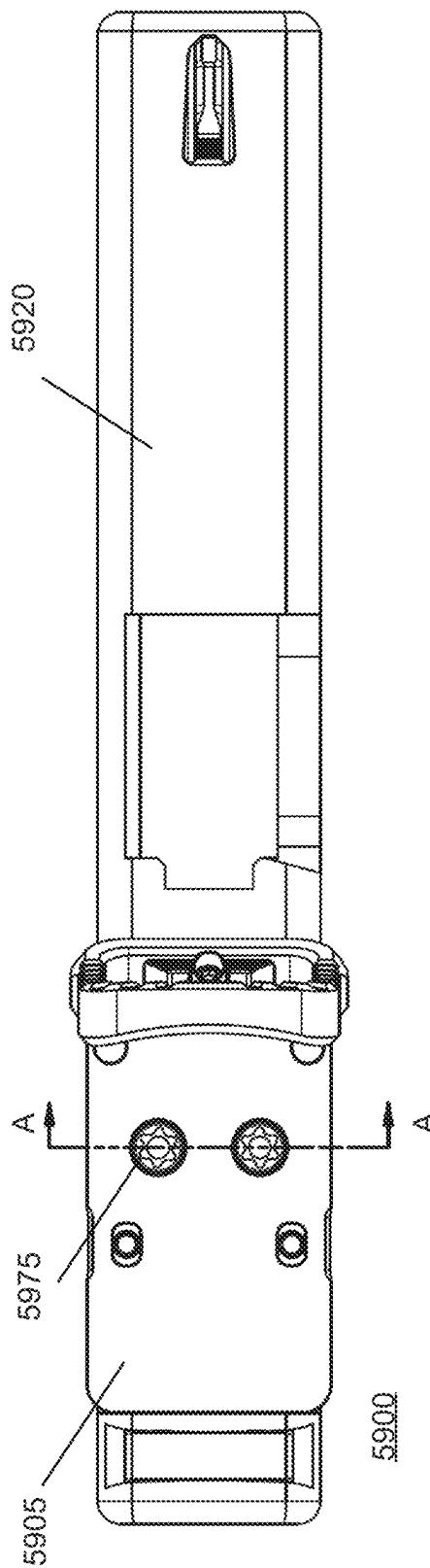
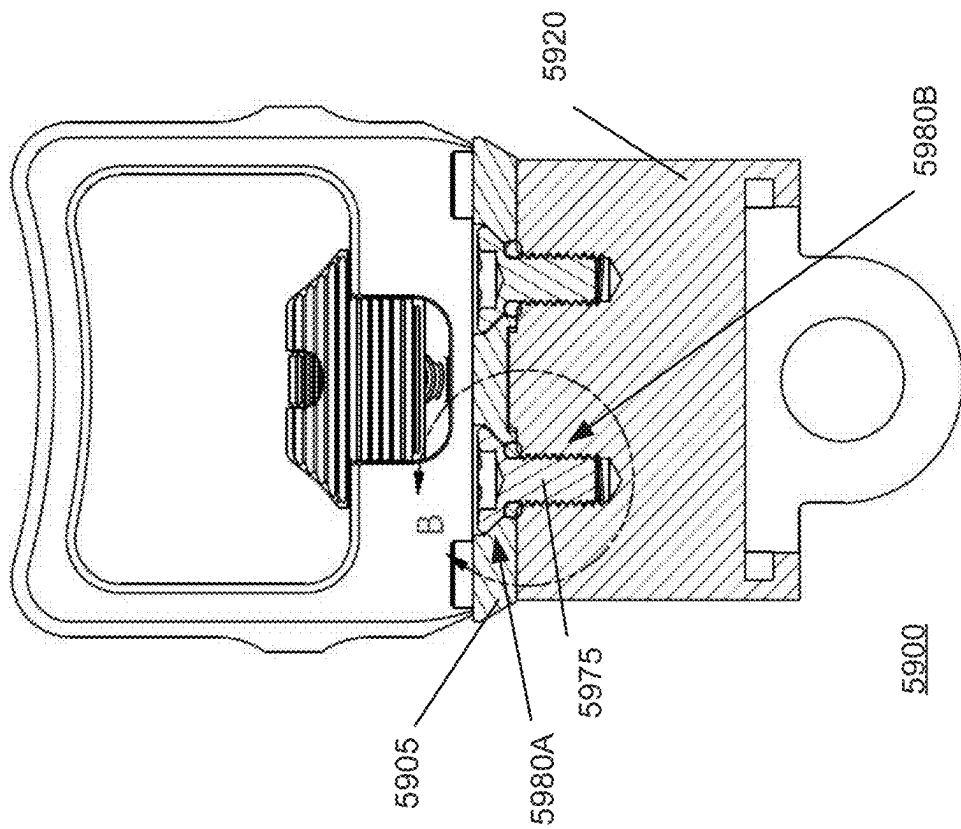
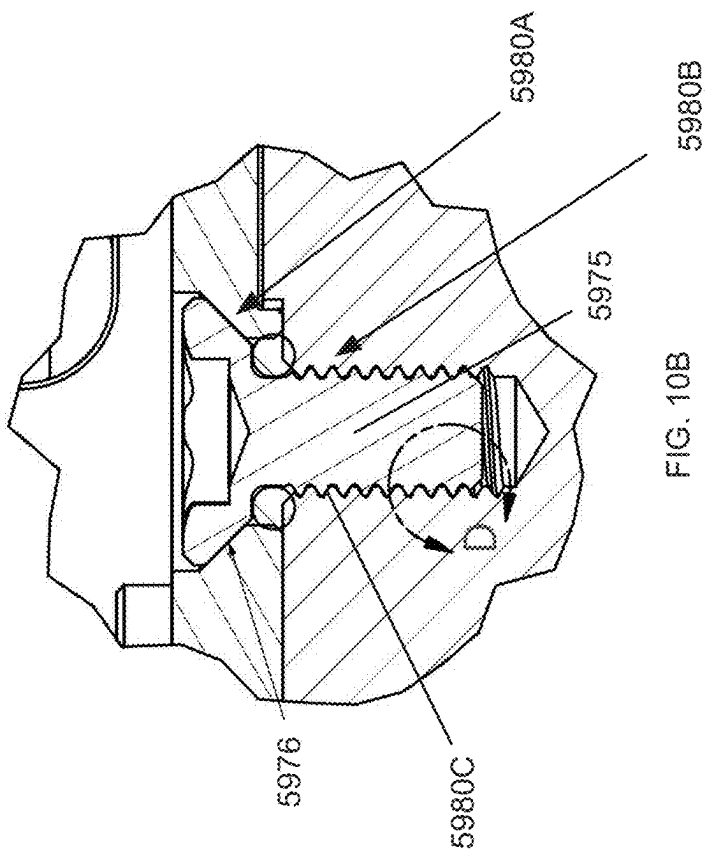


FIG. 9B



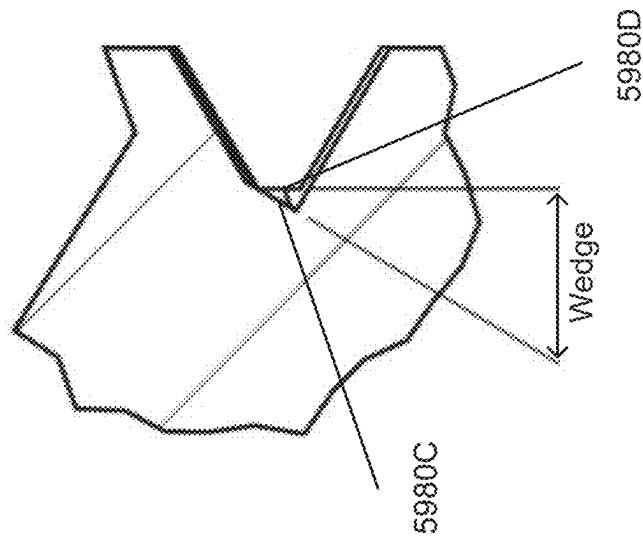
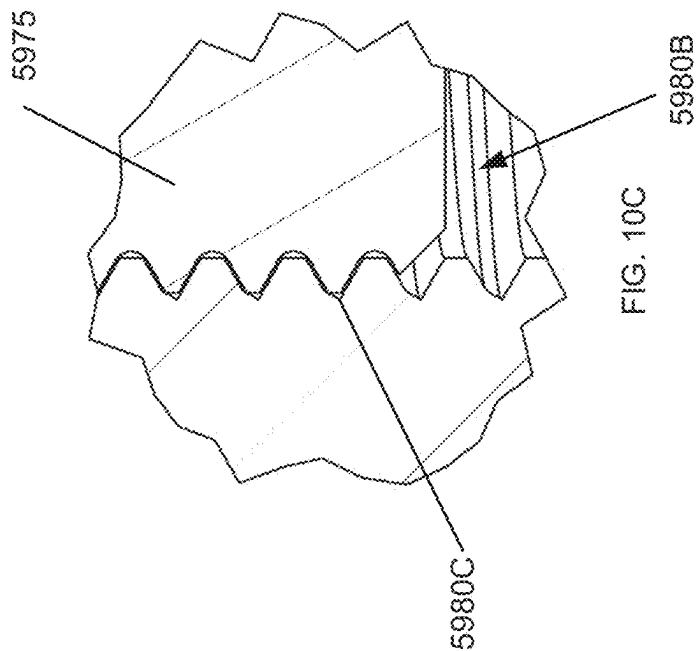


FIG. 10D

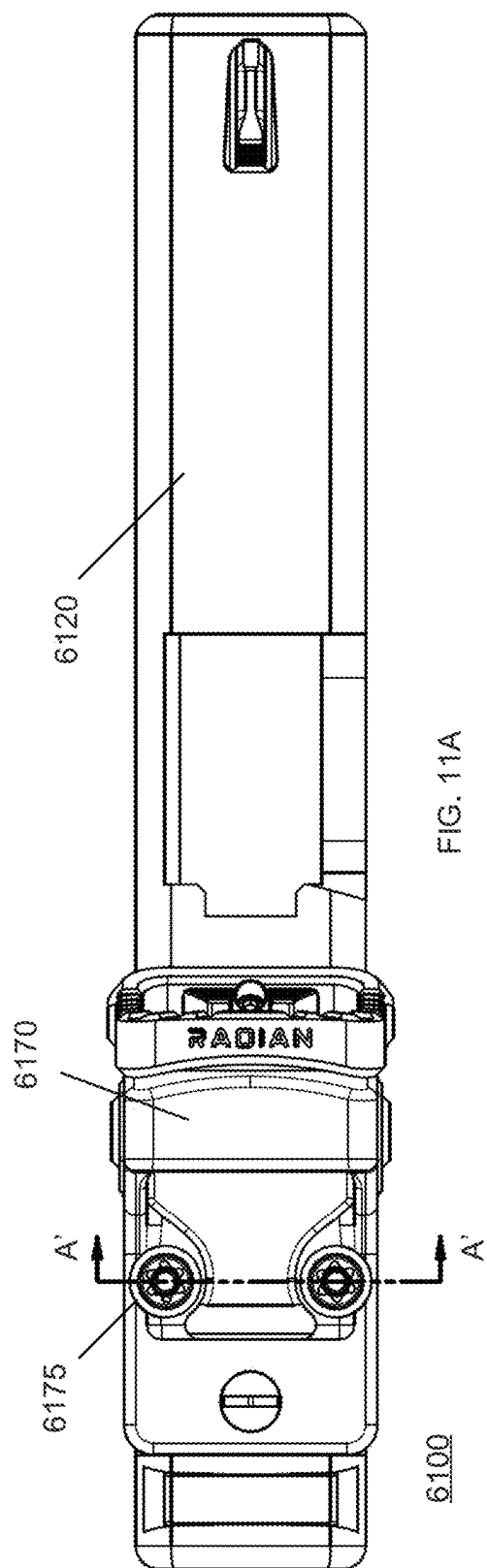


FIG. 11A

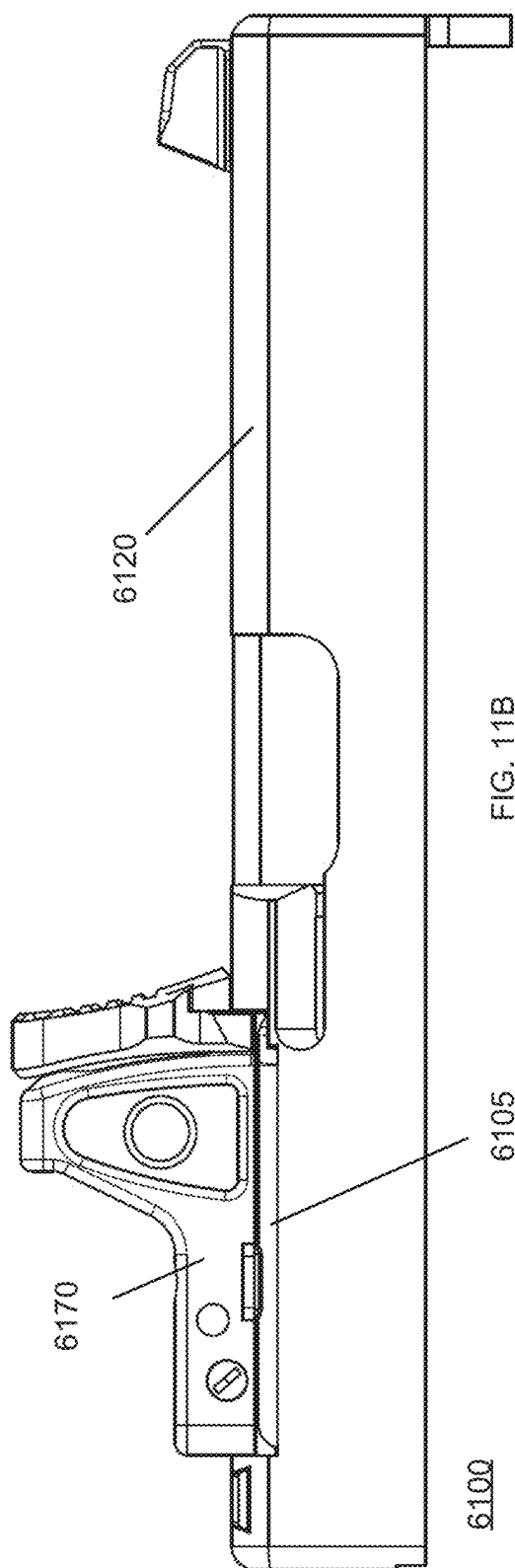


FIG. 11B

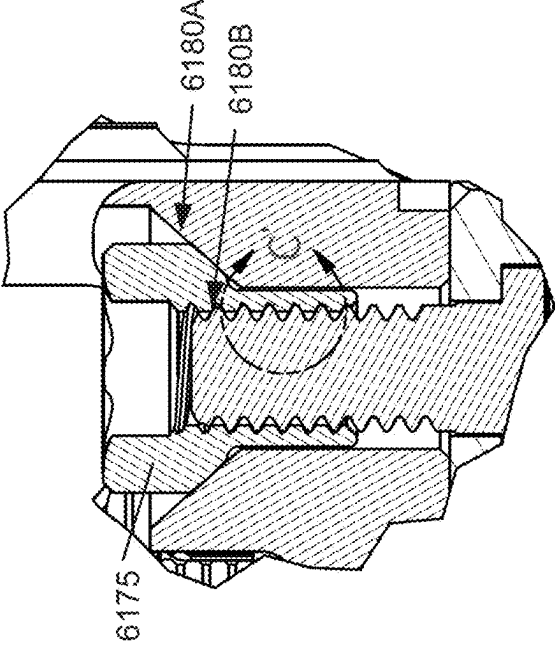


FIG. 12B

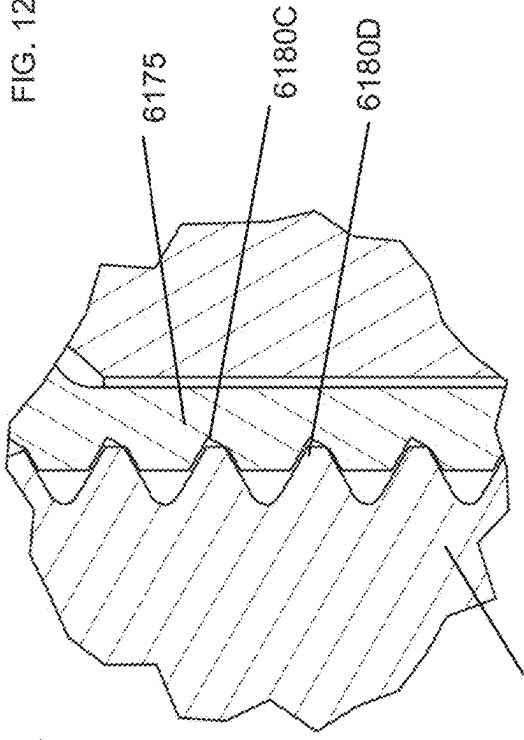


FIG. 12C

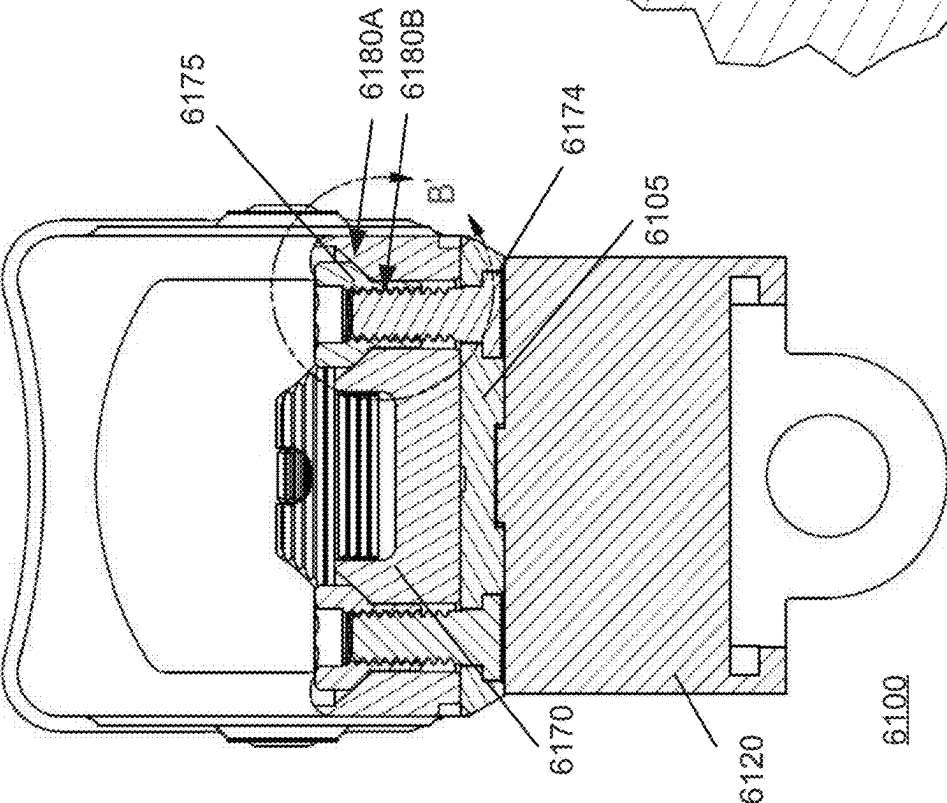
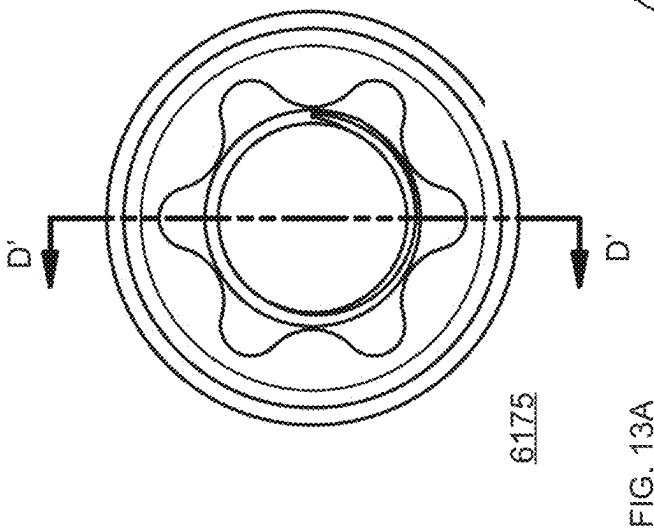
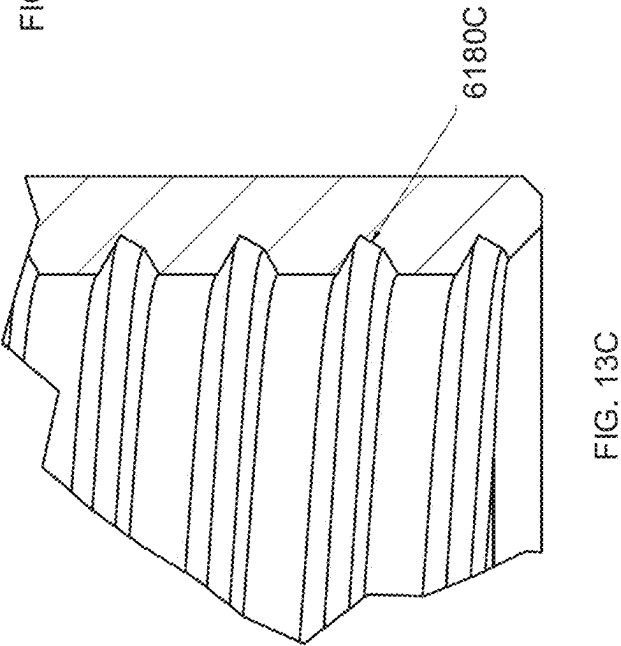
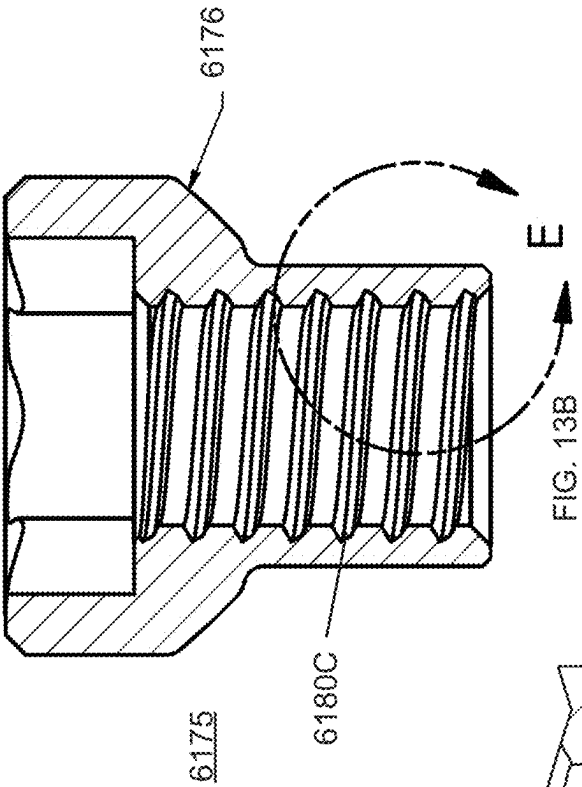


FIG. 12A



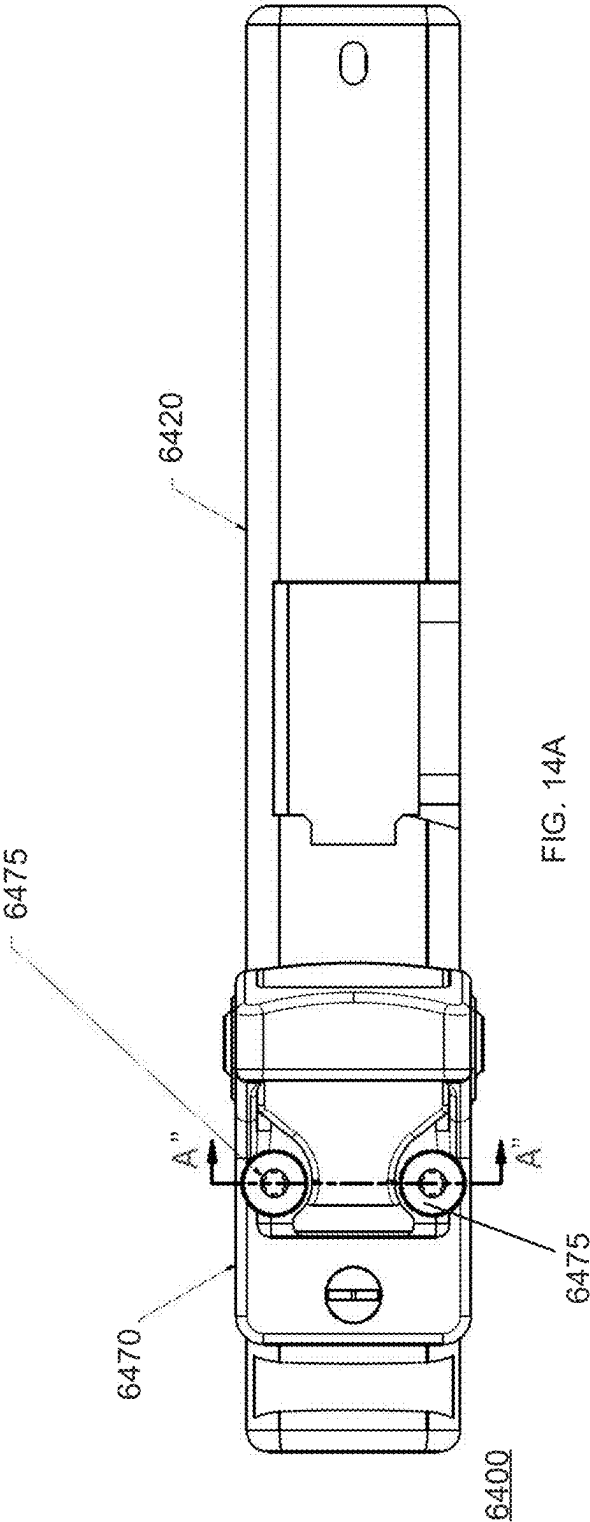


FIG. 14A

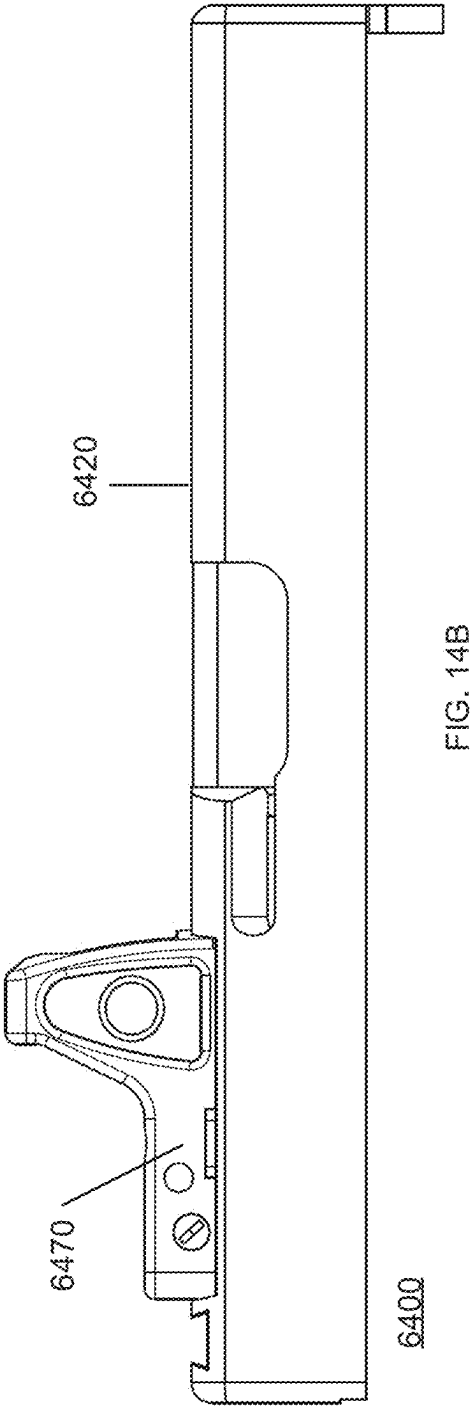


FIG. 14B

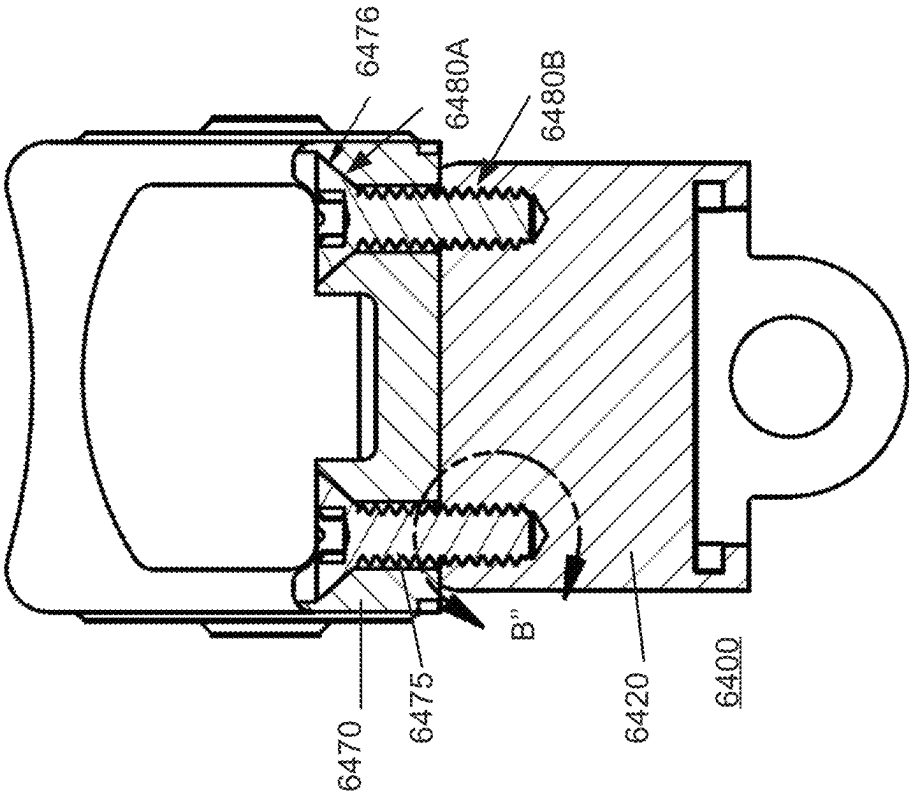


FIG. 15A

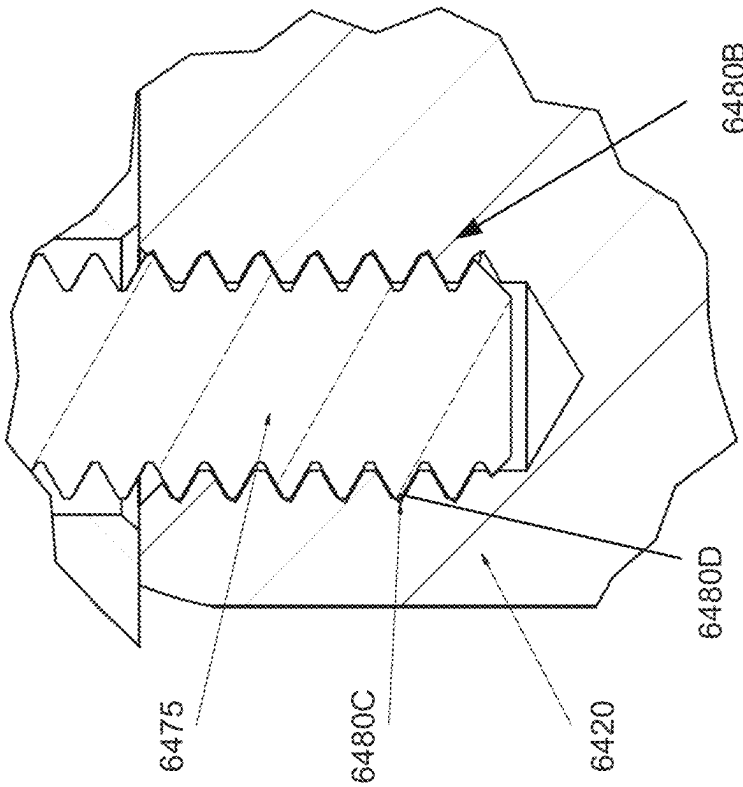


FIG. 15B

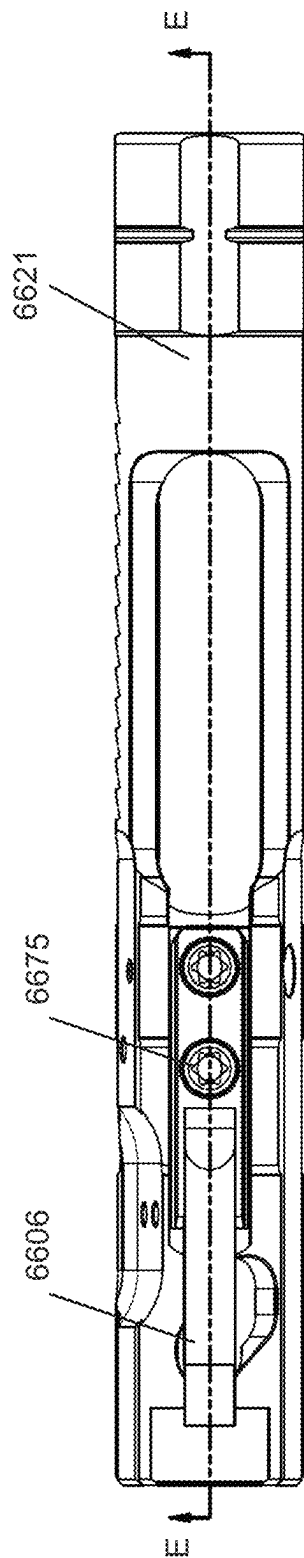


FIG. 16A

6600

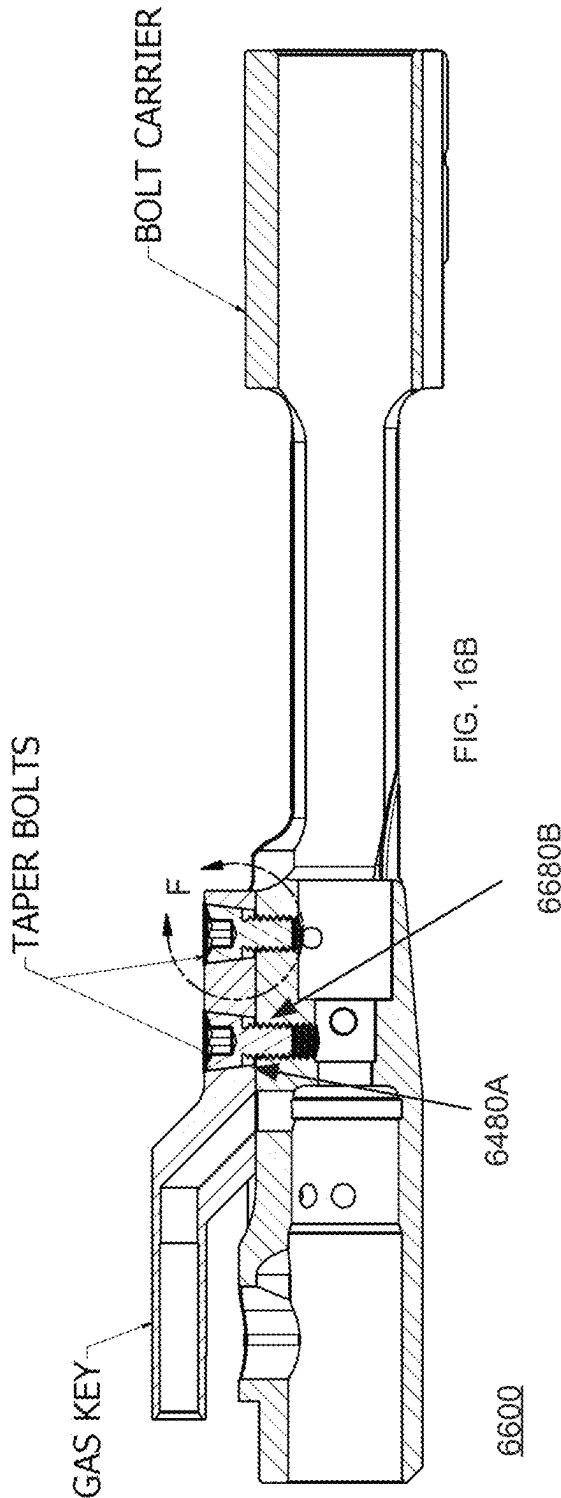
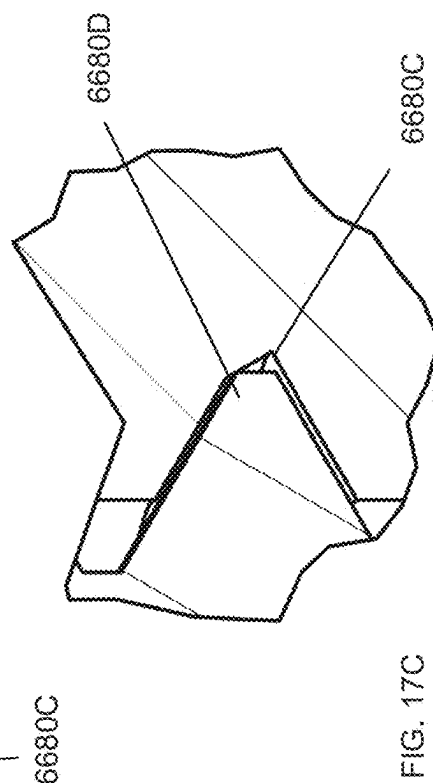
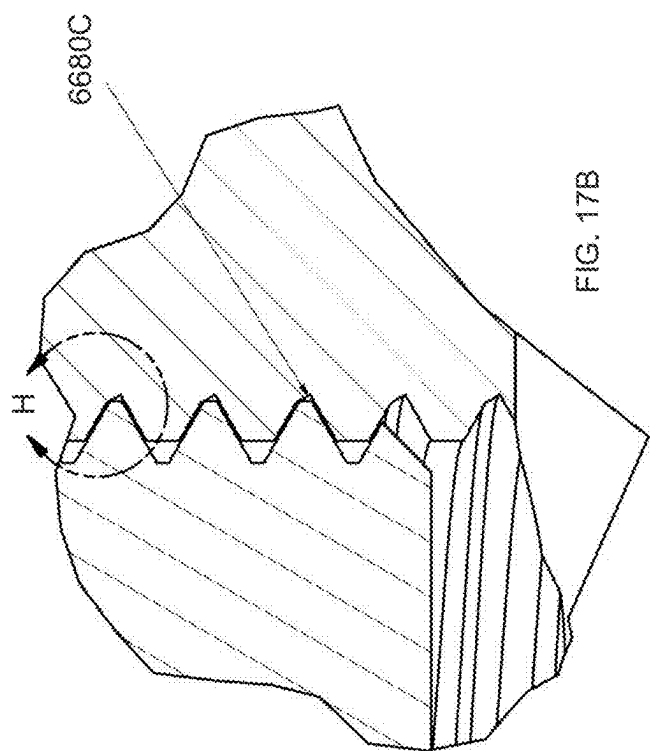
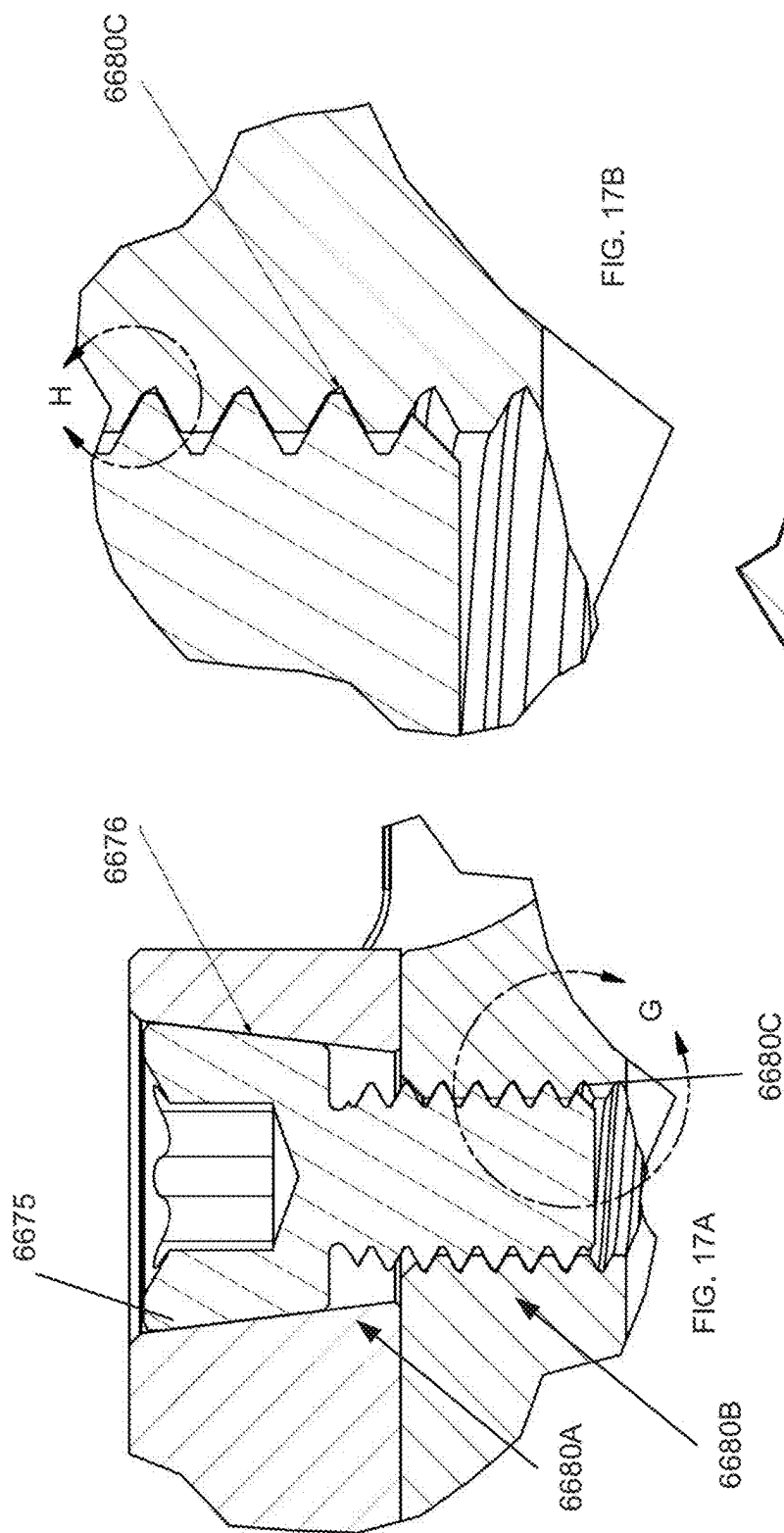
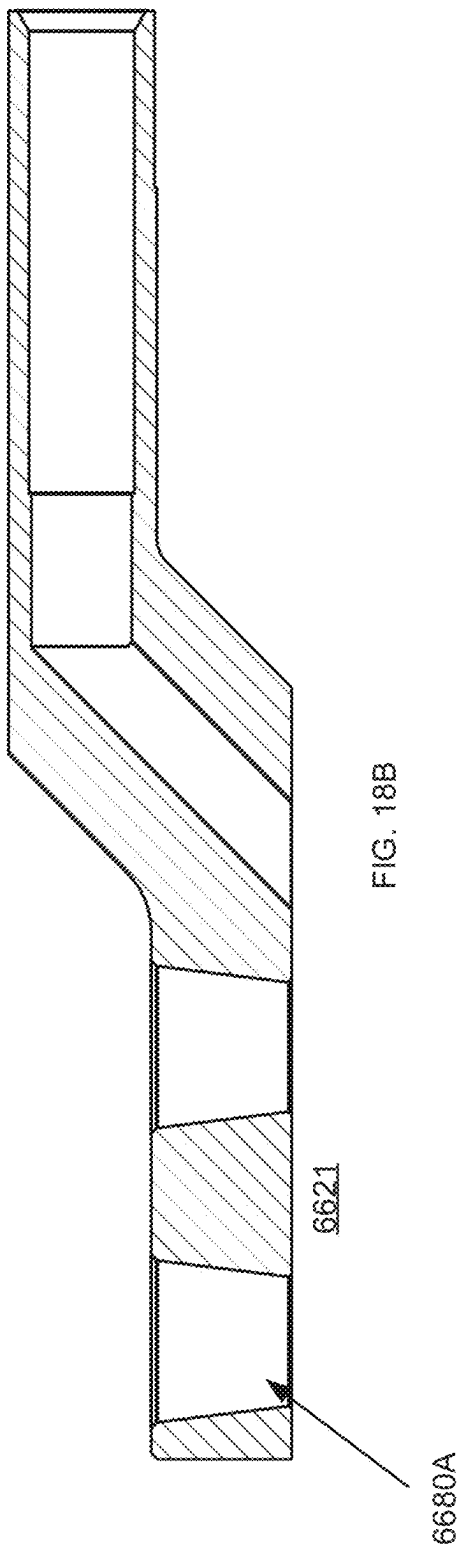
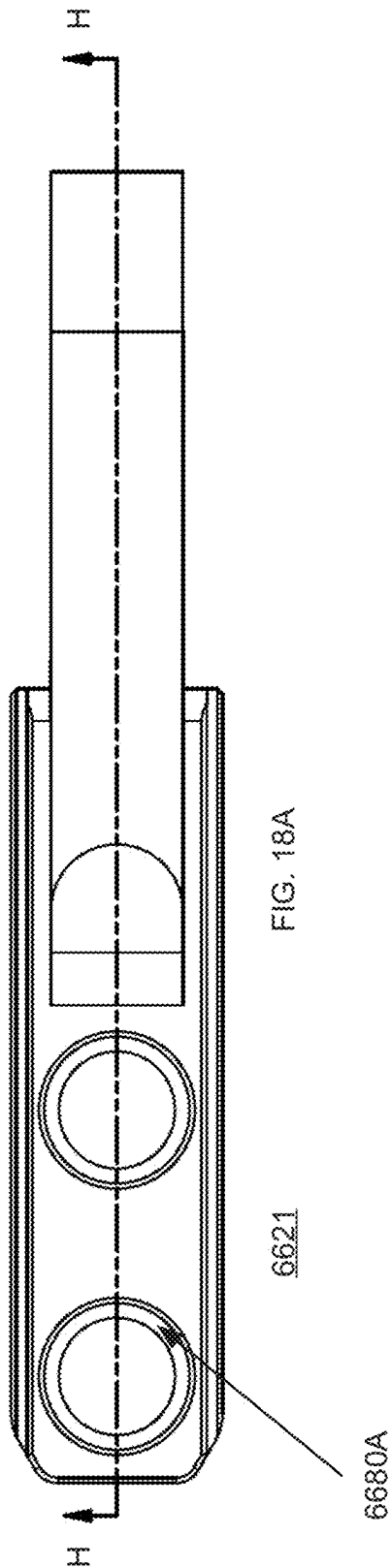


FIG. 16B

6600





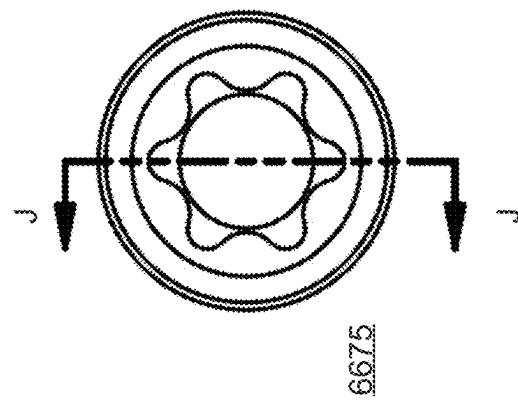
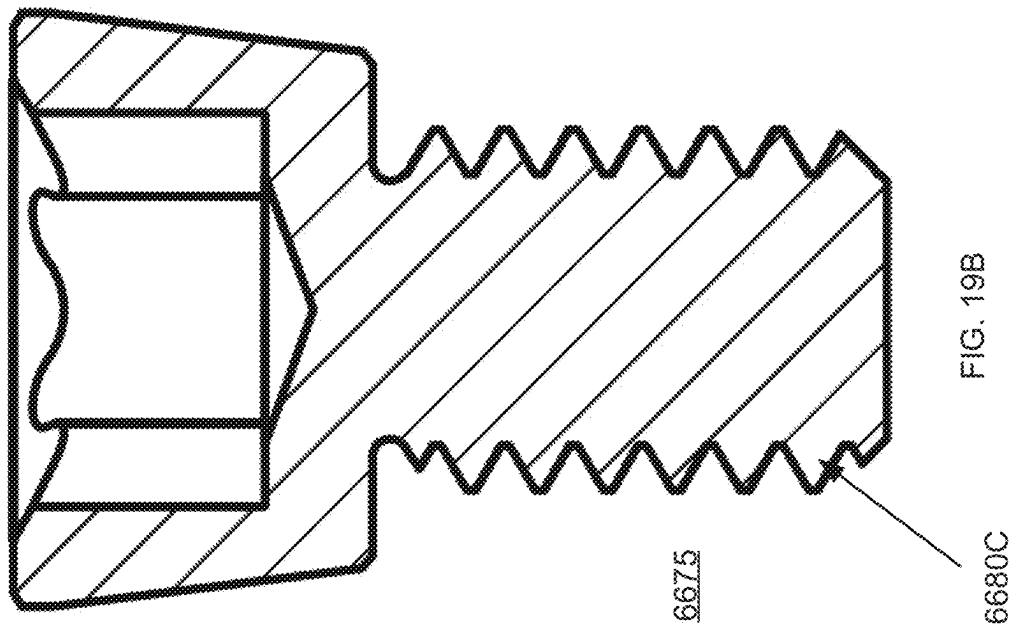


FIG. 19A

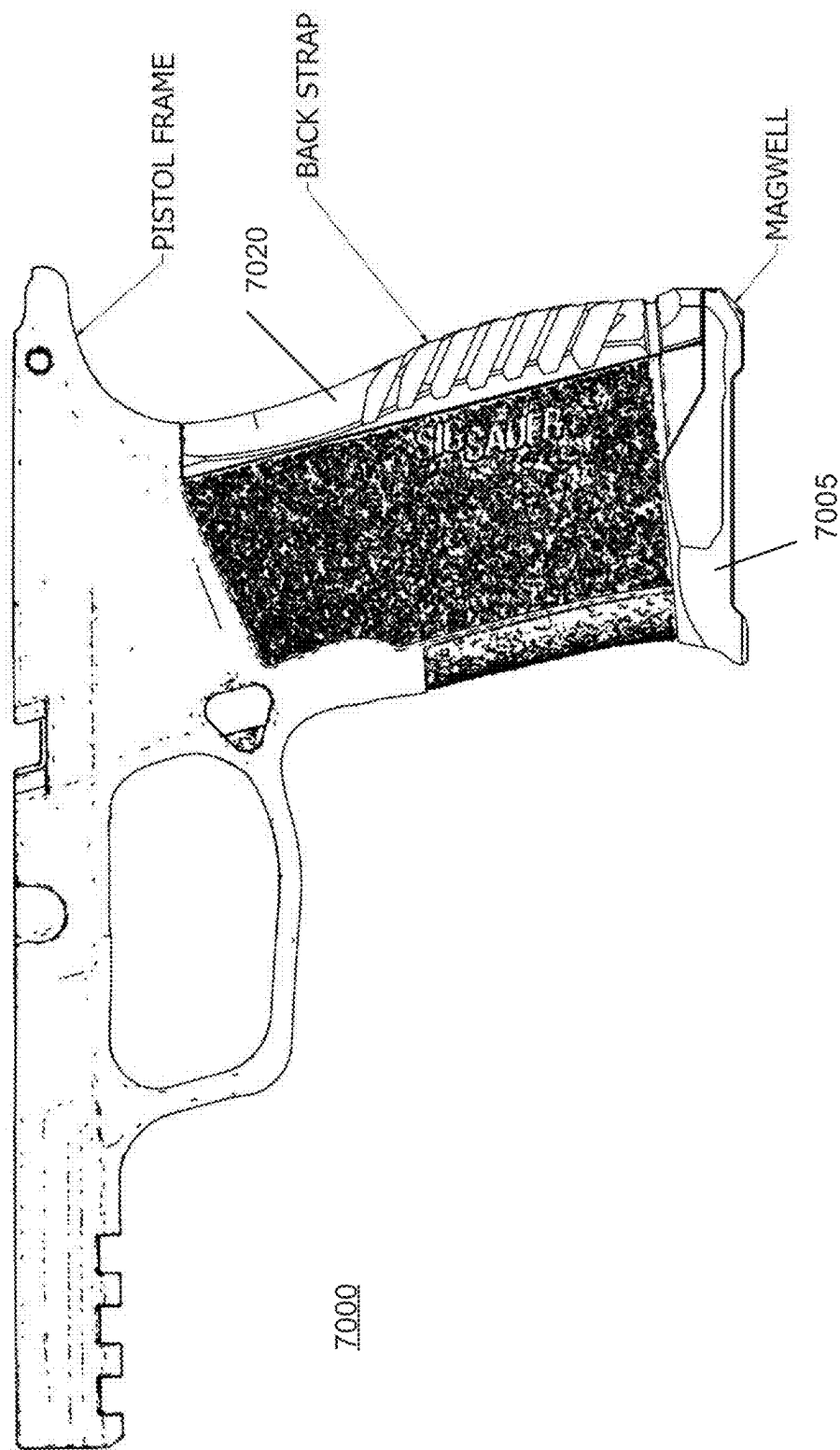
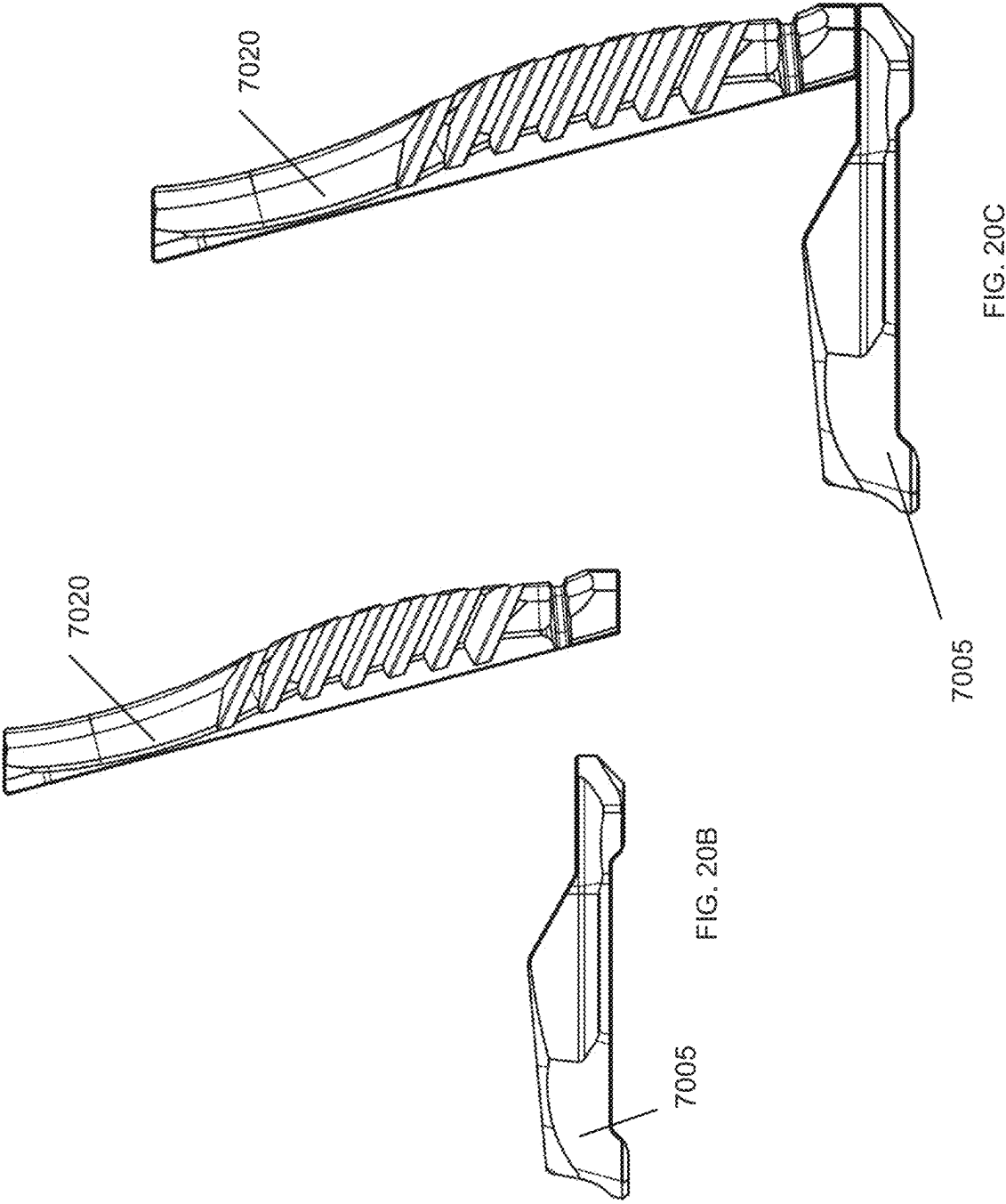
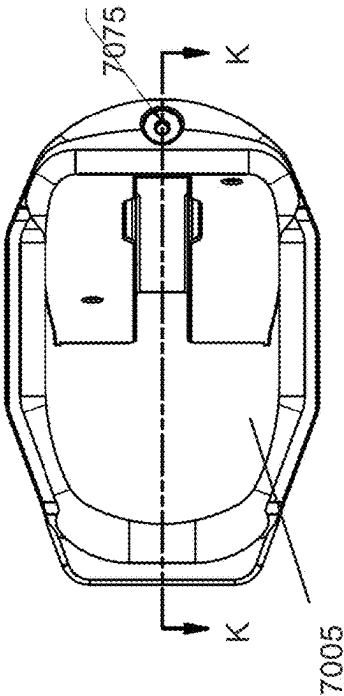
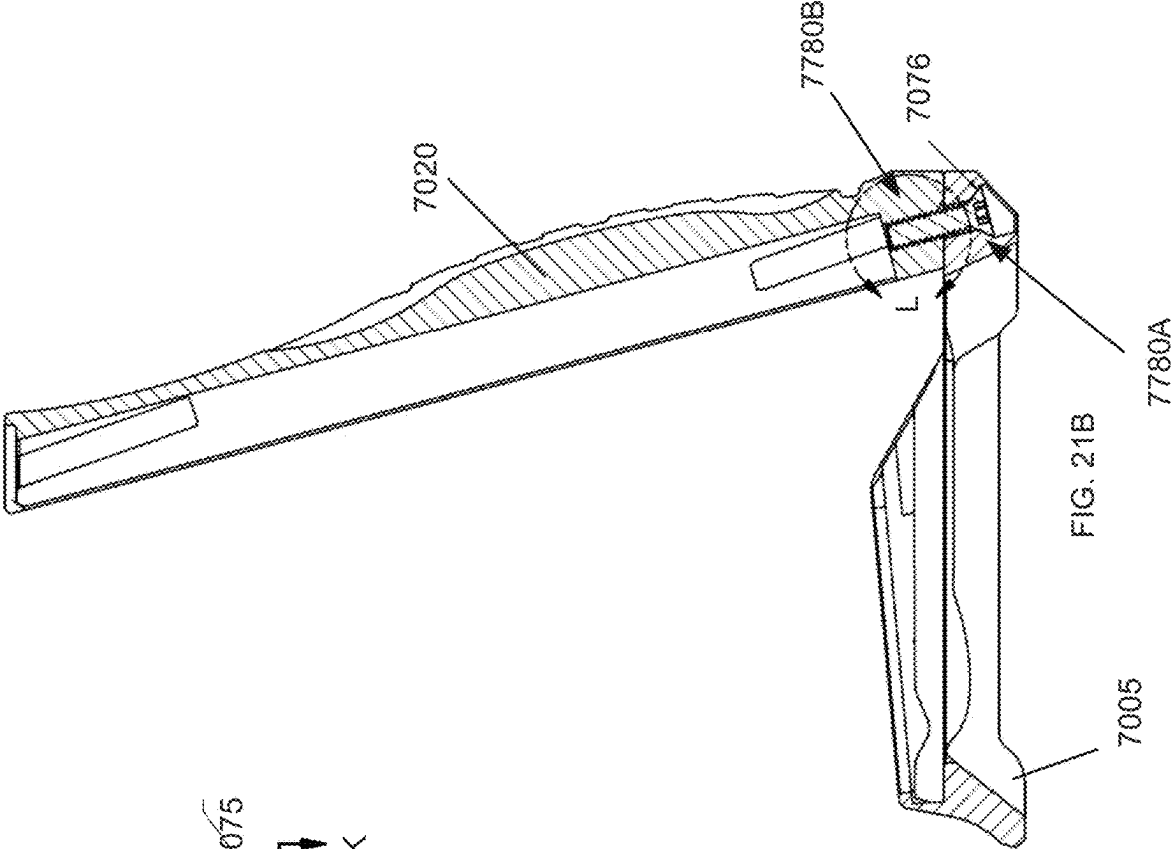


FIG. 20A





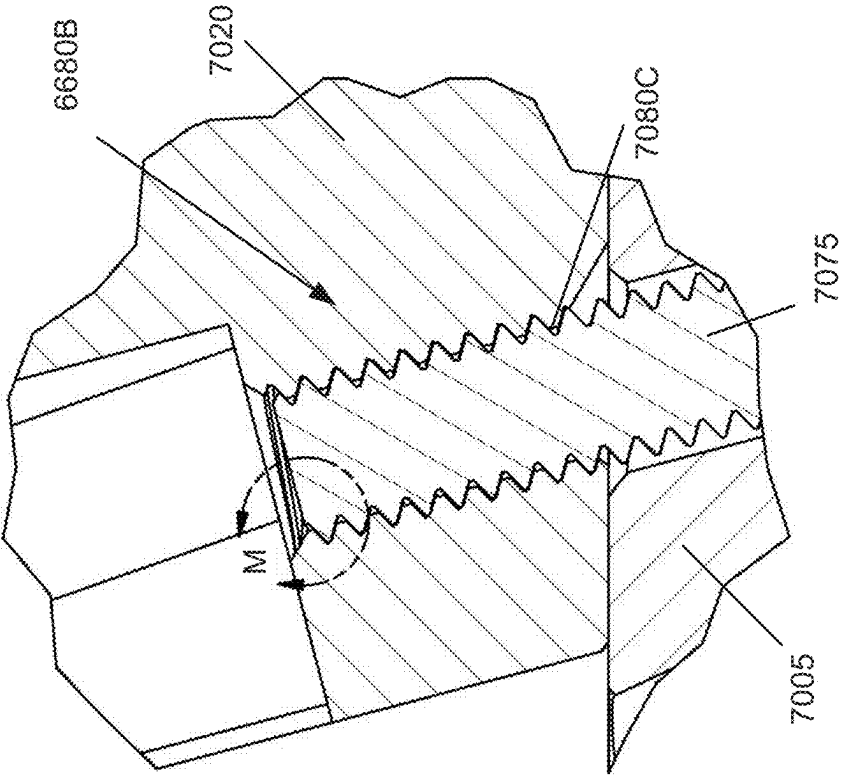


FIG. 22A

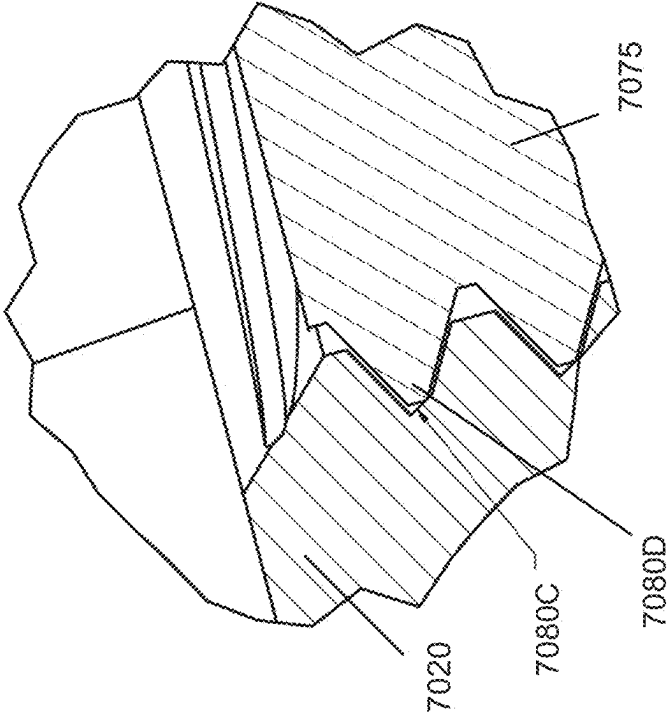


FIG. 22B

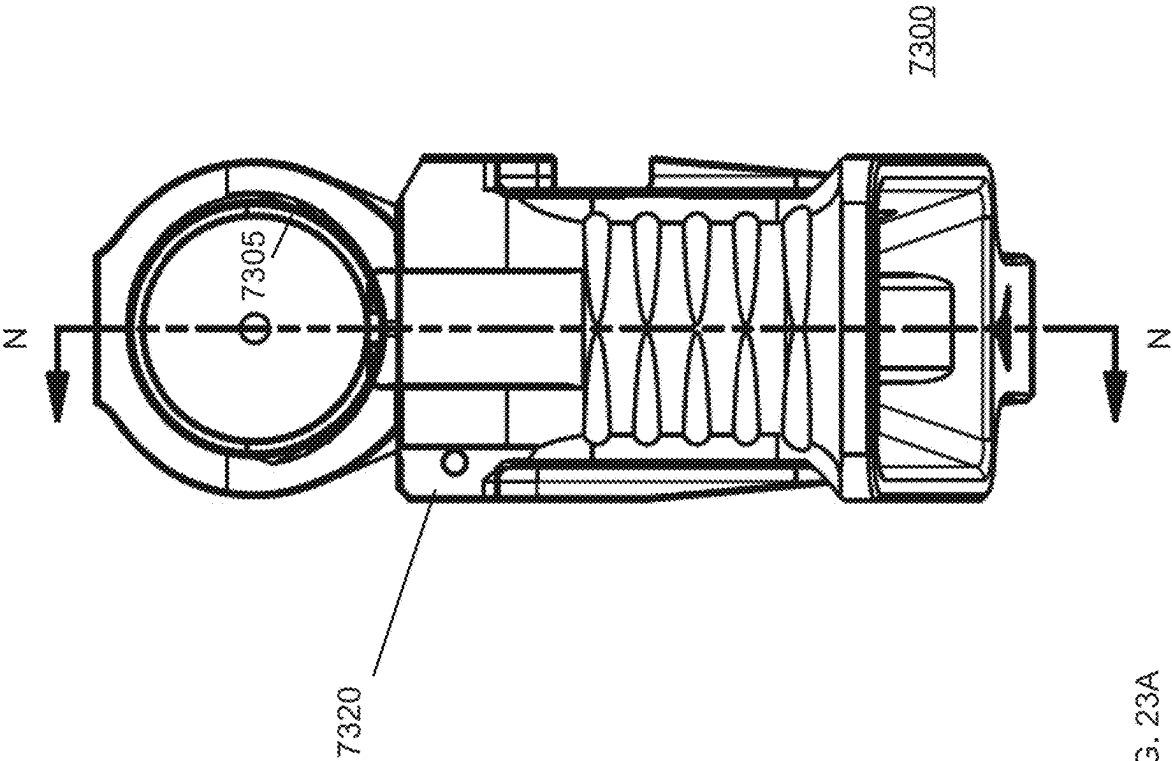
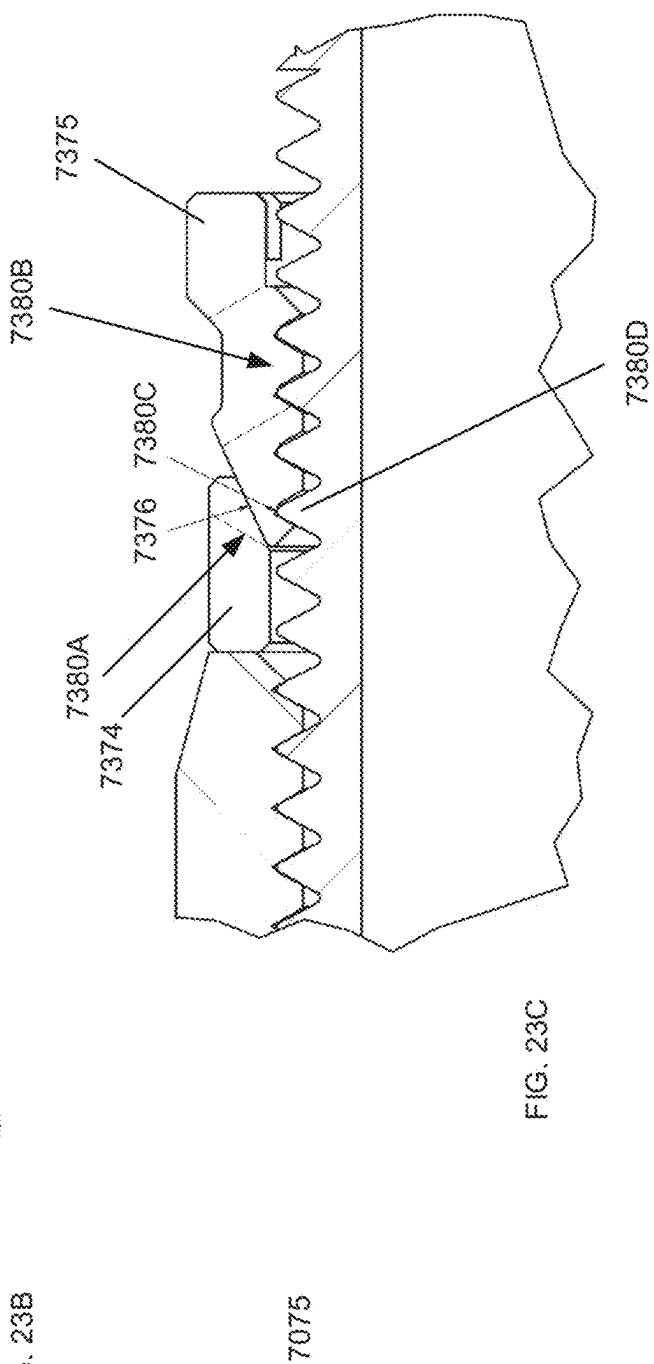
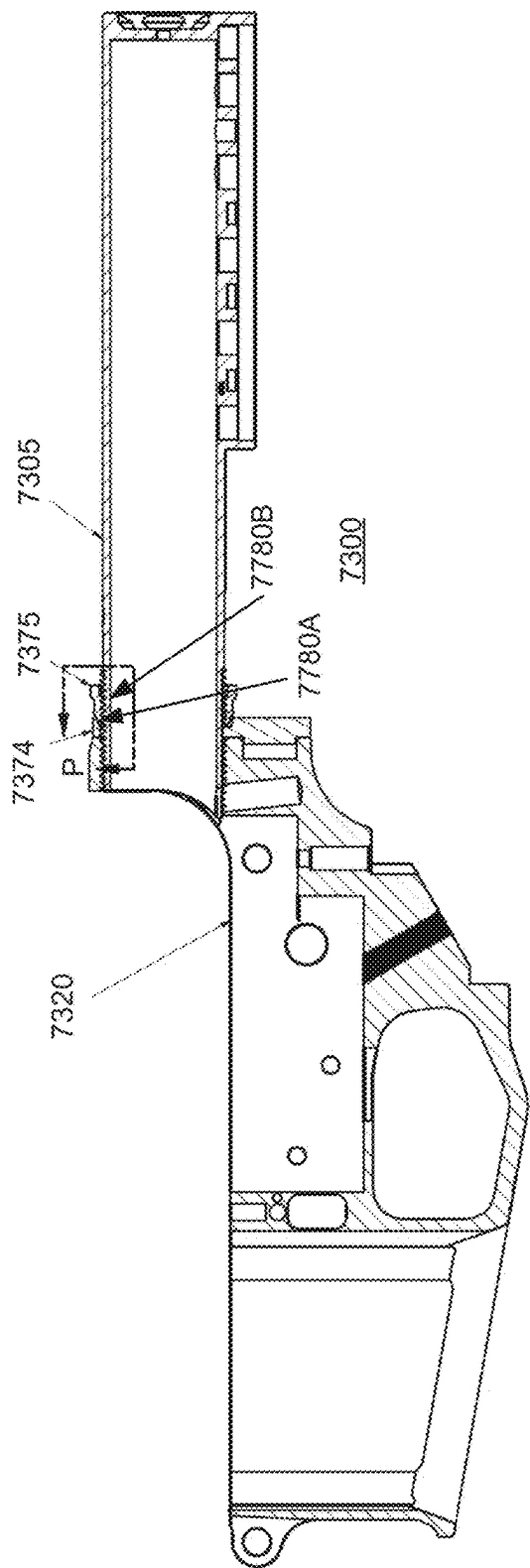


FIG. 23A



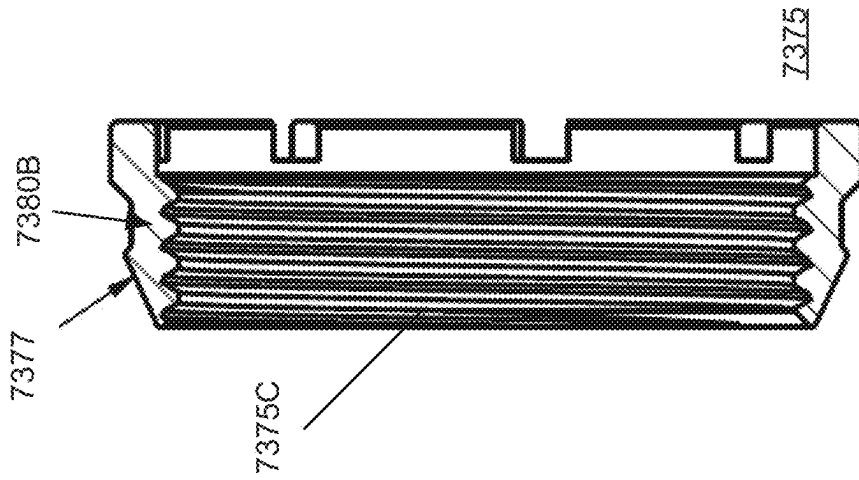


FIG. 24B

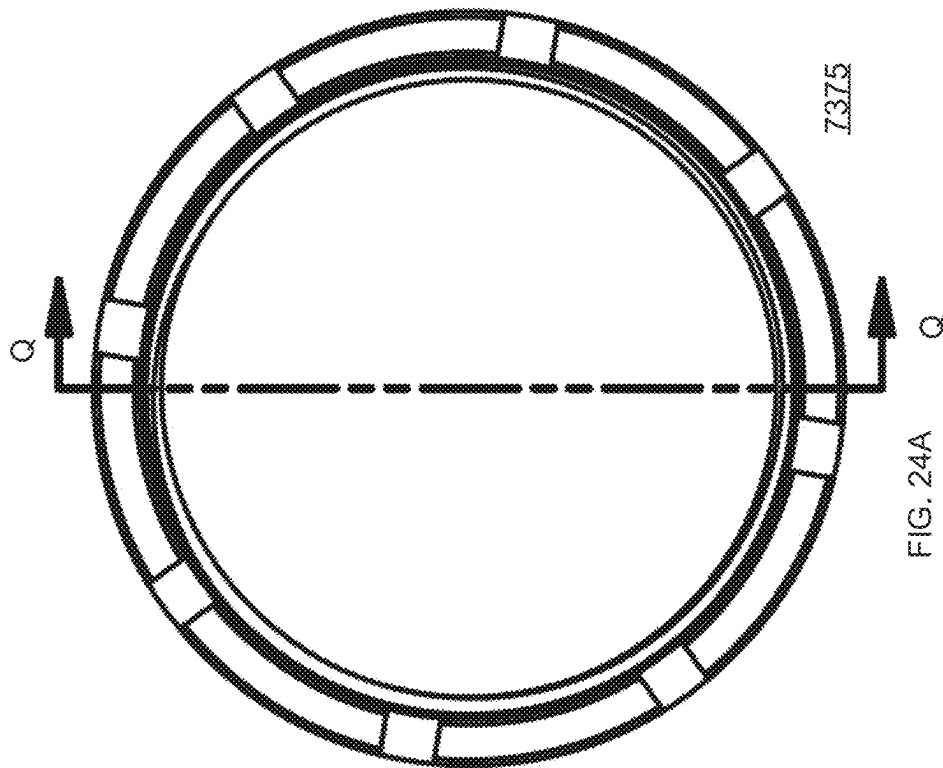
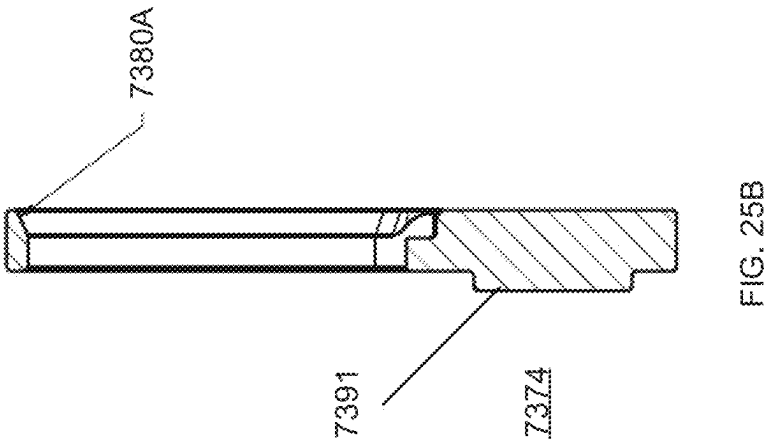
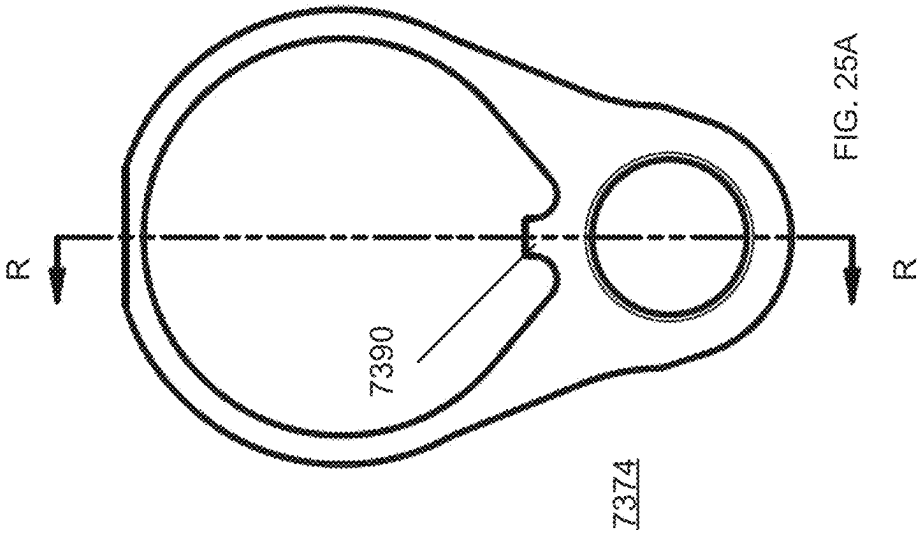
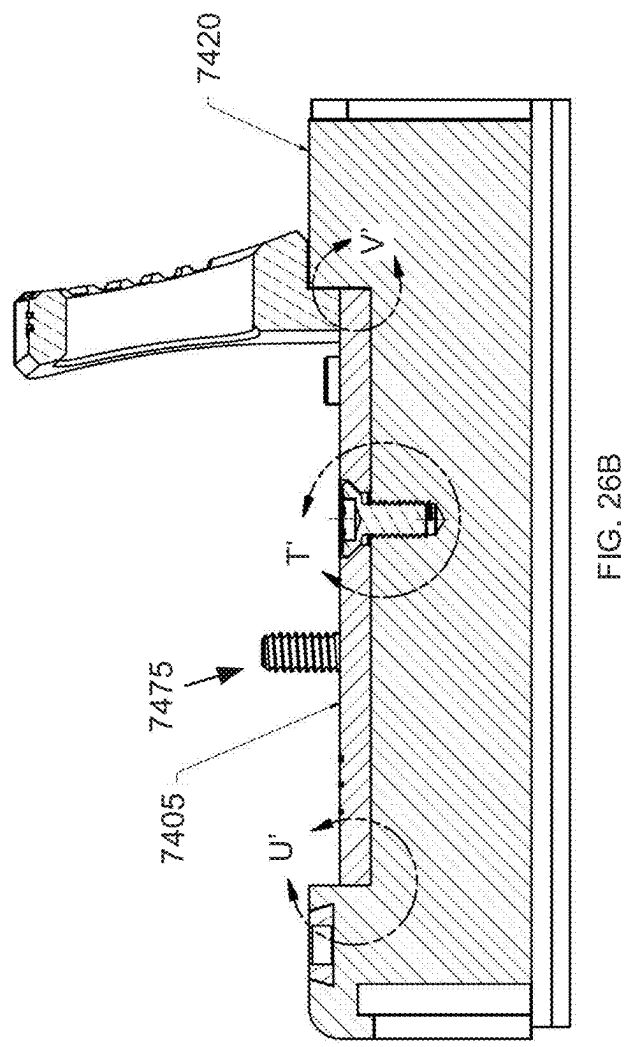
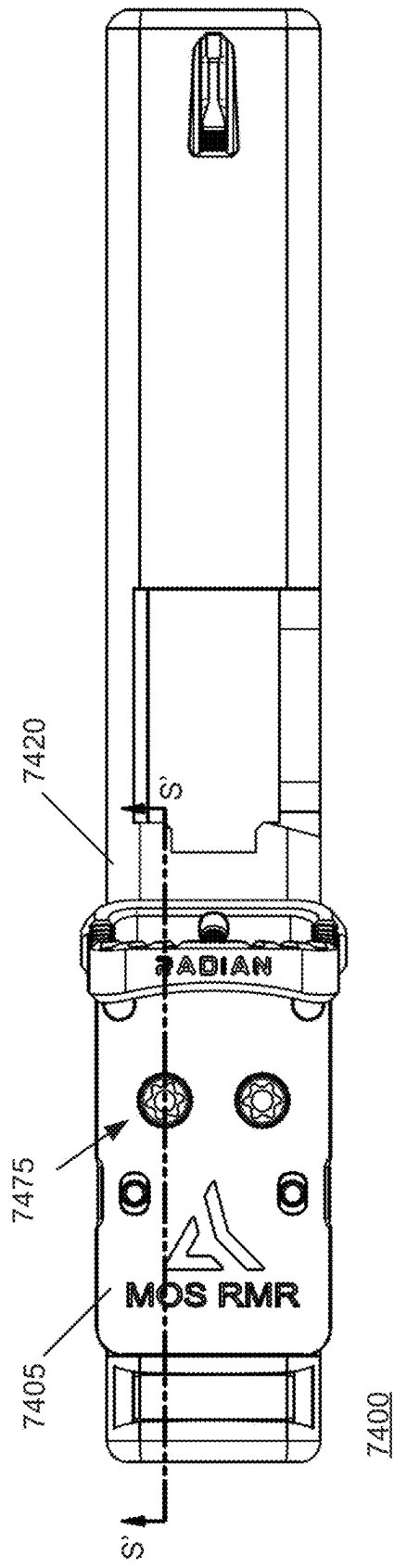
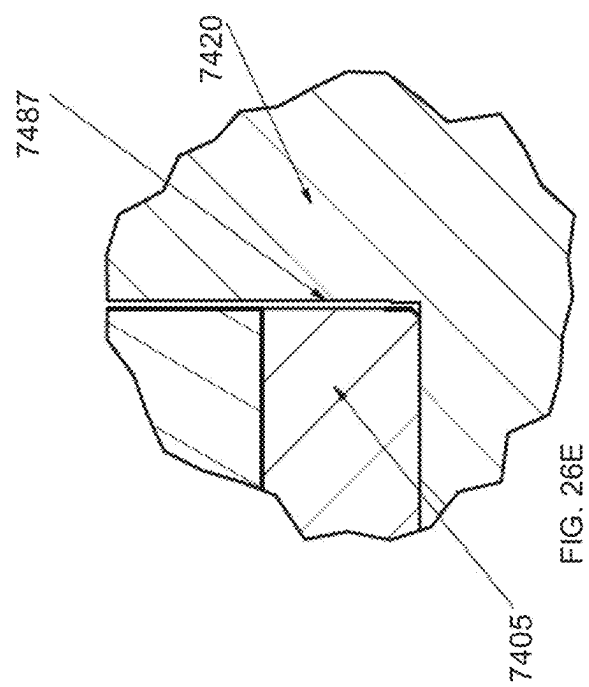
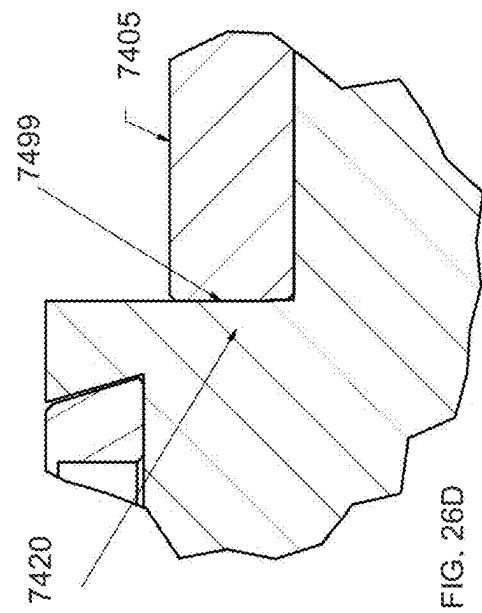
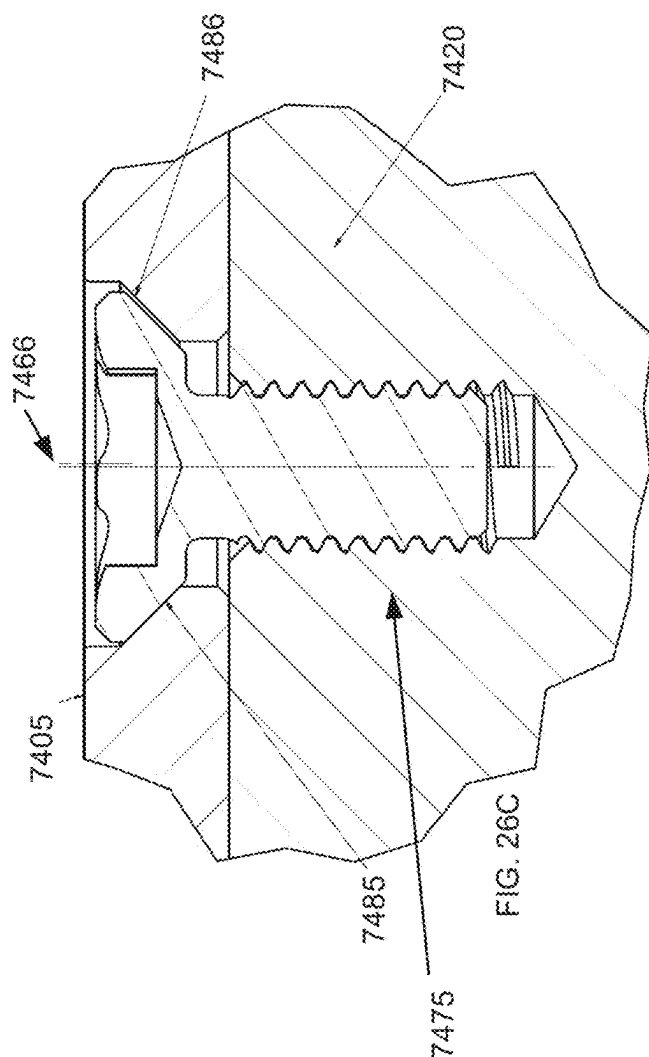


FIG. 24A







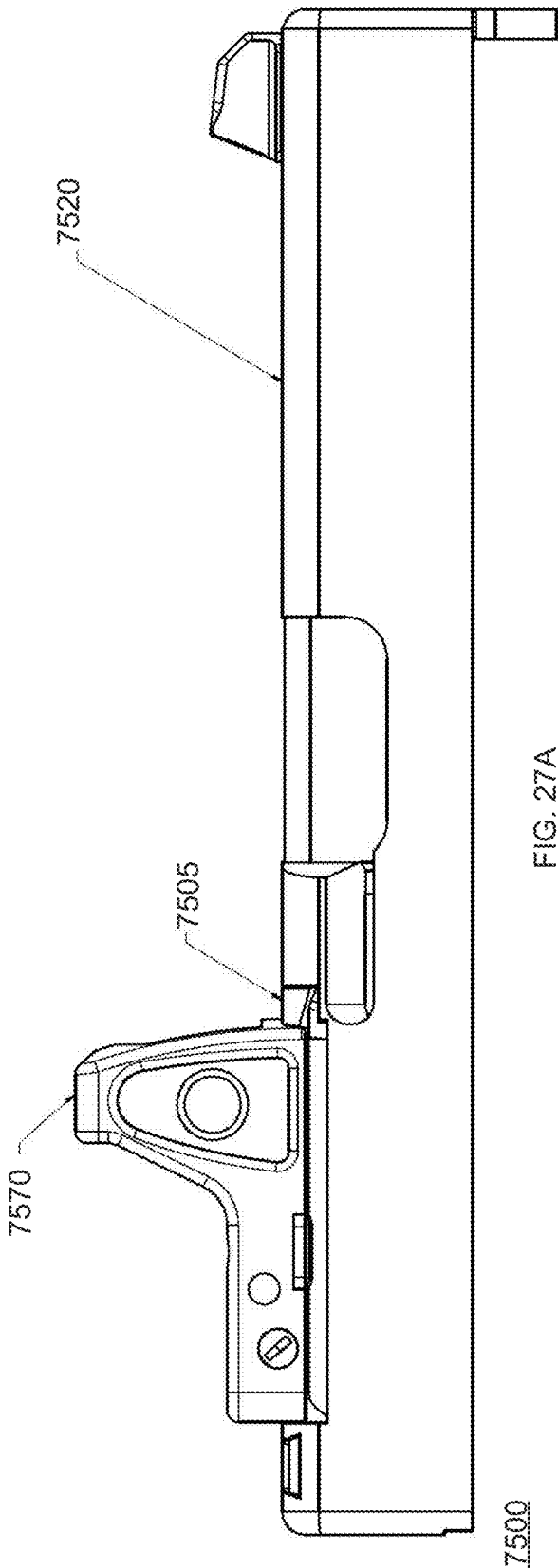


FIG. 27A

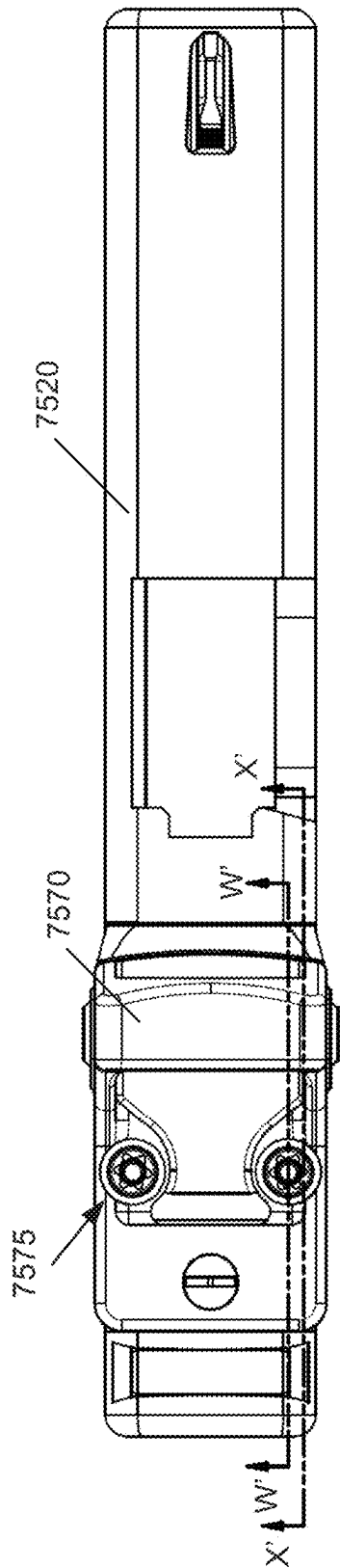
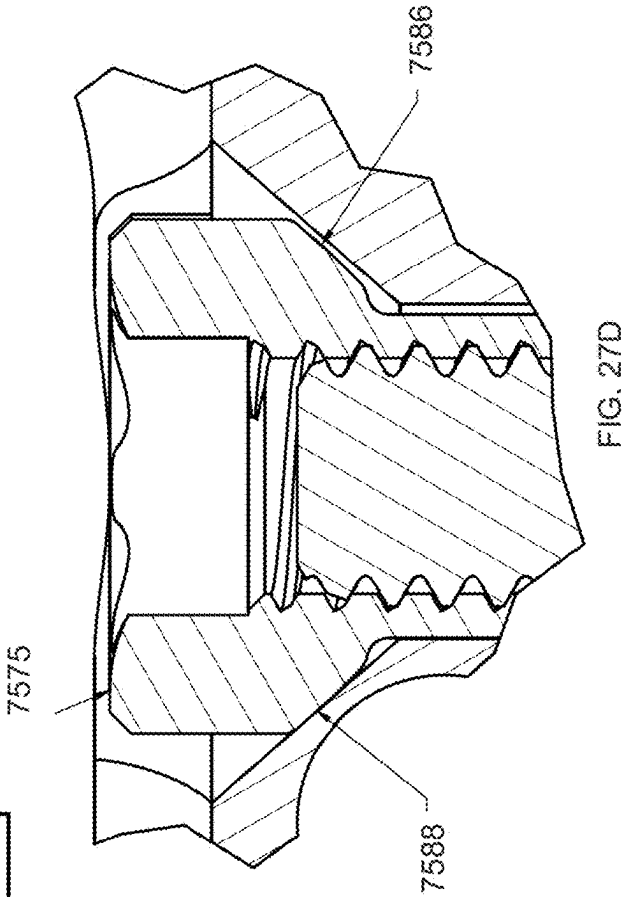
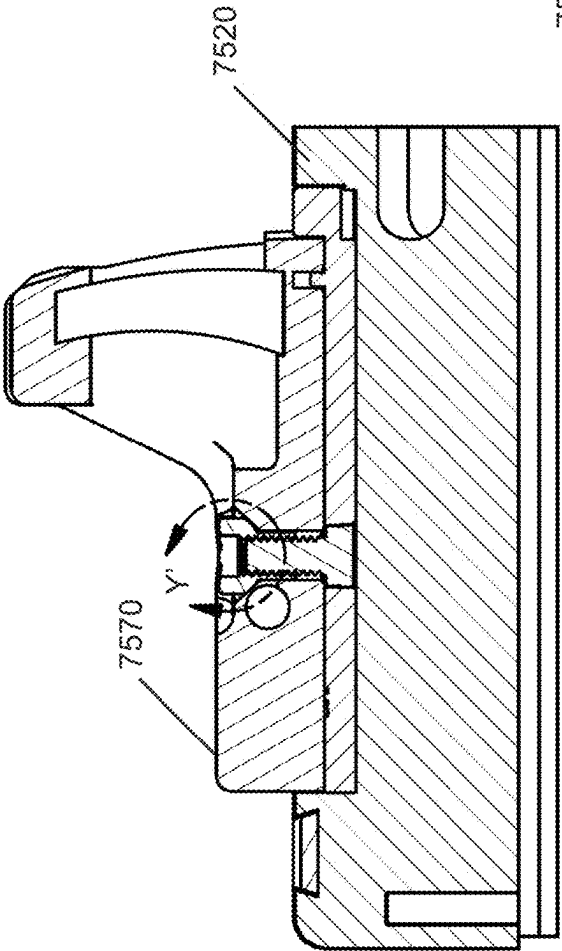
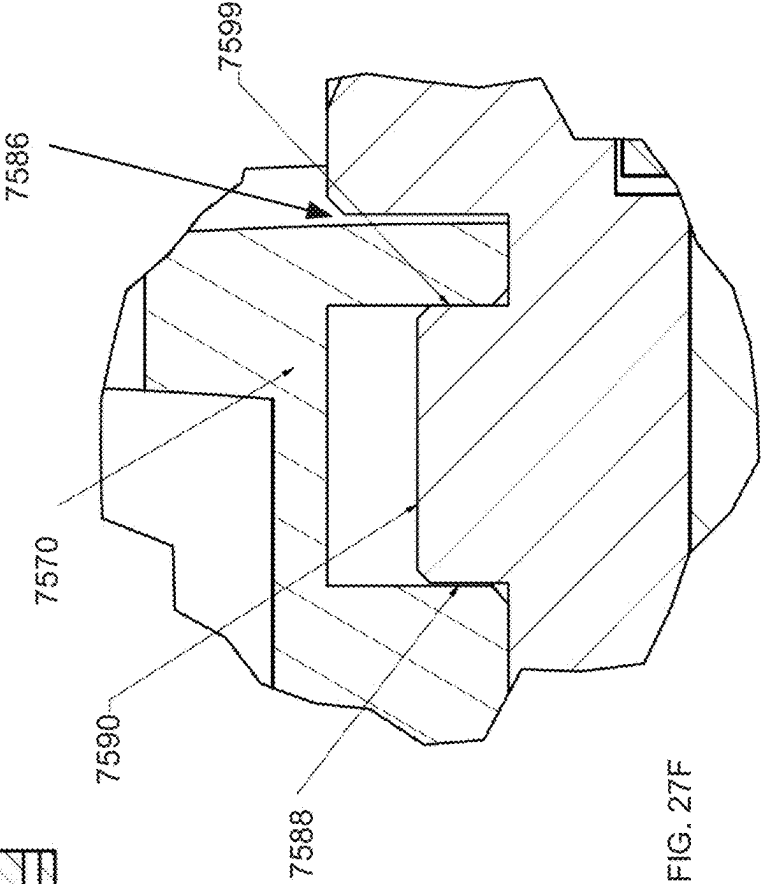
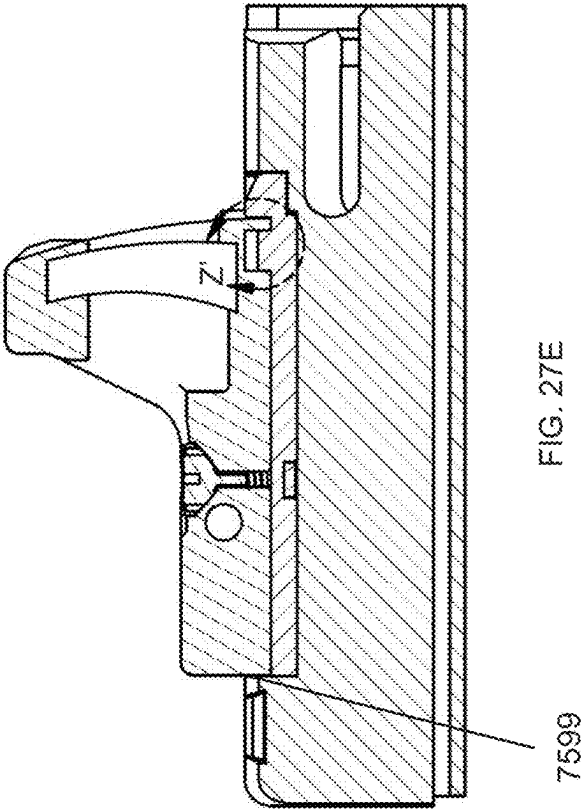
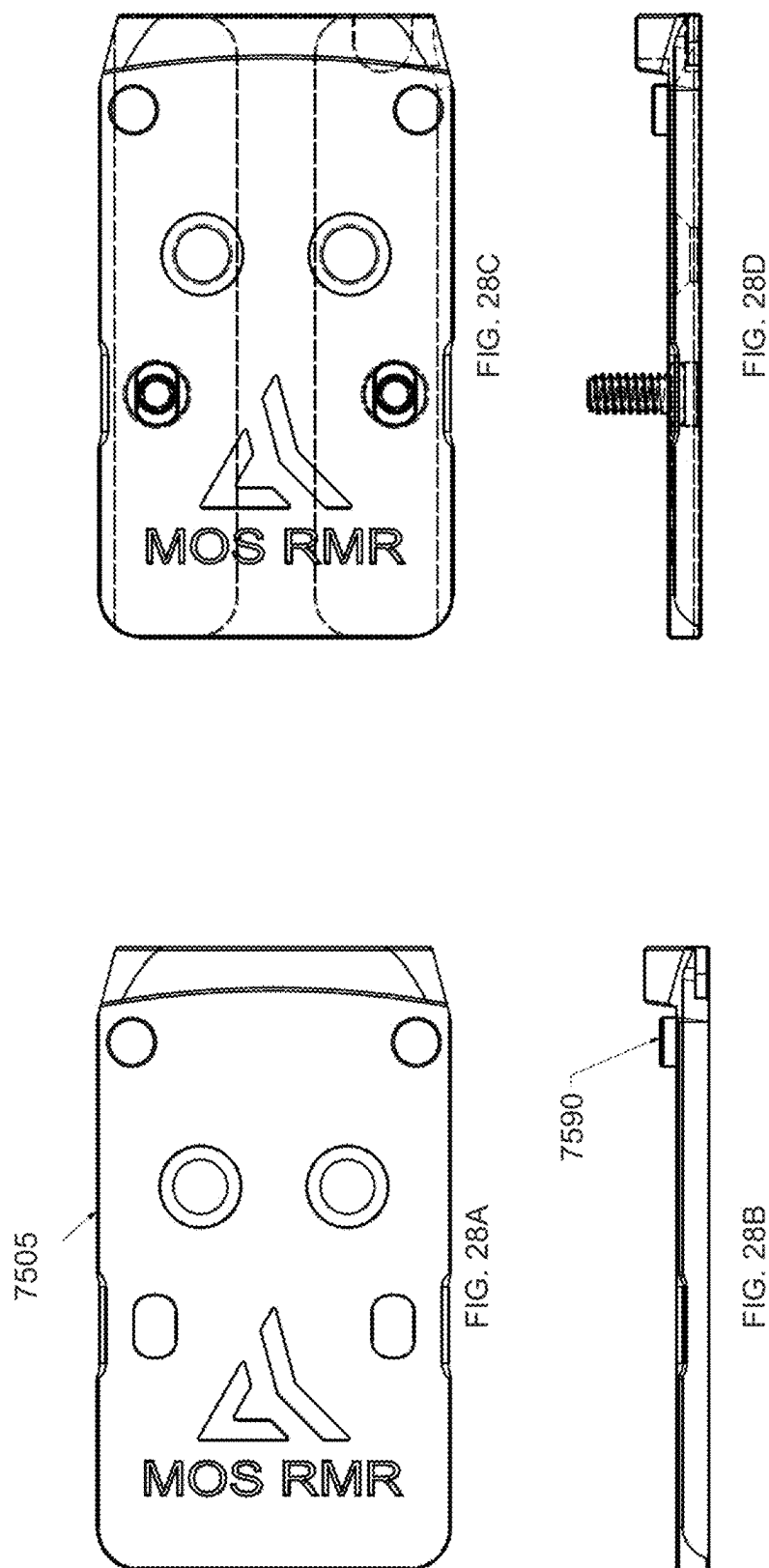
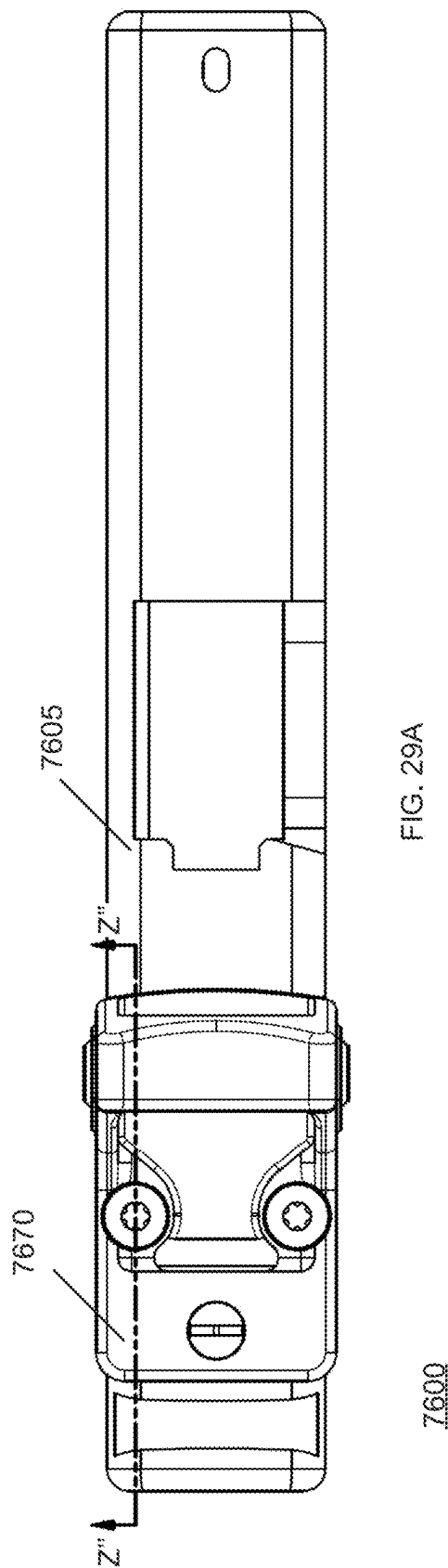


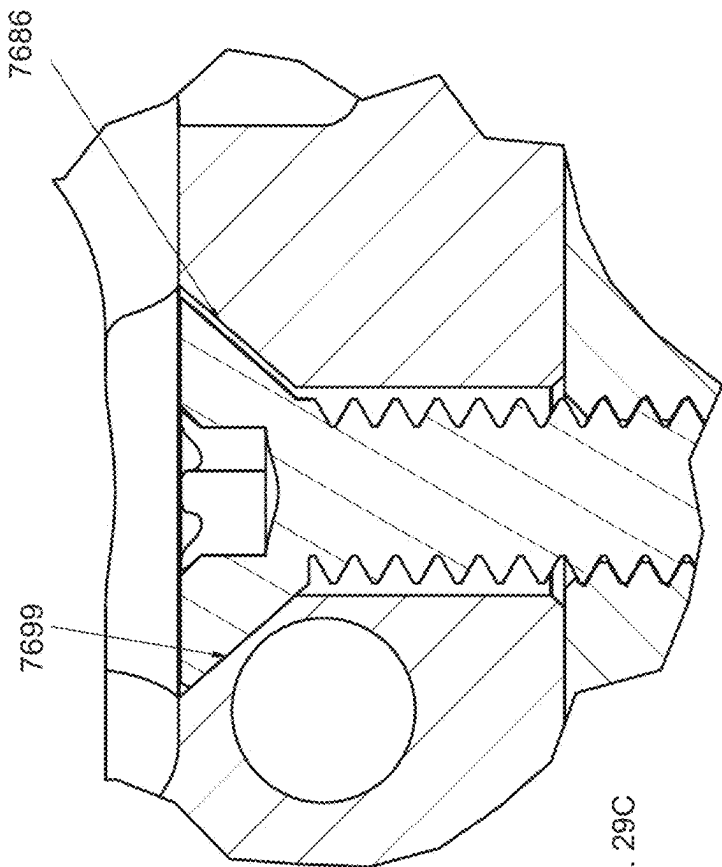
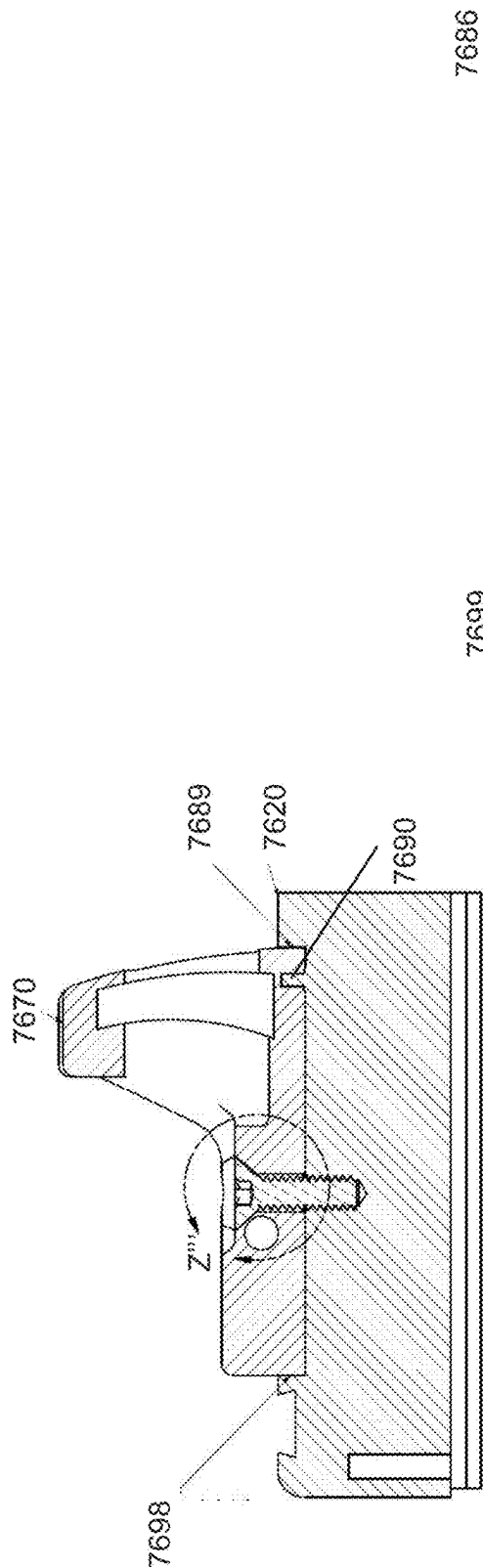
FIG. 27B











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LOCATING FASTENER FOR FIREARM OPTIC

RELATED APPLICATIONS

This non-provisional application claims benefit of U.S. Provisional Application No. 63/412,758, filed on Oct. 3, 2022 and entitled LOCATING FASTENER FOR FIREARM OPTIC, and U.S. Provisional Application No. 63/417,236, filed on Oct. 18, 2022 and entitled UNDERCUT ATTACHMENT INTERFACE FOR FIREARM OPTIC, each of which are incorporated by reference herein.

BACKGROUND

A firearm may be zeroed after an optic is attached thereto. However, with some known optic assemblies, the firearm may need to be re-zeroed after an optic is removed and re-attached to the firearm.

To avoid the need to re-zero the firearm, some firearms include rail-mounting assemblies. In these rail-mounting assemblies, an optic is slidably coupled to the firearm on a rail. Although these slidably coupling systems may eliminate the need to re-zero the firearm after the optic is removed and re-attached to the firearm, they may require a dedicated rail-mounted optic.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A illustrates an exploded isometric view of an optic assembly coupled to a locating optic assembly mount, according to various embodiments.

FIG. 1B illustrates a top view of the optic assembly of FIG. 1A.

FIG. 1C illustrates a section view taken along section line OO of FIG. 1B.

FIG. 1D illustrates a top view of the bracket of FIG. 1A, with the optic omitted for brevity.

FIG. 1E illustrates a section view taken along section line PP of FIG. 1D.

FIG. 1F illustrates a top view of the slide of FIG. 1A.

FIG. 1G illustrates a section view taken along section line QQ of FIG. 1F.

FIG. 1H illustrates a top view of the fastener of FIG. 1A.

FIG. 1I illustrates a section view taken along section line RR of FIG. 1H.

FIG. 2A illustrates a section view of an optic assembly coupled to a locating optic assembly mount, according to various embodiments.

FIG. 2B illustrates a detailed view SS of FIG. 2A.

FIG. 3A illustrates an exploded isometric view of an optic assembly coupled to a slide using an undercut attachment interface, according to various embodiments.

FIG. 3B illustrates a side view of the optic assembly of FIG. 3A coupled to the slide of FIG. 3A.

FIG. 3C illustrates an exploded isometric view showing a bottom of the optic assembly of FIG. 3A coupled to the slide of FIG. 3B.

FIG. 4A illustrates a top view of the optic assembly of FIG. 3A coupled to the slide of FIG. 3A.

FIG. 4B illustrates a section view taken along section line TT of FIG. 4A.

FIG. 4C illustrates a section view taken along section line UU of FIG. 4A.

FIG. 4D illustrates a detailed view VV of FIG. 4B.

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FIG. 5A-B illustrate installation of the bracket of the optic assembly of FIG. 2A on the slide in which the bracket is shown with transparency.

FIG. 5C illustrates a side view of the slide of FIG. 2A.

FIGS. 6A and 6B illustrate top and rear views, respectively, of the slide of FIG. 2A.

FIGS. 6C and 6D illustrate section views taken along lines WW and XX of FIG. 6A, respectively.

FIG. 7A illustrates a side view of the bracket of FIG. 2A.

FIG. 7B illustrates an isometric view showing a bottom of the bracket of FIG. 2A.

FIG. 7C illustrates a bottom view of the bracket of FIG. 2A.

FIG. 7D illustrates a section view taken along section line YY of FIG. 6C.

FIG. 8A illustrates an isometric view of the locating fastener of FIG. 2A.

FIG. 8B illustrates a top view of the locating fastener of FIG. 2A.

FIG. 8C illustrates a section view taken along section line ZZ of FIG. 7A.

FIG. 9A illustrates a side view of a slide assembly including an optic adapter self-lockingly fastened to a slide, according to various embodiments.

FIG. 9B illustrates a top view of the slide assembly of FIG. 9A.

FIG. 10A illustrates a section view taken along section line AA of FIG. 9B.

FIG. 10B illustrates a detailed view B of FIG. 10A.

FIG. 10C illustrates a detailed view D of FIG. 10B.

FIG. 10D illustrates a detailed view of a part of the self-locking threads of FIG. 10C.

FIG. 11A illustrates a top view of another a slide assembly including an optic self-lockingly fastened to an optic adapter, according to various embodiments.

FIG. 11B illustrates a side view of the slide assembly of FIG. 11A.

FIG. 12A illustrates a section view taken along section line A'A' of FIG. 11A.

FIG. 12B illustrates a detailed view B' of FIG. 12A.

FIG. 12C illustrates a detailed view C' of FIG. 12B.

FIG. 13A illustrates a top view of the nut of FIG. 11A.

FIG. 13B illustrates a section view taken along section line D'D' of FIG. 13A.

FIG. 13C illustrates a detailed view E' of FIG. 13B.

FIG. 14A illustrates a top view of another a slide assembly including an optic self-lockingly fastened to a slide, according to various embodiments.

FIG. 14B illustrates a side view of the slide assembly of FIG. 14A.

FIG. 15A illustrates a section view taken along section line AA" of FIG. 14A.

FIG. 15B illustrates a detailed view B" of FIG. 15A.

FIG. 16A illustrates a top view an assembly including a gas key self-lockingly fastened to a bolt carrier, according to various embodiments.

FIG. 16B illustrates a section view taken along section line EE of FIG. 16A.

FIG. 17A illustrates a detailed view F of FIG. 16B.

FIG. 17B illustrates a detailed view G of FIG. 17A.

FIG. 17C illustrates a detailed view of a part of the self-locking threads of FIG. 17B.

FIG. 18A illustrates a top view of the gas key of FIG. 16A.

FIG. 18B illustrates a section view taken along section line HH of FIG. 18A.

FIG. 19A illustrates a top view of the rotatably-driven part of FIG. 16A.

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FIG. 19B illustrates a section view taken along section line JJ of FIG. 19A.

FIG. 20A illustrates an assembly including a magwell self-lockingly fastened to a backstrap, according to various embodiments.

FIG. 20B illustrates an exploded view of the backstrap and magwell of FIG. 20A.

FIG. 20C illustrates a side view of the backstrap and the magwell of FIG. 20A.

FIG. 21A illustrates a bottom view of the assembly of FIG. 20C.

FIG. 21B illustrates a section view taken along section line KK of FIG. 21A.

FIG. 22A illustrates a detailed view L of FIG. 21B.

FIG. 22B illustrates a detailed view M of FIG. 22A.

FIG. 23A illustrates a front view of a firearm assembly including a buffer tube self-lockingly fastened to a lower receiver, according to various embodiments.

FIG. 23B illustrates a section view taken along section line NN of FIG. 23A.

FIG. 23C illustrates a detailed view P of FIG. 23B.

FIG. 24A illustrates a front view of the castle nut of FIGS. 23A-B.

FIG. 24B illustrates a section view taken along section line QQ of FIG. 24A.

FIG. 25A illustrates a front view of the end plate of FIG. 23A-B.

FIG. 25B illustrates a section view taken along section line RR of FIG. 25A.

FIG. 26A illustrates a top view of a slide assembly including a tapered opening to urge contact between a mounted component and a base component, according to various embodiments.

FIG. 26B illustrates a section view taken along section line S'S' of FIG. 26B.

FIG. 26C illustrates a detailed view T' of FIG. 26B.

FIG. 26D illustrates a detailed view U' of FIG. 26B.

FIG. 26E illustrates a detailed view V' of FIG. 26B.

FIG. 27A illustrates a side view of another slide assembly including a tapered opening to urge contact between a mounted component and a base component, according to various embodiments.

FIG. 27B illustrates a top view of the slide assembly of FIG. 27A.

FIG. 27C illustrates a section view taken along section line W'W' of FIG. 27A.

FIG. 27D illustrates a detailed view Y' of FIG. 27C.

FIG. 27E illustrates a section view taken along section line X'X' of FIG. 27A.

FIG. 27F illustrates a detailed view Z' taken of FIG. 27E.

FIG. 28A illustrates a top view of the optic adapter of FIG. 27A.

FIG. 28B illustrates a side view of the optic adapter of FIG. 27A.

FIG. 28C illustrates a top view of an optic adapter assembly including the optic adapter of FIG. 27A.

FIG. 28D illustrates a side view of the optic adapter assembly of FIG. 28C.

FIG. 29A illustrates a top view of another slide assembly including a tapered opening to urge contact between a mounted component and a base component, according to various embodiments.

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FIG. 29B illustrates a section view taken along section line Z''Z'' of FIG. 29A.

FIG. 29C illustrates a detailed view Z''' of FIG. 29B.

DETAILED DESCRIPTION

Locating Fastener for Firearm Optic

FIG. 1A illustrates an exploded isometric view of an optic assembly **3900** coupled to a locating optic assembly mount **3901** defined by a slide **3920**, according to various embodiments. FIG. 1B illustrates a top view of the optic assembly **3900** of FIG. 1A, with the optic omitted for brevity. FIG. 1C illustrates a section view taken along section line OO of FIG. 1A. FIG. 1D illustrates a top view of the bracket **3905** of FIG. 1A. FIG. 1E illustrates a section view taken along section line PP of FIG. 1D. FIG. 1F illustrates a top view of the slide **3920** of FIG. 1A. FIG. 1G illustrates a section view taken along section line QQ of FIG. 1F. FIG. 1H illustrates a top view of the locating fastener **3980A** of FIG. 1A. FIG. 1I illustrates a section view taken along section line RR of FIG. 1H.

The locating optic assembly mount **3901** is arranged for repeatably removing and alignably attaching an optic **3970** to the slide **3920**. It is known that a firearm may be zeroed after an optic is attached thereto. However, with some known optic assemblies, the firearm may need to be re-zeroed after an optic is removed and re-attached to the firearm.

To avoid the need to re-zero the firearm, some firearms include rail-mounting assemblies. In these rail-mounting assemblies, an optic is slidably coupled to the firearm on a rail. Although these slidably coupling systems may eliminate the need to re-zero the firearm after the optic is removed and re-attached to the firearm, they may require a dedicated rail-mounted optic.

Unlike the rail-mounting assemblies that may require the dedicated rail-mounted optic, locating optic assembly mount **3901** may be compatible with any style of optic (e.g., any style of pistol optic), such as the RMR optic **154** (FIG. 1F).

Referring to FIG. 1C, the locating optic assembly mount **3901** (FIG. 1A) may include locating fastener openings **3980B** that have a locating function and a fastening function. The locating function of the locating fastener openings **3980B** may provide repeatable locating of the optic **3970** relative to the slide **3920** (by providing repeatable locating of the bracket **3905** in a same position on the slide **3920**, in this embodiment). The fastening function may limit movement of the optic **3970** relative to the slide **3920** (by limiting “play” of the bracket **3905** relative to the slide **3920**, in this embodiment).

In this embodiment, referring to FIG. 1C, the locating fastener openings **3980B** include a counter opening **3980D** for a head of the shouldered fastener **3980A** (e.g., a counter sink), a counter opening **3980C** for the shoulder of the shouldered fastener **3980A**, and the threaded hole. The counter sink **3980D** may be arranged to provide a clearance fit for the head, while the counter bore **3080C** may be arranged to provide a slip fit for the shoulder (e.g., dimensioned to slip-fittingly receive the shoulder). The slip fit causes a center axis of the shouldered fastener **3980A** to precisely coincide with a center axis of the fastener opening **3080B** (the slip fit counter bore **3080C** provides the locating function). This contrasts with fastener openings in which the fastener may not be precisely aligned in a fastener hole due to tolerances in the threading (this threading tolerance allows screws to be rotationally driving into the hole).

When the shouldered fastener **3980A** is rotatably driven into the slide **3920** (e.g., into a threaded fastener opening), the slip fit may force the fastener **3980A** into a centered position in the locating fastener opening **3980B**. Although threading may be required to have sufficient tolerance to allow the shouldered fastener **3980A** to be rotatably driven into the threaded section of the locating fastener opening **3980B** (with no tolerance, driving a screw into a threaded hole may be difficult or impossible), the shoulder may be tight-fitting in the counter bore **3080C**, so that a center axis of the locating fastener **3980A** precisely coincides with a center axis of the locating fastener opening **3980B** when the shoulder is located in the tight-fitting counter bore **3080C**. As used herein, the term “locating counter opening” refers to a counter opening that performs this locating function by being tighter-fitting than the threading (e.g., dimensioned to slip-fittingly receive a corresponding part of a rotatably-driven part).

In some embodiments, tapered head may be less tight-fitting in the counter sink **3980D** than the shoulder is in the counter bore **3980C** (e.g., clearance fitting). This may optimize the ease of rotatably driving the locating fastener **3980A** into the locating fastener opening **3980B** on the final rotations.

Regarding the shoulder section **3980C**, tolerance of this section may be tighter than the tolerance of the threaded section of the locating fastener opening **3980B**, which may reduce or eliminate “play” of the bracket **3905** relative to the slide **3920**. Also, the shoulders may be wider than an outer diameter of the threading, which may resist sheering similar to any bolt shank described herein.

In this embodiment, an additional counter bore (centered on a same axis as the other counter openings) may provide a pocket for the seal **3981**. The seal **3981** and the pocket may be similar to any other seal and pocket described herein. The seal **3981** may prevent moisture from damaging the threading and/or threaded holes (in arrangements without the seal **3981** rust may form—binding up the threading in the threaded holes).

Referring again to FIG. 1A, in this embodiment the locating optic assembly mount **3901** uses fastener openings at opposite ends of the locating optic assembly mount. This may also may reduce or eliminate “play” of the bracket **3905** relative to the slide **3920**. In this embodiment, there are three threaded fastener openings, but other embodiments may have different numbers of fastener openings.

Also, in this embodiment, the locating and fastening features of the locating optic assembly mount **3901** may be formed using the same type of tooling (e.g., tooling for making counter openings). This may reduce manufacturing steps in forming the slide **3920** than other locating solutions on slides that may require different tooling for different regions of a locating structure on the slide (e.g., tooling for pins and tooling for grooves).

Referring again to FIG. 1A, the bolts **3964** and barrel nuts **3975** may be similar to any bolts and barrel nuts described herein. The optic **3970** may be similar to any optic described herein. The bracket **3905** includes a dovetail groove for a rear sight (the rear sight is shown installed in the dovetail groove). The bracket **3905** may include any features of any bracket described herein.

In the illustrated embodiment, the locating optic assembly mount is arranged to receive an optic assembly that includes the bracket **3905**. In other embodiments, a locating optic assembly mount may be arranged for direct-mounting an optic on a firearm (e.g., a slide include a locating optic assembly mount arranged for direct-mounting an optic).

In this embodiment, the coaxial counter openings are circular openings to mate with circular sections of the locating fastener **3980A**. In other embodiments, a locating fastener may have a faceted section in place of at least one of the circular sections (e.g., a square shoulder).

In this embodiment, the optic assembly **3900** includes the bracket **3905** and the optic **3970**. In other embodiments, it may be possible and practical to provide an optic assembly with an optic to mount directly to a locating optic assembly mount (e.g., without any bracket). In these embodiments, locating fasteners (similar in various respects to the locating fasteners **3980A**) may attach the optic directly to the locating optic assembly mount.

In this embodiment, the optic assembly **3900** is attached to the locating optic assembly mount **3901** using a homogeneous group of fasteners (e.g., three locating fasteners **3980A**). In other embodiments, an optic assembly may be attached to a locating optic assembly mount using a heterogeneous group of fasteners. Such a heterogeneous group of fasteners may include any number of locating fasteners (similar in various respects to locating fastener **3980A**) and any number of other fasteners now known or later developed. For example, a single locating fastener opening may be all that is required to achieve the locating function—and the fastening function may be provided in part using other fasteners now known or later developed (in one example, the single locating fastener opening is provided on one end of the locating optic assembly mount and threaded holes for any threaded fasteners now known or later developed may be provided at the other end of the locating optic assembly).

The embodiments described in the previous paragraph may use threaded openings on both ends of a locating optic assembly mount (e.g., using locating fastener openings exclusively or using any number of locating fastener opening(s) in combination with some other attachment feature that utilizes fasteners). However, in yet other embodiments, the other attachment feature may not require threading (or even separate fasteners). For example, in one embodiment, an optic assembly may include a bracket with a non-threaded structure (such as a tab) or recess to mate with a corresponding recess or non-threaded structure (such as the tab), respectively, on a firearm. The optic assembly may be installed by coupling the non-threaded structure with the recess on the one end, and then attaching the other end of the optic assembly to at least one locating fastener opening of the locating optic assembly mount. This is one example of an attachment feature for one end of an optic assembly in which the attachment feature does not use threading (or even separate fasteners). In various embodiments, an optic assembly may use any attachment feature now known or later developed in combination with at least one locating fastener opening (which may be similar in various respects to any locating fastener opening described herein).

FIG. 2A illustrates a section view of an optic assembly **4906** coupled to a locating optic assembly mount, according to various embodiments. This embodiment may be similar in any respect to the embodiment of FIG. 1A, but with a different fastener opening (both of the counter openings **4080C** and **4080D** of the locating fastener opening are counter bores) and a corresponding locating fastener **4080A**.

In this embodiment, the shoulder section **4080C** is shown with a slip fit arrangement (the spacing on the head section may be exaggerated to illustrate the contrasting clearance fit). As the fastener threading (which may be similar to fastener threading of FIG. 1A) is rotatably driven into the threaded section **4080B** of the locating fastener opening, the shoulder section **4080C** (which may have a tighter tolerance

than the threaded hole **4080B**) may precisely locate a center axis of the locating fastener **4080A** coincident with a center axis of the locating fastener opening. The shoulder section **4080C** may also may resist sheering similar to any bolt shank described herein.

The counter bore **4080D** may be arranged to be non-contacting a sidewall of the head of the locating fastener **4080A**, as illustrated. In this embodiment, the counter bores **4080C** and **4080D** are coaxial, but this is not required. In other examples, a counter bore (or other opening for a fastener head) may a center axis that is offset from a center axis of a locating counter opening.

In the embodiments illustrated herein, the mechanism (e.g., a socket) for rotating the locating fasteners is covered by an optic when the optic is coupled to a bracket. In other embodiments, the mechanism for rotating the locating fasteners may be exposed, which may allow the optic assembly to be removed from the firearm without removing the optic from the bracket. For example, in some embodiments a rear iron sight may be omitted from the bracket, and fastener opening(s) for the rear end of the optic assembly may be provided below this part of the bracket, so that the rotation mechanism of the rear fastener(s) may be exposed when the optic is attached to the mount. Fastener opening(s) for a front end of the optic assembly may also be exposed when the optic is attached to the mount. In other embodiments, one end of the optic assembly may attach to a firearm using a non-threaded attachment structure (such as a tab), and the other end of the optic assembly may attach to the firearm using one or more locating fasteners having rotating mechanisms that are exposed when the optic is attached to the bracket. In these embodiments, the optic assembly may be removed from the firearm without removing the optic from the bracket.

Undercut Attachment Interface for Firearm Optic

As described in the previous section, one end of an optic assembly may be arranged to attach to a firearm using a non-threaded attachment structure (such as a tab), and the other end of the optic assembly may attach to the firearm using one or more locating fasteners having rotating mechanisms that are exposed when the optic is attached to the bracket. In an optic assembly including a bracket and an optic, this may allow the optic assembly to be removed from the firearm without removing the optic from the bracket—which preserves zero. In the embodiments that follow, undercut structure(s) may be used to define a non-threaded attachment structure that may be used, in combination with one or more fasteners (e.g., one or more locating fasteners), to allow removal of the optic assembly from the firearm without removing the optic from the bracket.

FIG. 3A illustrates an exploded isometric view of an optic assembly **4100** coupled to a slide **4120** using an undercut attachment interface, according to various embodiments. FIG. 3B illustrates a side view of the optic assembly **4100** of FIG. 3A coupled to the slide **4120** of FIG. 3A. FIG. 3C illustrates an exploded isometric view showing a bottom of the optic assembly **4100** of FIG. 3A coupled to the slide **4120** of FIG. 3A.

A locating optic mount **4101** defined by a top of the slide **4120** includes an undercut structure **4199B** (FIG. 3A) to mate with a corresponding undercut structure **4199A** defined by an underside of the bracket **4105** (FIG. 3C). Referring briefly to FIGS. 6A, 6B, 6C, and 6D, in FIG. 6A a dashed line **4499** illustrates the undercut of undercut structure **4199B** (the undercut is also shown in FIG. 6C). The corre-

sponding undercut structure **4199A** (FIG. 3A) of the underside of the bracket **4105** is shown in FIGS. 7A, 7B, 7C, and 7D.

In this example, one of the undercut attachment structures **4199A** and **4199B** may include a first dovetail (e.g., the undercut attachment structure **4199B**, in this example), and the other of the undercut attachment structures **4199A** and **4199B** may include a second dovetail (e.g., the undercut attachment structure **4199A**, in this example) to mate with the first dovetail. However, in other examples undercut interfaces having any shape, dovetail or otherwise, may be used for an undercut attachment interface.

Referring again to FIG. 3A, the locating optic mount **4101** defined by the top of the slide **4120** may also include a locating fastener opening **4180B**, which may be similar in various respects to any locating fastener opening described herein. A locating fastener **4180A** includes threading and a shoulder to mate with the locating fastener opening **4180B**. However, the locating fastener **4180A** also includes a tapered head to mate with a tapered opening **4185** defined by a top surface of the bracket **4105**. The locating fastener **4180A** is illustrated in more detail FIGS. 8A, 8B, and 8C.

Referring again to FIG. 3A, the locating optic assembly mount **4101** may also include a fastener opening **4180B** to provide a fastening function, and the bracket **4105** may include a tapered opening **4185** to provide a locating function. The fastening function may limit movement of the optic assembly **4100** relative to the slide **4120** (by limiting “play” of the bracket **4105** relative to the slide **4120**, in this embodiment), similar to the locating fastener **3980A** (FIG. 1A). Regarding the locating function, the tapered head of the locating fastener **4180A** is arranged to engage with part of a tapered opening **4185** of the bracket **4105** to interlock the undercut structures **4199A** and **4199B**, which repeatedly locates the bracket **4105** (and therefore the optic **4170** installed thereon) in the same position on the slide **4120**.

Installation of the bracket **4105** on the locating optic mount **4101** is illustrated in FIGS. 5A and 5B. The slide **4120** includes recesses **4488** (FIG. 6A), and the bracket **4105** includes projections **4489** (FIG. 7B) that drop into the recesses **4488** to position the bracket **4105** on the slide **4120**, as shown in FIG. 5A. Next, a user slides the bracket **4105** forwardly to perform an initial alignment of the bracket **4105** on the slide **4120** (e.g., when the tapered opening **4185** and the locating fastener opening **4180B** align, as shown in FIG. 5B, which allows the locating fastener **4180A** (FIG. 3A) to be installed through the aligned openings **4185** and **4180B**).

As the locating fastener **4180A** is installed in the aligned openings **4180B** and **4185**, referring now to FIGS. 4A, 4B, 4C, and 4D, as shown in FIG. 4D a part of the tapered head **4180D** engages a part of the tapered opening **4185**. This engagement urges the bracket **4105** forward, which interlocks the undercut structure **4199B** (FIG. 3A) and the undercut structure **4199A** (FIG. 3C). FIG. 4D shows a gap **4186** (not to scale) on only one side of the tapered head **4180D** to emphasize the forward-urging caused by the taper-engagement. This interlocking performs a final alignment of the bracket **4105** on the slide **4120**.

Still referring to FIG. 4D, in this embodiment the locating fastener **4180A** includes a non-threaded shoulder **4180C**. In one embodiment, this non-threaded shoulder **4180C** may be slip fit with the locating opening **4180A** similar to any locating opener described herein. A part of the non-threaded shoulder **4180C** may be clearance fit with the straight section below the tapered opening **4185** (so that the only part of the locating fastener **4180A** that contacts the bracket **4105**

is the tapered head **4180D**, which is illustrated in the drawing), in various embodiments. Other embodiments may not include the non-threaded shoulder—instead the entire length of a locating fastener below a tapered head may be threaded.

The threading on the locating fastener **4180A** can be similar to any threading described herein. In some embodiments, the locating fastener opening **4180B** may have a counter bore for receiving a seal (not shown), similar to the counter bore **3982** (FIG. 1C). This may prevent corrosion—so that the threading does not bind up over time.

Referring again to FIG. 6A, a part of a sidewall of the undercut structure **4199B** (FIG. 3A) of the slide **4120** may have a pair of non-parallel straight sections **4479B** (FIG. 6A) located between curved sections **4479A** and **4479C**. This pair of non-parallel straight sections **4479B** interlocks with a corresponding pair of non-parallel straight sections on the undercut structure **4199A**—contributing, together with the tapered head **4180D**, to alignment (e.g., lateral and/or front/aft alignment) of the optic assembly **4100** in a repeatable position on the slide **4120**.

Referring again to FIG. 3A, the optic **4170** may be similar to any optic described herein. The fasteners **4175** that couple the optic **4170** to the bracket **4105** may also be similar to any fasteners described herein for attaching an optic to a bracket, or any known fastener. The rear sight **4189** may be fixably installed (e.g. press fit) to a rear end of the bracket **4105** using a dovetail groove, as illustrated.

In another embodiment, an optic may be installed more aftly on a bracket, and a rear sight may be fixably installed to a front end of the bracket using a dovetail groove. In another embodiment, an optic guard (e.g., any optic guard described herein) may be fixably or releasably installed to a front end of the bracket using a dovetail groove.

Self-Lockingly Fastened Components in a Recoil Environment

Various firearm assemblies have some degree of modularity. Sometimes modularity is provided by the original equipment manufacturer (OEM) of the firearm—an operator may have the option of configuring and/or operating the firearm with or without an optional component. The firearm may be manufactured with the optional component fixably installed on the attachment interface, in some examples. If the firearm is configured with the optional component, the operator may wish to uninstall the component of the firearm for some applications and re-install the component for other applications.

In other instances, modularity may be provided in the aftermarket. As a simple example, a user may replace the original barrel of a firearm with a threaded barrel to provide modularity. This essentially adds an additional attachment interface—the threaded part of the barrel. The user may then thread mating accessories, such as compensators, suppressors, or the like, onto the threaded barrel.

In either case, in some examples, the attachment interface is located on a reciprocating component of the firearm. An example is an optic attachment interface on a slide. This may expose the attachment interface to various recoil-based forces with each movement of the assembly including reciprocating component and the reciprocating-component mounted device.

To withstand the repeated action of the reciprocating-component, various known attachment interfaces may fixably install the reciprocating-component mounted device on the reciprocating component. In a fixable installation, a

non-temporary deformation occurs when installing and/or removing the reciprocating-component mounted device from the reciprocating component. In many examples of fixable installations generally, non-temporary deformation must occur at both installation and removal. For example, with riveting, the rivets are non-temporarily deformed on attachment and removal. In other examples, non-temporary deformation may occur in response to clamping, drilling, welding, or the like.

In other fixable installations, non-temporary deformation may occur at removal specifically. One example is an adhesive, such as a thread locking adhesive. Once the thread-locking adhesive cures, the cured adhesive must be broken in order to remove the reciprocating-component mounted device from the reciprocating component. A user may unintentionally damage the reciprocating component or the reciprocating-component mounted device when attempting to break the cured adhesive. Even when this damage is avoided, the user may still have to clean up the devices (i.e., remove the broken cured compound in order to apply the required new adhesive for fixable re-installation).

U.S. Pat. No. 11,480,414, which is incorporated by reference herein, describes a taper lock interface. This taper lock interface may be utilized to barrel-mount a firearm accessory. What is needed is an attachment interface that may be similar in some respects to the '414 embodiments. It would also be beneficial for such an attachment interface to have attributes consistent with self-lockingly fastening a wide variety of parts in a recoil environment (e.g., not limited to barrel-mounting, and having attributes consistent with use in various locations of a firearm assembly, such as compact dimensions, ease of use, etc.)

Various embodiments described herein include an assembly to operate in a recoil environment in which the assembly may be mounted on a device that produces or transmits recoil. At least one part of the assembly may be self-lockingly fastened to a mating part. In various embodiments, the mating part may be a slide or other reciprocating part of a recoil-producing device.

While various features described herein may be arranged to withstand the variety of forces involved when the mating component is a reciprocating component (e.g., a slide), any of the features described herein are also well-suited to self-lockingly fastening the assembly to some other mating part of a device that produces or transmits recoil (e.g., a non-reciprocating part such as a frame assembly of a firearm, or another part of an assembly in which both parts are mounted on a reciprocating or non-reciprocating part in a recoil environment). For example, the features described herein may be used for self-lockingly fastening a magwell to a backstop, self-lockingly fastening a gas key to a bolt carrier, self-lockingly fastening a buffer tube to a lower receiver, or self-lockingly fastening any other parts in a recoil environment.

In various embodiments, an apparatus including at least one part of an assembly self-lockingly fastened to a mating part may include a threaded opening defining self-locking threads, and a non-threaded opening defining: a tapered opening comprising a locking taper, or a tapered counter opening arranged to make contact (e.g., taper-to-taper contact) with a taper defined by a rotatably-driven part, wherein the threaded hole is arranged to mate with threading defined by the rotatably-driven part. In various embodiments, the at least one part of the assembly may be self-lockingly fastened to the mating part in more than one way (e.g., using the locking taper and self-locking threads).

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In some embodiments, the threaded and non-threaded openings may be part of a same fastener opening, and the rotatably-driven part may be a rotatably-driven fastener. The fastener opening may have a top section and a bottom section located below the top section in the fastener opening. The bottom section may define the threaded opening and the top section may define the non-threaded opening.

In various embodiments, one of the self-lockingly fastened parts may define the threaded opening and the other of the self-locking fastened parts may define at least a portion of the tapered counter opening.

In some embodiments, the rotatably-driven fastener may define the self-locking threads—such as in the case of a nut (e.g., a sleeve nut, a castle nut, or any other nut). In some embodiments, the tapered opening may mate with a tapered surface defined by an exterior of the nut, in which the threaded opening is defined by an interior of the nut.

FIG. 9A illustrates a side view of a slide assembly **5900** including an optic adapter **5905** self-lockingly fastened to a slide **5920**, according to various embodiments. FIG. 9B illustrates a top view of the slide assembly **5900** of FIG. 9A. FIG. 10A illustrates a section view taken along section line AA of FIG. 9B. FIG. 10B illustrates a detailed view B of FIG. 10A. FIG. 10C illustrates a detailed view D of FIG. 10B. FIG. 10D illustrates a detailed view of a part of the self-locking threads of FIG. 10C.

In this embodiment, the reciprocating-part mounted assembly includes an optic adapter **5905** (to receive an optic, not shown) and the illustrated fasteners, and the mating part is the slide **5920**. The optic adapter **5905** is self-lockingly fastened to the slide **5920**, rather than being fixably installed on a slide like other slide assemblies that may require adhesive (e.g., a thread locking compound).

Like other self-lockingly fastened embodiments described herein, the slide assembly **5900** includes a non-threaded opening and a threaded opening (**5980A** and **5980B**, respectively). In this embodiment, self-locking threads (FIG. 10C, **5980C**) are defined by the mating part (by the slide **5920** in this embodiment), and a tapered opening (e.g., a conical counter opening) is defined by the other part **5905**.

The tapered opening may have a slope that is corresponding to (e.g., matching) a slope of a tapered section of the rotatably-driven fastener **5975**, to provide taper-to-taper contact (as illustrated). Referring to FIG. 10B, a conical taper **5976** may make taper-to-taper contact with the tapered opening.

Depending on a number of factors including a slope of the tapers, this taper-to-taper contact may produce some resistance to the rotatably-driven fastener **5975** backing out due to a recoil impulse transmitted into the rotatably-driven fastener **5975** by, say, the mating part (which hereinafter may be referred to as ‘recoil-loosening resistance’). The slope of the conical taper **5976** is shallow compared to the conical taper **6676** of the locking taper illustrated in FIG. 17A. A degree of slope may be varied depending on design requirements. Referring again to FIG. 10B, in this embodiment using the shallow slope allows an optic to ‘sit lower’ on the pistol, because the shallow slope does not require increasing a height of the fastener head, which therefore does not require increasing the thickness of a plate of the optic adapter **5905**. An optic that sits lower on a pistol is, of course, an advantage for firearm operators. The slope may be varied from what is shown based on any application requirement such as:

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A total magnitude of recoil-loosening resistance required (based on the location of the attachment interface on the firearm, characteristics of the firearm, or the, or combinations thereof);

Sizing requirements for the attachment interface; or

A magnitude of total recoil-loosening resistance provided by self-locking threads (which may be impacted by the count of individual threads of self-locking threads used on the mating part, a number of threaded openings used for the attachment interface, a type of self-locking thread, etc.), or the like, or combinations thereof.

In embodiments in which the magnitude of the recoil-loosening resistance provided by self-locking threads alone is not sufficient for an application, a steep slope may be used to provide recoil-loosening resistance that is equal or greater than a threshold amount, such as a recoil-loosening resistance comparably provided by the self-locking threads (e.g., redundancy), e.g., a ‘locking taper’. In other embodiments, a conical taper may have a shallow slope which may provide some other non-zero amount of additional recoil-loosening resistance. In various embodiments, a steep slope (e.g., an included angle) may be thirty degrees or less, as just one example; however, this is just an example—since various factors may impact slope requirements, as previously discussed. In some examples, it may be possible and practical to use a locking taper and any threads now known, or later developed (e.g., no redundancy).

The self-locking threads **5980C** (FIGS. 10C and 10D) may be any self-locking threads now known, or later developed, that may provide a threshold recoil loosening resistance alone or in combination with any recoil resistance provided by the taper-to-taper contact, which does not fixably couple the self-locking fastened parts (e.g., do not result in non-temporary deformation on every fastening or unfastening, such as threads that require thread-locking compound to prevent loosening in a recoil environment).

In some examples, the self-locking female threads may have a thirty degree wedge ramp, as illustrated, may be created using threading tools such as the Emuge threading tools described on the Emuge-Fraken Group website. The male threading **5980D** may be any threading to mate with the self-locking female threads **5980C**.

Referring again to FIGS. 9A and 9B, the self-lockingly fastened parts **5905** and **5920** are non-fixably coupled. Unlike some known firearm assemblies that may fixably install an optic plate on a slide using a thread-locking compound, the optic adapter **5905** can be releasably removed from the slide **5920** and re-fastened to the slide **5920** by loosening and/or tightening the rotatably-driven fastener **5975**.

The optic adapter **5905** and slide **5920** may be similar in any respect to any other optic adapter and slide described herein. In various embodiments, the slide **5920** and/or the optic adapter **5905** may have any locating fastener opening described herein, which may allow the optic adapter **5905** and/or an optic attached thereto (not shown for brevity) to be released and re-fastened without having to re-zero the firearm. The locating fastener opening may be a different opening than the fastener opening illustrated in FIG. 9A, or the illustrated opening may perform the locating function similar to any locating fastener opening described herein. A locating fastener opening may have any threads and/or may or may not provide the same taper-to-taper contact described above (in the case of a locating fastener opening that is also relied upon for recoil-loosening resistance).

FIG. 11A illustrates a top view of another slide assembly **6100** including an optic **6170** self-lockingly fastened to an

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optic adapter **6105**, according to various embodiments. FIG. **11B** illustrates a side view of the slide assembly of FIG. **11A**. FIG. **12A** illustrates a section view taken along section line AA' of FIG. **11A**. FIG. **12B** illustrates a detailed view B' of FIG. **12A**. FIG. **12C** illustrates a detailed view C' of FIG. **12B**. FIG. **13A** illustrates a top view of the nut **6175** of FIG. **11A**. FIG. **13B** illustrates a section view taken along section line DD' of FIG. **13A**. FIG. **13C** illustrates a detailed view E' of FIG. **13B**.

Referring now to FIG. **11B**, the optic adapter **6105** may be similar to optic adapter **5905** in any respect, and may be self-lockingly fastened to the slide **6120** in a similar way that optic adapter **5905** is self-lockingly fastened to the slide **5920**. The slide **6120** may be similar in any respect to the slide **5920**.

In this embodiment, the rotatably driven part **6175** may be a nut (e.g., a sleeve nut, which may be similar in any respect to any sleeve nut described herein), which may include self-locking female threads **6180C** that may be similar to self-locking female threads **5980C** in any respect. Referring to FIG. **13B**, the nut **6175** may have a tapered section **6176** to provide taper-to-taper contact with the non-threaded opening **6180A** (FIG. **12B**) similar to any other taper-to-taper contact described herein.

The nut **6175** may include any features of any other nut described herein, in various embodiments. Also, although the illustrated nut **6175** (FIGS. **13A-13C**) is a sleeve nut in the illustrated embodiment, in other embodiments other types of nuts may be used in a similar way to the sleeve nut **6175**. In still other embodiments, other rotatably driven parts similar in any respect to the sleeve nut **6175** may be used in other embodiments in which an optic is self-lockingly fastened to a slide assembly.

In contrast to the previously described embodiments in which the threaded opening **5980A** is defined by the slide **5920**, the threaded opening **6180B** is a threaded hole of the sleeve nut **6175**. The non-threaded opening **6180A** may receive a head of the sleeve nut **6175**, and may provide taper-to-taper contact.

The optic **6170** and optic adapter **6105** may be similar in any respect to any other optic and optic adapter described herein. In various embodiments, the optic adapter **6105** and/or the optic **6170** may have any locating fastener opening described herein, which may allow the optic **6170** to be released and re-fastened to the slide assembly **6100** without having to re-zero the firearm. The locating fastener opening may be a different opening than the fastener opening illustrated in FIG. **12A**, or the illustrated opening may be modified to perform the locating function similar to any locating fastener opening described herein.

The fastener **6174** may be a bottom-up fastener **6174** (e.g., a bottom-up bolt), as illustrated. However, in other embodiments, a threaded post may be integrally formed on an optic adapter, to provide the threading **6180D** to mate with the self-locking female threads **6180C**. Any structure, now known or later developed, to provide threading to which the self-locking threads may be self-lockingly fastened to may be used.

FIG. **14A** illustrates a top view of another a slide assembly **6400** including an optic **6470** self-lockingly fastened to a slide **6420**, according to various embodiments. FIG. **14B** illustrates a side view of the slide assembly **6400** of FIG. **14A**. FIG. **15A** illustrates a section view taken along section line AA" of FIG. **14A**. FIG. **15B** illustrates a detailed view B" of FIG. **15A**.

This is a direct mount embodiment in which slide **6420** includes an optic mounting section to which the optic **6470**

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may be directly fastened, instead of using optic adapter. There may be a gasket/O-ring and/or a sealing plate (not shown) or the like, located between the directly fastened optic **6470** and the slide **6420**, in various embodiments.

In this embodiment, the threaded opening **6480B** may be defined by the slide **6420** and the non-threaded opening **6480A** (to provide taper-to-taper contact with the taper **6476** defined by the rotatably-driven part **6475**) may be defined by the optic **6470**. The rotatably-driven part **6475** may provide the threading **6480D** to mate with the self-locking threads **6480C**. In the illustration, the rotatably-driven part **6475** has threading all the way to its head, but this is not required (in some embodiments the rotatably-driven part **6475** may have a non-threaded shoulder between the threading **6480D** and its head).

FIG. **16A** illustrates a top view an assembly **6600** including a gas key **6606** self-lockingly fastened to a bolt carrier **6621**, according to various embodiments. FIG. **16B** illustrates a section view taken along section line EE of FIG. **16A**. FIG. **17A** illustrates a detailed view F of FIG. **16B**. FIG. **17B** illustrates a detailed view G of FIG. **17A**. FIG. **17C** illustrates a detailed view of a part of the threading of FIG. **17B**. FIG. **18A** illustrates a top view of the gas key **6606** of FIG. **16A**. FIG. **18B** illustrates a section view taken along section line HH of FIG. **18A**. FIG. **19A** illustrates a top view of the rotatably-driven part of FIG. **16A**. FIG. **19B** illustrates a section view taken along section line JJ of FIG. **19A**.

Referring now to FIG. **16B**, the mating part (the bolt carrier **6621**, which may be a reciprocating assembly that reciprocates on an inside of an upper receiver) may define the threaded opening **6680B**. The non-threaded opening **6680A** may be defined by the gas key **6606**. Referring now to FIG. **17A**, a taper **6676** to form a locking taper is illustrated here. This arrangement is also self-locking in more than one way, given the self-locking threads **6680C** to mate with the threading **6680D**. One or more of the recoil-loosening resistances provided by the different structures may individually provide a threshold recoil-loosening resistance (e.g., with the other providing some amount of redundancy), or may provide the threshold recoil-loosening resistance in combination.

FIG. **20A** illustrates an assembly **7000** including a magwell **7005** self-lockingly fastened to a backstrap **7020**, according to various embodiments. FIG. **20B** illustrates an exploded view of the backstrap **7020** and magwell **7005** of FIG. **20A**. FIG. **20C** illustrates a side view of the backstrap **7020** and the magwell **7005** of FIG. **20A**. FIG. **21A** illustrates a bottom view of the assembly of FIG. **20C**. FIG. **21B** illustrates a section view taken along section line KK of FIG. **21A**. FIG. **22A** illustrates a detailed view L of FIG. **21B**. FIG. **22B** illustrates a detailed view M of FIG. **22A**.

Referring now to FIGS. **21B**, **22A**, and **22B** in combination, the threaded opening **7780B** may be defined by the mating part (the backstrap **7020**). The magwell **7005** may define the non-threaded opening **7780A**. A conical taper **7076** of the rotatably-driven fastener has a shallower slope than the previously illustrated embodiment due to application requirements. The self-locking threads **6680C** may be defined by the backstrap **7020**, and the threading **7080D** may be defined by the rotatably-driven part **7075**.

FIG. **23A** illustrates a front view of a firearm assembly **7300** including a buffer tube **7305** self-lockingly fastened to a lower receiver **7320**, according to various embodiments. FIG. **23B** illustrates a section view taken along section line NN of FIG. **23A**. FIG. **23C** illustrates a detailed view P of FIG. **23B**. FIG. **24A** illustrates a front view of the castle nut

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7375 of FIGS. 23A-B. FIG. 24B illustrates a section view taken along section line QQ of FIG. 24A. FIG. 25A illustrates a front view of the end plate 7374 of FIG. 23A-B. FIG. 25B illustrates a section view taken along section line RR of FIG. 25A.

End plate 7374 includes a tab 7390 (FIG. 25A) that engages a corresponding notch in the bottom of the buffer tube 7305. The boss 7391 (FIG. 25B) engages a corresponding recess in the lower receiver 7320 (FIG. 23B). These fix rotational positions (relative to the lower receiver 7320) of the buffer tube 7305 and the endplate 7374, respectively.

This embodiment uses a nut 7375 (e.g., a castle nut) to self-lockingly fasten the buffer tube 7305 to a buffer tube tower of the lower receiver 7320. The threaded opening 7380B is defined by an interior of the nut 7375. In this embodiment, the mating part is a lower receiver assembly including the end plate 7374. The end plate 7374 defines the non-threaded opening 7380A including the tapered opening. A taper 7376 to form a locking taper is illustrated, formed by the taper-to-taper contact of a taper 7376 (FIG. 23B) on the nut 7375 and the steeply sloped non-threaded opening 7380A defined by the end plate 7374.

In this embodiment, the threading 7380D to mate with the self-locking threads 7380C is defined by the other part of the self-lockingly fastened parts (e.g., the buffer tube 7305). The buffer tube 7305 may be threaded to the end plate, and then the castle nut 7375 may be tightened to non-fixably couple the buffer tube 7305 to the mating part.

In any embodiments described herein in which a rotatably-driven part is a fastener, any locating function described herein may be also be performed by the rotatably-driven part, if desired/required. In one example, any self-locking fastener opening described herein may have a locating section located between any threaded and non-threaded openings described herein. The locating section may be dimensioned to slip-fittingly receive a non-threaded shoulder of the rotatably driven fastener to align a center axis of the rotatably-driven part with a center axis of the threaded opening defined by the mating part, similar to any locating fastener opening embodiments described herein. In another example, a conical taper head of a locating fastener may be received by a locating counter opening of the self-lockingly fastened parts to align a center axis of the rotatably-driven part with a center axis of the threaded opening defined by the mating part.

Any of the self-locking features described herein may be used in any embodiments of a locating fastener opening and/or undercut attachment interface. For example:

Any self-locking threads described herein may be added to any locating counter opening described herein (e.g., added to the locating counter opening illustrated in FIG. 2A);

The counter opening 4180B (FIG. 3A) may define self-locking threads similar to any self-locking threads described herein, to self-lockingly fasten the optic adapter 4105 to the slide 4120 in more than one way (i.e., in addition to the recoil-loosening resistance provided the locking taper illustrated by that figure).

Also, it is noted that any threaded opening described in the '414 patent can be modified to include any self-locking threads described herein. U.S. Non-provisional patent application Ser. No. 18/244,731, filed on Sep. 11, 2023 and entitled SELF-LOCKINGLY FASTENED COMPONENTS IN A RECOIL ENVIRONMENT, is incorporated by reference herein.

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Tapered Opening to Urge Contact Between a Mounted Component and a Base Component in a Recoil Environment

As explained herein in the section describing self-lockingly fastened parts, threaded fasteners may be used to attach a wide variety of parts in a recoil environment—including OEM parts and aftermarket parts. However, some known arrangements may expose the fasteners to a wide variety of shear forces. For instance, the fasteners may be exposed to these shear forces when the firearm receives an impact (such as from the ground or an object). Also, the fasteners may be exposed to these shear forces during normal operation of the firearm, particularly when the part is mounted to a reciprocating part, is part of an assembly mounted to the reciprocating part and/or when the firearm is charged by pushing a mounted component against an object.

These forces may cause the fasteners to flex, which may produce damage over time to the fasteners (e.g., stretching of the fasteners and/or micro fractures) and/or degradation of the fastening functioning (e.g., loosening). This loosening may induce play to the assembly, further allowing damage as the fasteners are allowed to flex more from impact or recoil. This play may be particularly undesirable when the mounted part is an aiming device, such as an optic or a mechanical sight because the play may produce some degree of inaccuracy. The damage may ultimately lead to the fastener failing completely (e.g., fracturing into two pieces).

FIGS. 5A-B illustrate an arrangement in which a tapered opening 4185 has a center axis that is offset with respect to a center axis of a locating fastener opening 4180B. As a fastener is tightened in this arrangement, the bracket 4105 is urged forward, interlocking the undercut structure 4199B (FIG. 3A) and the undercut structure 4199A (FIG. 3C).

In some known optic mounting devices, the only surface that may make contact with a firearm may be the bottom surface of an adapter (or a bottom surface of an optic in a direct mount arrangement). However, this known contact may not prevent the optic or the adapter from developing play in the fore/aft direction and/or lateral directions. This known contact also may not protect the fasteners during reciprocating action of the slide and/or during an impact (the fasteners may be exposed to sheering forces in the fore/aft direction and/or lateral directions).

Referring now to FIG. 5B, the urging caused by the installation of the locating fastener 4180A in the locating fastener opening 4180B and the tapered opening 4185 may result in contact between additional surface(s) of bracket 4105 and/or the slide 4120. In this particular example, the additional surface(s) of the bracket 4105 may include surfaces of sides of the bracket 4105 (e.g., undercut structure 4199A and/or an end such as a front end of the bracket 4105). This additional contact may protect the fastener 4180A from shear forces resulting from reciprocation of the slide 4120, impact to the bracket 4105, and/or impact to an optic 4170 mounted thereon. As such, fastener flex may be avoided and/or reduced, thus eliminating or reducing damage to fasteners and fastener function over time.

In various embodiments using a tapered opening to urge contact between a mounted component and a base component, with reference to FIG. 5A, an apparatus may include an assembly to operate in a recoil environment, in which at least one part of the assembly (e.g., bracket 4105) is fastened to at least one base part (e.g., slide 4120). The fastened parts may include a threaded opening (e.g., threaded hole 4180B), and a non-threaded opening (e.g., 4185). The non-threaded opening may include a tapered opening (e.g., 4185) to urge

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the at least one part of the assembly against at least one additional surface (e.g., the interlocking undercut structures), in response to fastening the at least one part of the assembly to the at least one base part using the threaded opening (e.g., using the threaded hole 4180B).

Although the contacting surface(s) of the mounted device (e.g., bracket 4105) are undercut structures in the embodiment illustrated in FIG. 5A, this is not required in other embodiments. The contacting surfaces may be any other surface of the mounted device, such as part of a back end and/or a front end of the mounted device, or part of a recoil post opening. Without limitation as to the location of contact, FIGS. 26A-29C illustrate various an embodiment in which the contacting surface(s) include a back of a mounted device and/or part of a recoil post opening.

In various embodiments, any contacting surface may be any other surface of the mounting device that may be urged into contact with any part of the firearm assembly (using a tapered opening) to prevent fastener flex and/or protect the fastener(s) from impact. In various embodiments, the contacting surface may be any surface-to-surface contact, including but not limited to:

- one or more curved surface(s) (e.g., a recoil post opening, or some other surface with one or more curved sections),
- a single flat (e.g., a surface having a single planar face),
- more than one flat (e.g., a surface with more than one face), or
- combinations thereof.

FIG. 26A illustrates a top view of a slide assembly 7400 including a tapered opening to urge contact between a mounted component and a base component, according to various embodiments. FIG. 26B illustrates a section view taken along section line S'S' of FIG. 26A. FIG. 26C illustrates a detailed view T' of FIG. 26B. FIG. 26D illustrates a detailed view U' of FIG. 26B. FIG. 26E illustrates a detailed view V' of FIG. 26B.

Referring to FIGS. 26A and 26B, in this embodiment the mounted device is an optic adapter 7405, which may be similar to any optic adapter described herein. The rotatably driven fastener 7475 may include a threaded section (e.g., a threaded length) and one or more non-threaded sections (e.g., a tapered head, and in some embodiment a non-threaded shoulder). The slide 7420 may be similar to any slide described herein.

Referring now to FIG. 26C, axes 7466 of the threaded hole in the slide 7420 and the tapered opening 7485 may be non-coaxial (e.g., an axis of the tapered opening 7485 may be offset relative to an axis of the threaded hole). Due to this offset, as the rotatably driven fastener 7475 is installed in the threaded hole, one side of the tapered head makes contact with the tapered fastener opening 7485. The other side of the tapered head may have no contact, in some embodiments (the illustration depicts a gap 7486 on an opposite side of the tapered head—the size of the gap may be exaggerated for emphasis and may not be to scale).

Referring to FIGS. 26C and 26D, the contact 7485 (FIG. 26C) by the tapered head and the tapered opening urges the optic adapter 7405 rearward causing the rear surface of the optic adapter 7405 to come into contact 7499 with the slide 7420. Referring to FIG. 26E, a gap 7487, also sized for emphasis, is also shown between a front end of the optic adapter 7405 and the slide 7420.

The contact of the surface 7499 resulting from the urging provided by the tapered opening 7485 may durably fix a position of the optic adapter 7405 relative to the slide 7420. Also, if the optic adapter 7405 takes an impact (e.g., by

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engaging the grip illustrated on the front end of the optic adapter 7405 with a surface to charge the firearm and/or by an impact to the frame of the front end of the optic adapter 7405), the contact of the surface 7499 may reduce an amount of shear force that may be applied to the rotatably driven fastener 7475 during the impact. Also, the contact of the surface 7499 may reduce an amount of shear force that may be applied to the rotatably driven fastener 7475 during reciprocating action of the slide 7420.

FIGS. 27A and 27B illustrates a side view of another slide assembly 7500 including a tapered opening to urge contact between a mounted component and a base component, according to various embodiments. FIG. 27B illustrates a top view of the slide assembly of FIG. 27A. FIG. 27C illustrates a section view taken along section line W'W' of FIG. 27A. FIG. 27D illustrates a detailed view Y' of FIG. 27C. FIG. 27E illustrates a section view taken along section line X'X' of FIG. 27A. FIG. 27F illustrates a detailed view Z' taken of FIG. 27E. FIG. 28A illustrates a top view of the optic adapter 7505 of FIG. 27A. FIG. 28B illustrates a side view of the optic adapter 7505 of FIG. 27A. FIG. 28C illustrates a top view of an optic adapter assembly including the optic adapter 7505 of FIG. 27A. FIG. 28D illustrates a side view of the optic adapter assembly of FIG. 28C.

The optic adapter 7505 may be similar in any respect to the optic adapter 7405 (FIG. 26A) or any other optic adapter described herein. The slide 7520 may be similar to any slide described herein. In this example, the rotatably driven fastener is a nut 7575, which may be similar to any nut described herein.

Referring to FIGS. 27C and 27D, in this embodiment the tapered opening is defined by the optic 7570. As a threaded section of the nut 7575 is installed onto a corresponding section of the slide 7520 (e.g., onto the threaded post of the bottom up fastener, in this example), contact 7588 by the non-threaded section of the nut 7575 and the tapered opening defined by the optic 7570 urges/drives the optic 7570 rearward, producing contact 7599 (FIG. 27E). A gap 7586 (FIG. 27F) may be similar to gap 7486 (FIG. 26C).

Referring now to FIGS. 27E and 27F, the urging also produces contact 7599 between the optic 7570 and part of a recoil post 7590 defined by the top side of the optic adapter 7505. This contact 7599 may fix a position of the optic 7570 relative to the optic adapter 7505 in a similar way that the contact 7499 (FIG. 26D) fixes a position of the optic adapter 7405 relative to the slide 7420. The contact 7599 may also protect the threaded post of the bottom-up fastener from shear forces in a similar way that the contact 7499 may protect the rotatably driven fastener 7475 from shear forces. The other side of the recoil post 7590 may define a gap 7586. FIGS. 28A-D illustrate the optic adapter 7520, and show a location of the recoil post 7590.

A tapered opening may also be used in a direct mount embodiment. FIG. 29A illustrates a top view of another slide assembly 7600 including a tapered opening to urge contact between a mounted component and a base component, according to various embodiments. FIG. 29B illustrates a section view taken along section line Z''Z'' of FIG. 29A. FIG. 29C illustrates a detailed view Z''' of FIG. 29B.

Referring to FIG. 29A, the optic 7670 may be similar in any respect to the optic 7570 (FIG. 27A). The slide 7605 may be similar in any respect to any slide described herein, and its top surface may be similar in any respect to any top surface of any optic adapter described herein. The top surface of the slide 7605 may define a recoil post 7690 (FIG. 29B), which may be similar to any recoil post described herein.

Referring now to FIGS. 29B and 29C, the when the fastener is tightened, contact 7699 of the head with the tapered opening may drive the optic 7670 in a rearward direction—a gap 7686 may also form. The forward urging may result in contact 7698 by a surface of a rear end of the optic 7670 with the slide 7620, which may be similar in any respect to the contact 7499 (FIG. 26D) by the surface of the rear end of the optic adapter 7405 and the slide 7420. The rearward urging may also result in contact by the recoil post 7690 and the optic 7670 similar in any respect to the contact by the recoil post 7590 (FIG. 27F) and the optic 7570. A position of the optic 7670 relative to the slide 7620 may be fixed based on the contact 7689 and/or recoil post contact. A gap 7689 may form in front of the optic 7670.

In any of the embodiments described above, an apparatus may include an assembly to operate in a recoil environment in which the assembly is mounted on a device that produces or transmits recoil. At least one part of the assembly (e.g., an optic, an optic adapter, or some other at least one body) may be fastened to at least one surface of at least one base part, the at least one base part comprising another part of the assembly (e.g., an optic adapter or a slide), a reciprocating part of a recoil-producing device (e.g., a slide), or another part of the recoil-producing device. The at least one part and the at least one base part may include a threaded opening (e.g., a threaded hole defined by a slide, or a threaded opening defined by a nut), and a non-threaded opening defining: a tapered opening to urge the at least one part of the assembly against one or more additional surfaces (e.g., against a surface defined by a recoil post, or against a surface defined by a slide), in response to installing a threaded fastener into the threaded opening (e.g., into a threaded hole) or in response to installing the threaded opening onto a threaded length (e.g., a nut onto a threaded post or other threaded length).

In the above-described embodiments, a tapered opening is used to urge the mounted component against a surface. In other embodiments, it may be possible to use some other counter opening having a center axis that is offset relative to a center axis of a threaded opening. For instance, FIG. 2B illustrates one example of such a counter opening (a counter opening for a flat headed rotatably driven fastener). During installation of the rotatably driven fastener, a bias can be applied by a person and/or a tool, to urge the mounted component against a surface while tightening the flat headed rotatably driven fastener.

Any non-threaded opening in an embodiment with a tapered opening to urge contact between a mounted component and a base component may include any locating features described herein, such as a non-threaded shoulder. In some embodiments, the base component may be a slide or other single part. However, this is not required. In various embodiments, the base component may be an assembly of parts (the surface that the mounted component may be urged against is not required to be the same part as the base component).

Examples in Urging Contact Between a Mounted Component and a Base Component in a Recoil Environment

Example 1: An apparatus, comprising:

an assembly mountable on a firearm, wherein at least one part of the assembly is fastenable to at least one surface of at least one base part;
the at least one part and the at least one base part including:

a threaded opening, and

a non-threaded opening defining a tapered opening arranged to urge the at least one part of the assembly against at least one additional surface, in response to fastening the at least one part of the assembly to the at least one base part using the threaded opening.

Example 2: The apparatus of example 1 or any other example herein, wherein the at least one additional surface is located above the at least one surfaces.

Example 3: The apparatus of example 1 or any other example herein, wherein the at least one additional surface comprises a sidewall of an opening defined by the at least one base part, wherein the at least one part of the assembly is located in the opening.

Example 4: The apparatus of example 1 or any other example herein, wherein the at least one part comprises a slide or other single part, wherein the at least one additional surface is defined by the slide or single part.

Example 5: The apparatus of example 1 or any other example herein, wherein the base part comprises another part of the assembly, a reciprocating part of the firearm, or another part of the firearm.

Example 6: The apparatus of example 1 or any other example herein, wherein the threaded opening comprises a threaded hole defined by the base part.

Example 7: The apparatus of example 1 or any other example herein, wherein the assembly includes a nut, wherein the threaded opening is defined by the nut.

Example 8: The apparatus of example 1 or any other example herein, wherein the threaded opening receives a threaded post to attach the at least one part to the at least one base part.

Example 9: The apparatus of example 1 or any other example herein, wherein the at least one surface is defined by a slide or an optic adapter, and wherein the at least one additional surface comprises part of a recoil post defined by the at least one base part or another part of the assembly.

Example 10: The apparatus of example 1 or any other example herein, wherein the assembly includes a rotatably driven fastener, a non-threaded part of which is located in the tapered opening, wherein the tapered opening is arranged to urge the at least one part of the assembly in a direction that is non-parallel with a center axis of the rotatably driven fastener.

Example 11: An apparatus, comprising:

at least one body to operate in a recoil environment in which the assembly is mounted on at least one surface of a part or assembly that produces or transmits recoil;
a rotatably driven fastener including at least one non-threaded section and at least one threaded section, wherein the at least one threaded section comprising an interior of the rotatably driven fastener or an exterior of the rotatably driven fastener;

the at least one body including:

an attachment interface arranged to attach the at least one body to the at least one surface, the attachment interface including a tapered opening to receive the at least one non-threaded section of the rotatably driven fastener;

wherein the attachment interface is arranged to urge the at least one body against at least one additional surface, in response to tightening the rotatably driven fastener.

Example 12: The apparatus of example 11 or any other example herein, wherein the part or assembly that produces or transmits recoil comprises a reciprocating part, or at least one part mountable on the reciprocating part.

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Example 13, The apparatus of example 12 or any other example herein, wherein the reciprocating part is a slide.

Example 14: The apparatus of example 11 or any other example herein, wherein the at least one body comprises an optic or an optic adapter.

Example 15: The apparatus of example 11 or any other example herein, wherein the rotatably driven fastener comprises a nut.

Example 16: The apparatus of example 11 or any other example herein, wherein the at least one threaded section of the rotatably driven fastener comprises an external thread.

We claim all modifications and variations coming within the spirit and scope of the following claims.

The invention claimed is:

1. An apparatus usable in a firearm system, the firearm system having a front end including an opening a fired projectile exits from, and a rear end opposite the front end, the apparatus comprising:

an assembly mountable on the firearm system, wherein at least one part of the assembly is fastenable to at least one surface of at least one base part, using one or more threaded lengths associated with the at least one surface of the at least one base part;

the at least one part and the at least one base part including:

a threaded opening, and
a non-threaded opening defining a tapered opening arranged to rearwardly urge the at least one part of the assembly against at least one buttressing surface, in response to fastening the at least one part of the assembly to the at least one base part using the threaded opening, wherein the at least one buttressing surface is not part of the one or more threaded lengths;

the rearward urging against the at least one buttressing surface to protect the one or more threaded lengths from sheering forces when a rearward force, in a direction from the front end of the firearm system to a rear end of the firearm system, is applied to the firearm system.

2. The apparatus of claim 1, wherein the at least one buttressing surface is located above the at least one surface of the at least one base part.

3. The apparatus of claim 1, wherein the at least one buttressing surface comprises a sidewall of an opening defined by the at least one base part, wherein the at least one part of the assembly is located in the opening defined by the at least one base part.

4. The apparatus of claim 1, wherein the at least one part comprises a slide or another single part, wherein the at least one buttressing surface is defined by the slide or the other single part.

5. The apparatus of claim 1, wherein the at least one base part comprises another part of the assembly, a reciprocating part of the firearm system, or another part of the firearm system.

6. The apparatus of claim 1, wherein the threaded opening comprises a threaded hole defined by the at least one base part.

7. The apparatus of claim 1, wherein the assembly includes a nut, wherein the threaded opening is defined by the nut.

8. The apparatus of claim 1, wherein a threaded length of the one or more threaded lengths is part of a threaded post, wherein the threaded opening receives the threaded post to attach the at least one part to the at least one base part.

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9. The apparatus of claim 1, wherein the at least one surface is defined by a slide or an optic adapter, and wherein the at least one buttressing surface comprises part of a recoil post defined by the at least one base part or another part of the assembly.

10. The apparatus of claim 1, wherein a threaded length of the one or more threaded lengths is part of a rotatably driven fastener, a non-threaded part of which is located in the tapered opening, wherein the tapered opening is arranged to urge the at least one part of the assembly in a direction that is non-parallel with a center axis of the rotatably driven fastener.

11. An apparatus including at least one body to operate with a firearm, the at least one body to couple to at least one surface of a mating body using one or more threaded lengths, in which the mating body is another body of the at least one part, a reciprocating body of the firearm, or another body of the firearm, the apparatus comprising:

a rotatably driven part including 1) at least one non-threaded section including a tapered portion and 2) at least one threaded section, the at least one threaded section to mate with a threaded section associated with the mating body;

the at least one body including a non-threaded opening locatable over the threaded section associated with the mating body when the bodies are coupled;

the non-threaded opening comprising an offset counter opening, the offset counter opening to receive the tapered portion as the rotatably driven part is tightened, wherein the offset counter opening is urged relative to the mating body based on contact of the tapered portion with the offset counter opening as the rotatably driven part is tightened;

wherein the urge relative to the mating body is against at least one buttressing surface to protect the at least one threaded length from sheering forces when, a rearward force, in a direction from a front end of the firearm to a rear end of the firearm, the front end including an opening a projectile exits from, the rear end opposite the front end, and wherein the at least one buttressing surface is not of the one or more threaded lengths.

12. A slide assembly comprising the apparatus of claim 11, the slide assembly including a slide, wherein the slide comprises the reciprocating part.

13. An optic assembly comprising the apparatus of claim 11, wherein the at least one body includes an optic sight of the optic assembly or an optic adapter of the optic assembly.

14. The apparatus of claim 11, wherein the rotatably driven fastener comprises a nut.

15. The apparatus of claim 11, wherein the at least one threaded section of the rotatably driven fastener comprises an external thread.

16. The apparatus of claim 11, wherein the rotatably driven part threads on a threaded length of the one or more threaded lengths.

17. The apparatus of claim 11, wherein the at least one threaded section of the rotatably driven part comprise a threaded length of the one or more threaded lengths.

18. An apparatus usable in a firearm system, the firearm system having a front end including an opening a fired projectile exits from, and a rear end opposite the front end, the apparatus comprising:

an assembly mountable on the firearm system, wherein at least one part of the assembly is fastenable to at least one surface of at least one base part, using one or more threaded lengths associated with the at least one surface of the at least one base part;

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the at least one part and the at least one base part including:

- a threaded opening, and
- a non-threaded opening defining a tapered opening arranged to urge, in a direction, the at least one part of the assembly relative to the at least one surface of the at least one base part, in response to fastening the at least one part of the assembly to the at least one base part using the threaded opening;

wherein the direction is one of 1) an aft direction and 2) a fore direction, in which:

- the aft direction from the front end of the firearm system to the rear end of the firearm system,
- the fore direction from the rear end of the firearm system to the front end of the firearm system,

wherein in the case the direction is the fore direction:

- the urge interlocks an undercut and a projection, one of the undercut and the projection being a portion of the at least one part, and the other of the undercut and the projection being a portion of the at least one base part.

19. The apparatus of claim 18, wherein the direction is the aft direction; and

- wherein the urge is against at least one buttressing surface, the at least one buttressing surface located above the at least one surface of the at least one base part.

20. The apparatus of claim 19, wherein the at least one buttressing surface comprises a sidewall of an opening defined by the at least one base part, wherein the at least one part of the assembly is located in the opening defined by the at least one base part.

21. The apparatus of claim 19, wherein the at least one part comprises a slide or another single part, wherein the at least one buttressing surface is defined by the slide or the other single part.

22. The apparatus of claim 19, wherein the at least one base part comprises another part of the assembly, a reciprocating part of the firearm system, or another part of the firearm system.

23. The apparatus of claim 18, wherein the threaded opening comprises a threaded hole defined by the at least one base part.

24. The apparatus of claim 18, wherein the assembly includes a nut, wherein the threaded opening is defined by the nut.

25. The apparatus of claim 18, wherein a threaded length of the one or more threaded lengths is part of a threaded post, wherein the threaded opening receives the threaded post to attach the at least one part to the at least one base part.

26. The apparatus of claim 19, wherein the at least one surface is defined by a slide or an optic adapter, and wherein the at least one buttressing surface comprises part of a recoil post defined by the at least one base part or another part of the assembly.

27. The apparatus of claim 18, wherein a threaded length of the one or more threaded lengths is part of a rotatably

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driven fastener, a non-threaded part of which is located in the tapered opening, wherein the tapered opening is arranged to urge the at least one part of the assembly in a direction that is non-parallel with a center axis of the rotatably driven fastener.

28. An apparatus including at least one body to operate with a firearm, the at least one body to couple to at least one surface of a mating body using one or more threaded lengths, in which the mating body is another body of the at least one part, a reciprocating body of the firearm, or another body of the firearm, the apparatus comprising:

- a rotatably driven part including 1) at least one non-threaded section including a tapered portion and 2) at least one threaded section, the at least one threaded section to mate with a threaded section associated with the mating body;

- the at least one body including a non-threaded opening locatable over the threaded section associated with the mating body when the bodies are coupled;

- the non-threaded opening comprising an offset counter opening, the offset counter opening to receive the tapered portion as the rotatably driven part is tightened, wherein the offset counter opening is urged relative to the mating body based on contact of the tapered portion with the offset counter opening as the rotatably driven part is tightened,

wherein the urge of the at least one part relative to the mating body:

- 1) is against at least one buttressing surface to protect the one or more threaded lengths from sheering forces when, a rearward force, in a direction from a front end of the firearm, the front end including an opening a fired projectile exits from, to a rear end of the firearm, the rear end opposite the front end, wherein the at least one buttressing surface is not of the one or more threaded lengths, or

- 2) interlocks an undercut and a projection, wherein the at least one part includes one of the undercut and the projection, and the other of the undercut and the projection is defined by the mating body.

29. The apparatus of claim 28, wherein the tapered portion comprises a length including a sloped surface on one side and another surface on an opposite side of the length;

- wherein when the at least one part is urged against the at least one buttressing surface, or when the undercut and the projection are interlocked, 1) at least a portion the sloped surface is in contact with the non-threaded opening and 2) a corresponding at least a portion of the other surface is not in contact with the non-threaded opening.

30. The apparatus of claim 28, wherein the urge of the at least one part relative to the mating body is against the at least one buttressing surface, and

- wherein the at least one buttressing surface is located above the at least one surface of the mating body.

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